# Introduction to Probability

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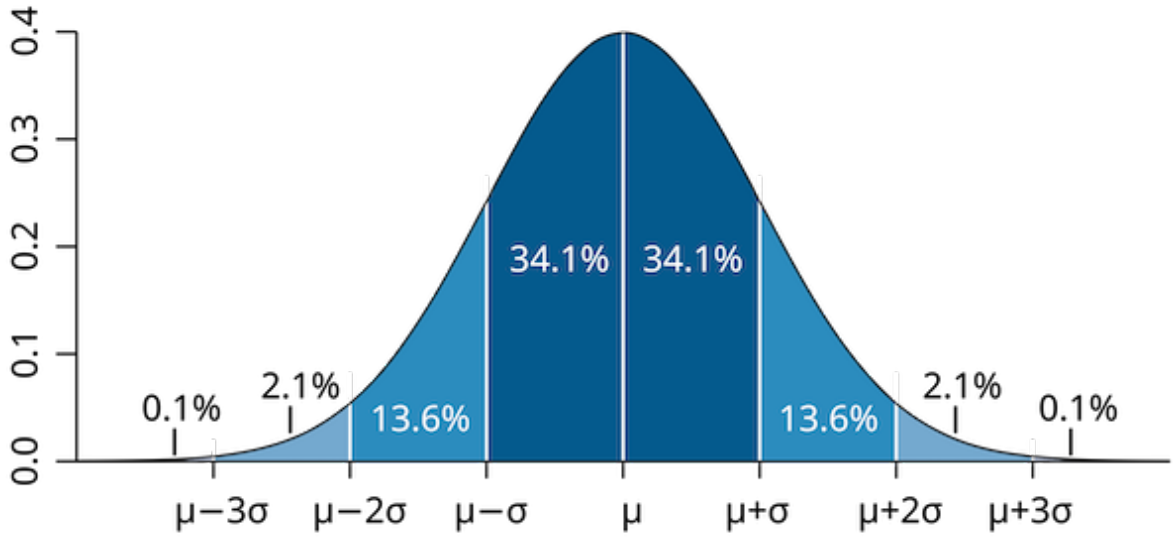
# Overview




This book is adapted from *MAT921: Probability* at Southwest University of Finance and Economics (RIEM). It is an introductory probability course that aims to be not boring. In this course, we try to blend **conventional teaching**, **interesting puzzles** and **data-oriented practical skills**.

Course Instructor's email: gamma12@126.com. Please indicate your class and student ID when you email me.

## Syllabus

**Topic 1: Classical probabilities**
How likely were some of your classmates born on the same day as you?

**Topic 2: Data and random variables**
Why is your exam score in this class a random variable?

**Topic 3: Discrete distributions**
How many earthquakes are likely to happen in a random year?

**Topic 4: Expectation and variance**
How old are you expected to live?

**Topic 5: Continuous distributions**
How long are you expected to wait in the queue at a restaurant?

**Topic 6: Limiting theorems**
Why a lottery company never loses?

**Topic 7: Sampling distribution**
How do I know I am taller than an average person?

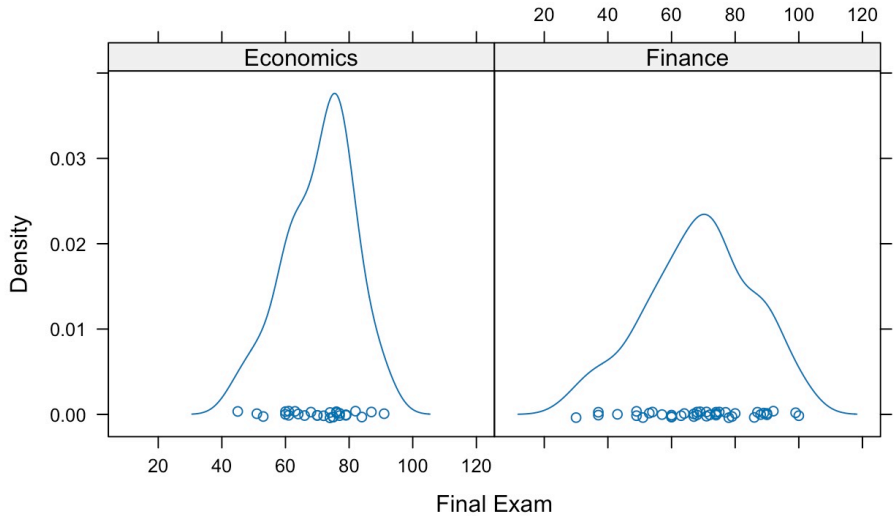
## Assessment

**Quiz (25%).** There will be an arbitrary number of in-class quizzes. The date for each quiz will be announced in advance. Each quiz will consist of 1-2 questions based on material covered in previous weeks. Every quiz is mandatory; there will be no make-up quizzes under any circumstances.

**Project (25%).** The goal of the projects is to encourage students to apply the knowledge learned in this course to solve practical problems. Projects are usually open-ended and may involve data analysis, simulations, or exploring real-world applications of probability. Essays that present innovative perspectives and use the data persuasively to support their conclusions will receive higher marks. Selective students may be invited to present their findings to the class.

**Final exam (50%).** The final exam will be a closed-book, paper-and-pencil exam scheduled for Week 17. It will not simply repeat lecture material but will assess your ability to apply the knowledge you have gained to solve novel problems. To perform well, you must have a deep understanding of the concepts and acquire some degree of problem-solving skills. The average score of the past exam is 69 with a standard deviation 15. The pass rate (>=60) is about 80%.

**Class participation (5%).** Additional 5 marks for class participation on top of the above. Regular attendance and constructive participation are are encouraged (though not mandatory) and will be recognized. Active engagement in class discussions or asking insightful questions will
also be considered.




## Lecture notes

All lecture materials will be published through this online website. You are not required to read any textbook. For students who insist on a textbook, it would be:

- DeGroot and Schervish's *Probability and Statistics (4th edition).*

It is recommended to use the textbook as a supplement not a replacement of the lecture note. For students who prefer to read the textbook. There are two key differences between this lecture note and the textbook. First, the sections are arranged differently. Second, the examples and exercises are entirely different despite the key definitions and theorems are the same.

## Homework

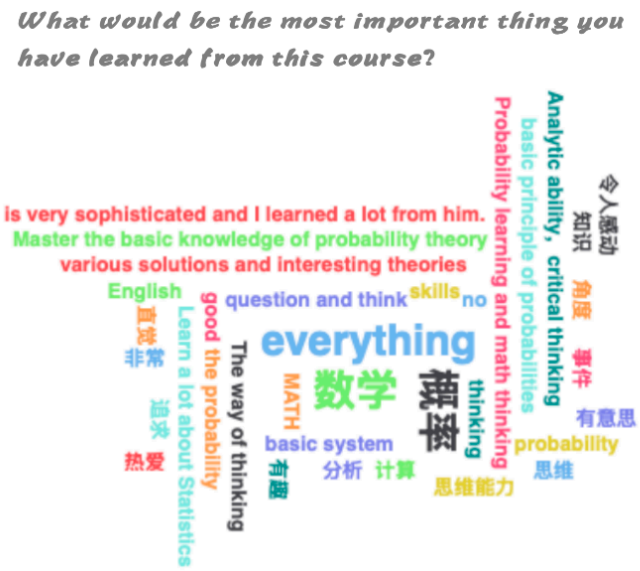
There is no homework assignment in this course. We will do in-class exercises instead. However, problem solving is essential for learning math. You are encouraged to practice the exercises in DeGroot and Schervish's textbook after class. But it is not mandatory.

## Statistical software

Statistical software is indispensable for modern statistics. For practical consideration, it is beneficial to start learn it as early as possible. We will demonstrate how to do statistics in R, which is a widely-used open-source statistical programming language. It is highly recommended that you try it yourself while learning this course, though it is not mandatory.

## Students' evaluation

If you have not decided whether to enroll in this course or not. Here are some surveys from the past students for your reference. In general, this is not an easy course especially for freshmen. We will deal with serious maths, though I will try to convey the beauty of the subject as much as possible.




*What would be the most important thing you have learned from this course?*

*What would you suggest the lecturer to improve?*

## Reference


1. Schervish, M. J., & DeGroot, M. H. (2014). *Probability and statistics.* Pearson Education.
2. Blitzstein, J. K., & Hwang, J. (2019). *Introduction to probability.* Chapman and Hall/CRC.
3. Hansen, B. (2022). *Probability and statistics for economists.* Princeton University Press.

4. Grimmett, G., & Stirzaker, D. (2020). *Probability and random processes.* Oxford University Press.

## Online playground

Probability Playground: Interactive Probability Distributions

StatKey: Statistics Unlocking the Power of Data

MathIsFun: Standard Normal Distribution Table

Central Limit Theorem Demo

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# Preface

Embarking on the study of probability is both an opportunity and a challenge. This course represents a critical foundation for all subsequent studies in econometrics and data science. As we navigate an era of big data and artificial intelligence, proficiency in probability theory is indispensable. It not only builds the foundations for higher level courses but also cultivates a mindset geared towards probabilistic reasoning, which is essential for mitigating cognitive biases and making more informed decisions. Human cognition is often not well-equipped to handle probabilistic thinking. As we will see, human intuition and heuristics are mostly wrong about probabilistic events. This course seeks to provide a mathematical framework for properly understanding and applying probability.

Probability theory is more than just a mathematical discipline; it is a vital tool for making sense of uncertainty in the real world. Consider the myriad of questions that we encounter daily: Will it rain tomorrow? What is the expected return on an investment? What are the odds of a particular political party winning an election? How can a business optimize its customer service strategy when customer arrival times are unpredictable? Probability theory provides a scientific approach to answering these questions, enabling us to model and analyze uncertainties with mathematical tools.

However, this journey is not without its difficulties. For freshmen, particularly those new to calculus and linear algebra, the course presents a steep learning curve. The breadth of new concepts—such as random variables, expectations, and various distributions—can be overwhelming if encountered for the first time. Additionally, the use of advanced calculus, particularly integrals, may pose challenges for those who are still familiarizing themselves with these mathematical tools.

Despite these challenges, the rewards of studying probability are substantial. Gaining a deep understanding of probability will not only enhance your knowledge base but also fundamentally transform your approach to problem-solving. The principles you will learn are applicable to a wide range of fields beyond econometrics and data science, including engineering, finance, and social sciences. You will learn the tools to approach these problems systematically and make informed decisions based on statistical evidence and probability. So be prepared for a challenging and rewarding journey!

## Learning objectives

- **Review fundamental concepts:** Revisit the probability and calculus concepts learned in high school to ensure a solid foundation for more advanced topics.
- **Understand core probability theory:** Gain a thorough understanding of key concepts and theorems in probability theory, including random variables, expectations, covariances, and so on.
- **Develop probabilistic thinking:** Learn to approach problems with a probabilistic mindset and use random variables to describe and analyze uncertain outcomes.
- **Model real-world events:** Identify and apply important probability distributions to model and interpret real-world phenomena effectively.
- **Enjoy and have fun:** Discover and appreciate the inherent elegance of mathematics and the beauty of probability theory.

## Study tips

- **Limit electronic distractions:** While digital tools like slides and tablets are convenient, traditional paper and pencil methods remain the most effective way to engage with and learn mathematics. Writing out problems and solutions helps reinforce concepts and improve retention.
- **Focus on key concepts:** College courses are often much more intensive than high school classes, and it is not feasible to master every detail. Concentrate on understanding the core ideas and principles, and don't get overwhelmed by the technical details.
- **Understand the "why":** In mathematics, understanding the underlying reasons and logic behind methods is more important than just knowing how to do computations. The "why" helps you grasp the broader implications and applications of the techniques you learn.
- **Gain practical experience:** Although this course emphasizes theoretical understanding and does not require programming, experimenting with statistical software such as R can be highly beneficial. Practical experience with data manipulation and analysis will enhance your comprehension and stimulate interest in the subject.
- **Engage with the material:** I will strive to make the course engaging and less boring. However, if this course is not your primary interest, focus on the aspects of the material that intrigue you. Try to have a general impression of the major concepts even though you do not remember any detail.
- **Exams are important, but more important is to enjoy the course.**

This content of this book is organized or follows. We start with probabilities based on counting, which should be familiar to high school graduates. Though rudimentary, they often yield

surprising results, as rigorous calculations frequently challenge our intuitions about the likelihood of events. Special emphasis is placed on conditional probability, as conditional thinking is crucial both in academic studies as well as in daily life.

Next, we introduce the core concept of the random variable, which forms the foundation of all probability distributions and statistical theory. Random variables are essential tools that allow us to mathematically model uncertainty. We introduce two types of random variables: discrete and continuous. We begin with discrete random variables because they do not require calculus, offering a smoother learning curve for beginners. Key concepts such as expectations, variance, and covariance are introduced alongside well-known discrete distributions such as the Binomial, Geometric, and Poisson distributions. This arrangement ensures that students can grasp these important concepts without being overwhelmed by calculus. We also demonstrate how these fundamental distributions can be applied to solve real-world problems.

Following this, we move on to continuous distributions. We will see that the formulas from discrete distributions extend naturally to continuous distributions with the aid of calculus—essentially replacing summation with integration. We cover some of the most important continuous distributions, such as the Normal, Exponential, and Gamma distributions, and explore the interconnections between them. We also extend the concepts of expectations, variance, and joint distributions to their continuous forms.

The book concludes with a discussion on sampling and statistical inference. Since we cannot observe entire distribution, it becomes necessary to infer distribution properties from finite samples. We introduce two of the most important theorems in probability and statistical theory—the Law of Large Numbers and the Central Limit Theorem. The breadth and generality of these theorems are remarkable. But their most significant contribution to statistical applications is they allow us to gauge how close our sample estimates are to the true parameter values. The final chapter also includes a brief discussion on estimator accuracy, confidence intervals, and hypothesis testing. These topics are introduced briefly, as they serve primarily to prepare students for more advanced courses, such as econometrics.

The chapters are organized logically, with each chapter building on the knowledge presented in the previous ones. Therefore, it is recommended to follow the sequence of chapters rather than reading them independently. However, advanced readers who are already familiar with the topics may feel free to skip between chapters as needed. This manuscript is written tersely, serving as a skeleton to complement lecture materials. It is not intended as a substitute for lectures or comprehensive textbooks. Students who wish to learn the course material solely by reading are encouraged to consult a formal textbook.

This manuscript is a preliminary version, and while efforts have been made to ensure accuracy, errors may still be present. Your feedback on any mistakes or inaccuracies is greatly appreciated and will help improve the material.

# Part I

# Probability Basics

# 1 Logic of uncertainty

> Probability is the most important concept in modern science, especially as nobody has the slightest notion of what it means. —— Bertrand Russell

What is probability? We all talk about probabilities in everyday life, but mostly in vague languages. This course is to introduce probability as a logical framework for quantifying uncertainty and randomness.

> Mathematics is the logic of certainty; probability is the logic of uncertainty.

The earliest development of probability is rooted in gambling. For instance, the renowned Monte Carlo method in statistics, invented by Stanislaw Ulam in the late 1940s, takes its name from the *Monte Carlo Casino* in Monaco, where Ulam's uncle would borrow money from relatives to gamble. Probability theories still apply today to analyze gambling odds, but their applications have expanded to nearly every field. It is the foundation of statistics, machine learning, and artificial intelligence. It also plays a crucial role in everyday decision-making, from stock investments to effective strategies to combat an infectious disease.

Probability is a concept that is intuitive to understand but very hard to define formally. Perhaps, the first formal definition of probability is often attributed to Pierre-Simon Laplace in the 18th century. In his work Theorie analytique des probabilites, published in 1812,

> The probability of an event is the ratio of the number of cases favorable to it, to the number of all cases possible when nothing leads us to expect that any one of these cases should occur more than any other, which renders them, for us, equally possible.

This definition is outdated, as we will soon discover. But before we explore the modern definition of probability, let's first clarify some preliminary concepts (based on sets), which is the mathematical language we use to describe probabilistic events.

# 2 Event and sample space

**Definition 2.1.** We use *sets* to build the foundational concepts in probability:

- **Experiment**: a procedure with an uncertain outcome
- **Event** $A$: a set of possible outcomes
- **Sample space** $S$: the set of all possible outcomes

Anything (a gamble, an exam, a financial year, ...) with uncertain outcomes can be an experiment. The sample space can be finite, countably infinite, or uncountably infinite. It is convenient to visualize events with *Venn diagrams*.

> **Don's confuse events with outcomes**
>
> Outcomes are individual results, while events are groups of outcomes that we are interested in. Outcomes are elements of the sample space, and events are subsets of this space.

**Example 2.1.** A coin is flipped twice. We write "H" if a coin lands Head, and "T" if a coin lands Tail.

- The sample space (all possible outcomes): $S = \{HH, HT, TH, TT\}$.
- Let $A_1$ be the event that the first flip is Heads, $A_1 = \{HH, HT\}$.
- Let $A_2$ be the event that the second flip is Heads, $A_2 = \{HH, TH\}$.
- Let $B$ be the event that at least one flip is Heads, $B = A_1 \cup A_2$.
- Let $C$ be the event that all the flips are Heads, $C = A_1 \cap A_2$.
- Let $D$ be the event that no flip is Heads, $D = B^c$.

### Use set language to describe events

| English | Sets |
|---|---|
| sample space | $S$ |
| $s$ is a possible outcome | $s \in S$ |
| $A$ is an event | $A \subseteq S$ |
| $A$ occurs | $s_{\text{actual}} \in A$ |

| English | Sets |
|---|---|
| $A$ or $B$ occurs | $A \cup B$ |
| Both $A$ and $B$ occur | $A \cap B$ |
| $A$ does not occur | $A^c$ |
| at least one of $A_1, \dots, A_n$ | $A_1 \cup \cdots \cup A_n$ |
| all of $A_1, \dots, A_n$ | $A_1 \cap \cdots \cap A_n$ |
| $A$ implies $B$ | $A \subseteq B$ |
| $A$ and $B$ are disjoint (mutually exclusive) | $A \cap B = \phi$ |
| $A_1, \dots, A_n$are a partition of $S$ | $A_1 \cup \cdots \cup A_n = S$ and $A_i \cap A_j = \phi$ for $i \neq j$ |

# 3 Classical probability

**Definition 3.1.** Classical (naive) definition of probability:

$$P(A) = \frac{|A|}{|S|} = \frac{\text{number of outcomes favorable to A}}{\text{total number of outcomes in A}}$$

assuming the outcomes are finite and equally likely.

> **Don's misuse the classical definition**
>
> The classical probability is very restrictive. It only applies to scenarios such as flipping coins or rolling dice where the outcomes are equally likely. It has often been misapplied by people who assume equally likely outcomes without justification. For example, if one wants calculate the probability of a rainy day, it would be misleading to assume every day is equally likely to rain and compute $\frac{\text{rainy days}}{365}$.

Calculating the naive probability of an event $A$ often involves counting the number of outcomes in $A$ and the number of outcomes in the sample space $S$, which usually involve some counting methods. We now review some of the counting methods (multiplications, factorials, permutations, combinations) that was introduced in high schools.

## Counting methods

**Multiplications.** Consider a compound experiment consisting of two sub-experiments, Experiment A and Experiment B. Suppose that Experiment A has $a$ possible outcomes, and for each of those outcomes Experiment B has $b$ possible outcomes. Then the compound experiment has $a \times b$ possible outcomes.

**Exponentiation.** Consider $n$ objects and making $k$ choices from them, one at a time with replacement. Then there are $n^k$ possible outcomes.

**Factorials.** Consider $n$ objects $1, 2, \ldots, n$. A permutation of $1, 2, \ldots, n$ is an arrangement of them in some order, e.g., $3, 5, 1, 2, 4$ is a permutation of $1, 2, 3, 4, 5$. The are $n!$ permutations of $1, 2, \ldots, n$.

**Permutations**. Consider $n$ objects and making $k$ choices from them, one at a time without replacement. Then there are $P_n^k = n(n-1)\cdots(n-k+1)$ possible outcomes, for $k \leq n$. (Ordering matters in this case, e.g. $1, 2, 3$ is considered different from $2, 3, 1$)

**Combinations**. Consider $n$ objects and making $k$ choices from them, one at a time without replacement, without distinguishing between the different orders in which they could be chosen (e.g. $1, 2, 3$ is considered no different from $2, 3, 1$). Then there are $C_n^k = \frac{n(n-1)\cdots(n-k+1)}{k!}$ possible outcomes. In modern math, we prefer the notation $\binom{n}{k} := C_n^k$, which reads as "$n$ choose $k$".

The following table summarizes the counting methods.

| | order matters | order doesn't matter |
|---|---|---|
| with replacement | $n^k$ | $C_{n+k-1}^k$ |
| non-replacement | $P_n^k$ | $C_n^k$ |

The upper-right corner case is equivalent to putting $k$ indistinguishable balls into $n$ distinguishable baskets (e.g. two balls in Basket 3 means the 3rd object is chosen twice). Therefore, the number of possible arrangements is $C_{k+n-1}^{n-1}$.

## Binomial coefficient

The Binomial coefficient $\binom{n}{k}$ counts the number of subsets of size $k$ for a set of size $n$. It is also the coefficient of $x^k$ when expanding the polynomial $(x+y)^n$.

**Theorem 3.1** (Binomial theorem)**.**

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}.$$

**"Binomial Expansion"**

| | |
|---|---|
| $(a+b)^0 =$ | $1$ |
| $(a+b)^1 =$ | $a+b$ |
| $(a+b)^2 =$ | $a^2 + 2ab + b^2$ |
| $(a+b)^3 =$ | $a^3 + 3a^2b + 3ab^2 + b^3$ |
| $(a+b)^4 =$ | $a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$ |
| $(a+b)^5 =$ | $a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + b^5$ |

The coefficients form an infinite triangle called the **Pascal triangle**. By observing the patterns in the triangle, it is not hard to conclude the following recursive formula:

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

The value of the binomial coefficient is only defined for non-negative integers $n$ and $k$ with $0 \leq k \leq n$. But mathematics is about **generalization**. We can generalize the notion of "$n$ choose $k$" for negative $n$:

$$\begin{aligned}
\binom{-n}{k} &= \prod_{i=0}^{k-1} \frac{-n-i}{k-i} = (-1)^k \prod_{i=0}^{k-1} \frac{n+i}{k-i} \\
&= (-1)^k \frac{n(n+1)\ldots(n+k-1)}{k!} \\
&= (-1)^k \frac{(n+k-1)!}{k!(n-1)!} \\
&= (-1)^k \binom{n+k-1}{k}
\end{aligned}$$

We can also extend the formula to real numbers and even complex numbers:

$$\binom{x}{y} = \frac{\Gamma(x+1)}{\Gamma(y+1)\Gamma(x-y+1)}$$

where $\Gamma(x+1) = x!$ is a generalization of factorials. We will come back to the Gamma function when we discuss Gamma distributions.

# 4 Gambling problems

**Example 4.1.** Texas hold'em is one of the most popular variant of the card game of poker. Essentially, the players in the game bet on the rankings of their hand of five cards (illustrated in the figure below). For the game to be fair, a hand of higher values must have lower probability than a hand of lower values. In this example, we compute the probabilities of different patterns of cards (assuming a total of 52 cards, excluding the 2 jokers).

*Solution.* To compute the probabilities of different five-card poker hands, we first determine the total number of possible five-card hands from a standard 52-card deck. The total number

of combinations is:

$$\binom{52}{5} = \frac{52 \times 51 \times 50 \times 49 \times 48}{5 \times 4 \times 3 \times 2 \times 1} = 2,598,960$$

Now, we find the number of combinations for each hand type and use this to compute the probabilities.

1. Straight Flush

   A straight flush consists of five consecutive cards of the same suit. There are 10 possible sequences (A-5 through 10-A), and each sequence can occur in any of the 4 suits. Thus, the total number of straight flushes is: $4 \times 10 = 40$. The probability is:

   $$P(\text{Straight Flush}) = \frac{40}{2598960} \approx 0.00154\%$$

2. Four of a Kind (Quads)

   Four of a kind consists of four cards of the same rank plus one additional card. There are 13 possible ranks, and for each rank, there are 4 cards that form the quads. The fifth card can be any of the remaining 48 cards. Therefore, the total number of quads is: $13 \times 48 = 624$. The probability is:

   $$P(\text{Four of a Kind}) = \frac{624}{2598960} \approx 0.024\%$$

3. Full House

   A full house consists of three cards of one rank and two cards of another rank. The number of ways to choose the rank for the triplet is $\binom{13}{1} = 13$ , and the number of ways to choose 3 cards of that rank is $\binom{4}{3} = 4$ . The number of ways to choose the rank for the pair is $\binom{12}{1} = 12$ , and the number of ways to choose 2 cards of that rank is $\binom{4}{2} = 6$ . Therefore, the total number of full houses is: $13 \times 4 \times 12 \times 6 = 3744$. The probability is:

   $$P(\text{Full House}) = \frac{3744}{2598960} \approx 0.144\%$$

4. Flush

   A flush consists of five cards of the same suit that are not in sequence. The number of ways to choose a suit is $\binom{4}{1} = 4$ , and the number of ways to choose 5 cards from that suit is $\binom{13}{5} = 1287$ . Since this includes straight flushes, we subtract 40 (the number of straight flushes): $4 \times 1287 - 40 = 5148$. The probability is:

   $$P(\text{Flush}) = \frac{5148}{2598960} \approx 0.198\%$$

5. Straight

   A straight consists of five consecutive cards of different suits. There are 10 possible sequences, and for each sequence, there are $4^5 = 1024$ combinations. Subtracting the 40 straight flushes gives: $10 \times 1024 - 40 = 10200$. The probability is:

$$P(\text{Straight}) = \frac{10200}{2598960} \approx 0.392\%$$

6. Three of a kind

   Three of a kind consists of three cards of the same rank plus two unrelated cards. The number of ways to choose the rank for the triplet is $\binom{13}{1} = 13$ , and the number of ways to choose 3 cards of that rank is $\binom{4}{3} = 4$ . The two additional ranks are chosen from the remaining 12, with one card selected from each: $13 \times 4 \times \binom{12}{2} \times 4^2 = 13 \times 4 \times 66 \times 16 = 54912$. The probability is:

$$P(\text{Three of a Kind}) = \frac{54912}{2598960} \approx 2.11\%$$

7. Two pairs

   Two pair consists of two cards of one rank, two cards of another rank, and one unrelated card. The number of ways to choose 2 ranks is $\binom{13}{2} = 78$ , and the number of ways to choose 2 cards from each rank is $\binom{4}{2} = 6$ . The fifth card is chosen from the remaining 44: $78 \times 6^2 \times 44 = 123552$. The probability is:

$$P(\text{Two Pair}) = \frac{123552}{2598960} \approx 4.75\%$$

8. One pair

   One pair consists of two cards of one rank and three unrelated cards. The number of ways to choose the rank for the pair is $\binom{13}{1} = 13$ , and the number of ways to choose 2 cards from that rank is $\binom{4}{2} = 6$ . The three unrelated ranks are chosen from the remaining 12, with one card selected from each: $13 \times 6 \times \binom{12}{3} \times 4^3 = 13 \times 6 \times 220 \times 64 = 1098240$. The probability is:

$$P(\text{One Pair}) = \frac{1098240}{2598960} \approx 42.26\%$$

**Example 4.2.** (Newton-Pepys problem) The Newton-Pepys problem is a classical probability problem involving dice rolls, posed in correspondence between Samuel Pepys, a famous diarist and government official, and Isaac Newton in 1693. The problem concerns which of the following three events is most likely when rolling fair dice:

1. At least one 6 appears when 6 fair dice are rolled.
2. At least two 6's appear when 12 fair dice are rolled.

3. At least three 6's appear when 18 fair dice are rolled.

*Solution.* We compute the probabilities for each scenario:

1. Probability of at least one 6 in six rolls of a single die:

   The probability of not rolling a 6 in six rolls is:

$$P(\text{no 6 in six rolls}) = \frac{5^6}{6^6}$$

   Thus, the probability of at least one 6 in six rolls is:

$$P(\text{at least one 6}) = 1 - \frac{5^6}{6^6} \approx 0.67$$

2. Probability of at least two 6s in twelve rolls of a single die:

   We use the complement, finding the probabilities of getting 0 or 1 six. The probability of getting exactly 0 sixes in twelve rolls is similar as above. The probability of getting exactly 1 six is:

$$P(1 \text{ six}) = \binom{12}{1}\frac{5^{11}}{6^{12}}$$

   Thus, the probability of at least two 6s is:

$$P(\text{at least two 6s}) = 1 - P(0 \text{ six}) - P(1 \text{ six}) \approx 0.62$$

3. Probability of at least three 6s in eighteen rolls of a single die:

   Similarly, we calculate the complement, finding the probabilities of getting fewer than 3 sixes.

$$P(\text{at least three 6s}) = 1 - P(0 \text{ six}) - P(1 \text{ six}) - P(2 \text{ sixes})$$

$$= 1 - \frac{5^{18} + \binom{18}{1}5^{17} + \binom{18}{2}5^{16}}{6^{18}} \approx 0.60$$

Thus, $P(\text{one 6 in 6 rolls}) > P(\text{two 6s in 12 rolls}) > P(\text{three 6s in 18 rolls})$.

Intuitively, this is because as the number of dice increases, the likelihood of matching higher thresholds does not keep pace with the probability of rolling individual sixes. This is perhaps contrary to most people's common sense: the more dice rolled, the more likely to see the sixes.

# 5 Birthday paradox

The birthday problem is a classic probability puzzle that demonstrates how likely it is for at least two people in a group to share the same birthday. While it might seem intuitive that the probability is low in small groups, the results are surprising.

**Example 5.1.** In a group of $k$ people, what is the probability that at least two people share the same birthday? Assume (1) there are 365 possible birthdays; (2) birthdays are evenly distributed across the year; (3) people are equally likely to be born on any given day.

*Solution.* If $k > 365$, the probability is 1. Assume $k \leq 365$ for the rest. Instead of directly calculating the probability of at least two people sharing a birthday, we first compute the complementary probability, $P(\text{no match})$ , where no two people in the group have the same birthday.

For the first person, there are 365 choices for their birthday. For the second person, to avoid a shared birthday, there are 364 remaining choices. For the third person, there are 363 choices, and so on. For $k$ people, the total number of arrangements (no shared birthday) is:

$$365 \times 364 \times 363 \times \cdots \times (365 - k + 1)$$

The total number of possible arrangements (with or without shared birthdays) is $365^k$. Thus, the probability of no shared birthday is:

$$P(\text{no match}) = \frac{365 \cdot 364 \cdots (365 - k + 1)}{365^k}$$

Thus, the probability of at least one matched birthday is:

$$P(\text{match}) = 1 - P(\text{no match}) = \begin{cases} 50.7\% & k = 23 \\ 70.6\% & k = 30 \\ 97.0\% & k = 50 \\ 99.9\% & k = 70 \end{cases}$$

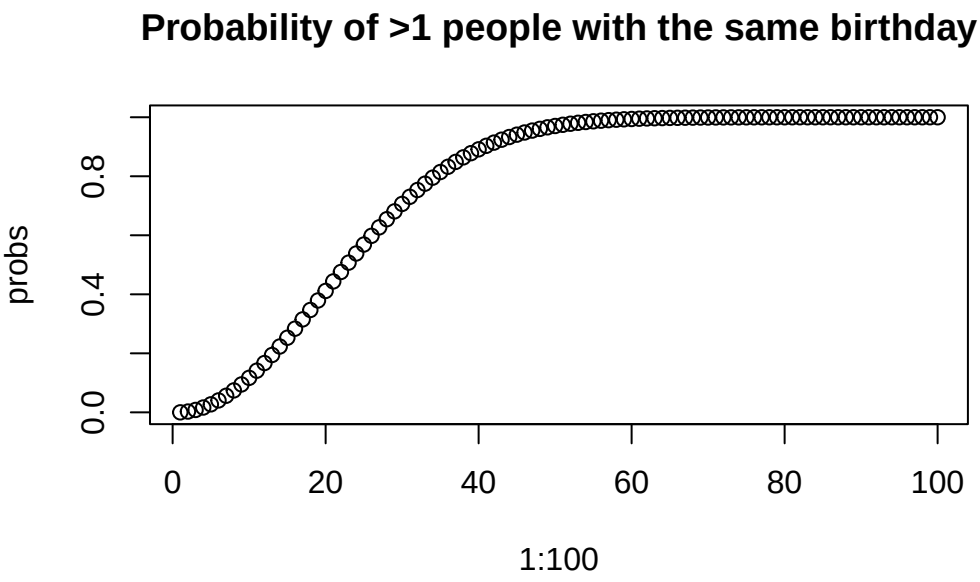
## R simulation

```r
# the birthday experiment with k people
birthday_experiment <- function(k = 30) {

  # draw k random numbers from 1 to 365 with replacement
  birthdays <- sample(1:365, k, replace = T)

  # number of duplicated birthdays
  dups <- length(birthdays) - length(unique(birthdays))

  # return the value
  dups
}

# repeat the experiment 1000 times
results <- replicate(1000, birthday_experiment(k = 30))

# count frequencies
table(results) |> knitr::kable()
```

| results | Freq |
|---|---|
| 0 | 292 |
| 1 | 375 |
| 2 | 239 |
| 3 | 77 |
| 4 | 15 |
| 6 | 2 |

There is even a built-in function `pbirthday` that computes the probability of birthday coincidence. We may utilize this function to plot the probability as the number of people increases.

```r
# compute the probability of birthday match for 30 people
prob <- pbirthday(30)

# compute a vector of probabilities for 1,2...100 people
probs <- sapply(1:100, pbirthday)

# make a plot
plot(1:100, probs, main = "Probability of >1 people with the same birthday")
```


Probability of >1 people with the same birthday
probs
0.0
0.4
0.8
0
20
40
60
80
100
1:100

# 6 Axiomatic probability

**Definition 6.1.** A *probability space* consists of $S$ and $P$, where $S$ is a sample space, and $P$ is a function which takes an event $A \subseteq S$ as input and returns $P(A) \in [0, 1]$ such that

- $P(\phi) = 0$,
- $P(S) = 1$,
- $P(\cup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} P(A_n)$ if $A_1, A_2, \dots, A_n$ are disjoint.

Note that this Definition does not imply any particular interpretation of probability. In fact, any function $P$ that satisfies the axioms are valid "probabilities". Thus, the theories of probability do not depend on any particular interpretation. It is purely axiomatic. From the three axioms, we can derive any property of probabilities. The interpretation also matters, but it is more of a philosophical debate.

> **Two interpretations of probability**
>
> - The *frequentist* view of probability is that it represents a long-run frequency over a large number of repetitions of an experiment: if we say a coin has probability 1/2 of Heads, that means the coin would land Heads 50% of the time if we tossed it over and over and over.
> - The *Bayesian* view of probability is that it represents a degree of belief about the event in question, so we can assign probabilities to hypotheses like "candidate A will win the election" or "the defendant is guilty" even if it isn't possible to repeat the same election or the same crime over and over again.

**Proposition 6.1.** *For any events $A$ and $B$, we have*

- $P(A^c) = 1 - P(A)$
- *If $A \subseteq B$, then $P(A) \leq P(B)$.*
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

*Proof.* We prove the three properties by just using the Axioms.

1) Since $A$ and $A^c$ are disjoint and their union is $S$, apply the third axiom:

$$P(S) = P(A \cup A^c) = P(A) + P(A^c);$$

By the second axiom, $P(S) = 1$. So $P(A) + P(A^c) = 1$.

2) The key is to break up the set into disjoint sets. If $A \subseteq B$, then $B = A \cup (B \cap A^c)$ where $A$ and $B \cap A^c$ are disjoint (draw a Venn diagram for intuition). By the third axiom, we have
$$P(B) = P(A \cup (B \cap A^c)) = P(A) + P(B \cap A^c) \geq P(A).$$

3) We can write $A \cup B$ as the union of the disjoint set $A$ and $B \cap A^c$. Then by the third axiom,
$$P(A \cup B) = P(A \cup (B \cap A^c)) = P(A) + P(B \cap A^c).$$
It suffices to show that $P(B \cap A^c) = P(B) - P(A \cap B)$. Since $B \cap A$ and $B \cap A^c$ are disjoint, we have
$$P(B) = P(B \cap A) + P(B \cap A^c).$$
So $P(B \cap A^c) = P(B) - P(A \cap B)$ as desired.

□

The last property is a very useful formula for finding the probability of a union of events when the events are not necessarily disjoint. We can generalize it to $n$ events.

**Theorem 6.1.** *For any events $A_1, A_2, \ldots, A_n$, it holds that*
$$P(A_1 \cup A_2 \cdots \cup A_n) = \sum_{j=1}^{n} P(A_j) - \sum_{i<j} P(A_i \cap A_j) + \sum_{i<j<k} P(A_i \cap A_j \cap A_k) - \cdots$$
$$(-1)^{n+1} P(A_1 \cap \cdots \cap A_n).$$

This formula can be proved by induction using the axioms. Below is a famous application (known as de Montmort's problem, named after French mathematician Pierre Remond de Montmort) of the inclusion-exclusion theorem.

**Example 6.1.** (Matching problem) Suppose there are $n$ people who each check in a hat at a party. The hats are randomly returned to them without any concern for whose hat is whose. What is the probability that at least one person gets their own hat back?

*Solution.* Let $A_j$ be the event: the $j$-th person gets his own hat. The problem is equivalent to find $P(A_1 \cup A_2 \cup \cdots \cup A_n)$.

Since all position are equally likely, $P(A_j) = \frac{1}{n}$. The probability of there being two matches is: $P(A_1 \cap A_2) = \frac{(n-2)!}{n!} = \frac{1}{n(n-1)}$. Similarly, the probability of there being $k$ matches is:

$P(A_1 \cap \cdots \cap A_k) = \frac{(n-k)!}{n!} = \frac{1}{n(n-1)\cdots(n-k+1)}$. Using the property of the union of events,

$$\begin{aligned}
P(A_1 \cup A_2 \cup \cdots \cup A_n) =& n \cdot \frac{1}{n} - \binom{n}{2}\frac{1}{n(n-1)} + \binom{n}{3}\frac{1}{n(n-1)(n-2)} - \cdots \\
=& 1 - \frac{1}{2!} + \frac{1}{3!} - \frac{1}{4!} + \cdots + (-1)^{n+1}\frac{1}{n!} \\
\approx& 1 - \frac{1}{e}.
\end{aligned}$$

**Pattern matching with Taylor series**

Pattern matching is a very useful technique. In the last step, we recognize that the Taylor series of $e^x$ is

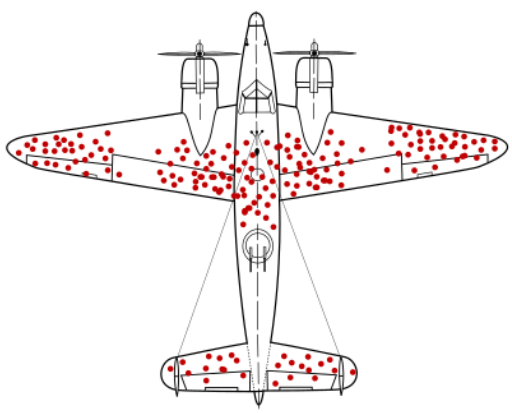
$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

Therefore, $e^{-1} = 1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \cdots$

# 7 Conditional probability

**Thinking conditionally**

Abraham Wald, the renowned statistician, was hired by the Statistical Research Group (SRG) at Columbia University to figure out how to minimize the damage to bomber aircraft. The data they had comprised aircraft returning from missions with bullet holes on their bodies. If asked which parts of the aircraft should be armored to enhance survivability, the obvious answer seemed to be to armor the damaged parts. However, Wald suggested the exact opposite—to armor the parts that were not damaged. Why? Because the observed damage was conditioned on the aircraft returning. If an aircraft had been damaged on other parts, it likely would not have returned. Thinking conditionally completely changes the answer!



See The Soul of Statistics by Professor Joseph Blitzstein.

The probability of A **conditioned on** B is the updated probability of event A after we learn that event B has occurred. Since events contain information, the occurring of a certain event may change our believes on probabilities of other relevant events. The updated probability of event A after we learn that event B has occurred is the conditional probability of A given B.

**Definition 7.1.** If $A$ and $B$ are events with $P(B) > 0$, then the conditional probability of $A$

given $B$ is defined as

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

**Don't confuse P(A|B) with P(A,B)**

$P(A|B)$ is the probability of $A$ occurring given that $B$ has already occurred. $P(A, B) := P(A \cap B)$ is the probability that $A$ and $B$ occur simultaneously.

**Proposition 7.1.** *Properties of conditional probability:*

- $P(A \cap B) = P(B)P(A|B) = P(A)P(B|A)$
- $P(A_1 \cap \cdots \cap A_n) = P(A_1)P(A_2|A_1)P(A_3|A_1, A_2) \cdots P(A_n|A_1 \ldots A_{n-1})$

**Theorem 7.1.** *(Bayes' rule) Assume $P(B) > 0$, we have*

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

**History of Bayes' rule**

Bayes' rule, quantifies how to update probabilities based on new evidence. It is named after Thomas Bayes in the 18th century. It gained prominence posthumously through Richard Price's publication of Bayes' work in 1763. The rule calculates the probability of a hypothesis based on prior knowledge and new data, foundation for Bayesian statistics. Historically, Bayes studied the problem in order to prove David Hume wrong. Hume argued that we cannot directly observe causation; instead, we infer it from patterns of events. Bayes' rule allows for a systematic way to update our beliefs about causal relationships as new evidence emerges, thereby bridging the gap between empirical observation and theoretical inference. This approach counters Hume's skepticism by providing a method for rationally assessing the likelihood of causes based on observed effects.
See The Christian roots of Bayesian statistics.

**Theorem 7.2.** *(Law of total probability). Let $A_1, \ldots, A_n$ be a partition of the sample space $S$ (i.e., the $A_i$ are disjoint events and their union is $S$), with $P(A_i) > 0$ for all $i$. Then*

$$P(B) = \sum_{i=1}^{n} P(B|A_i)P(A_i).$$

**Example 7.1.** Get a random 2-card hand from a standard deck. Find the probability of getting another ace conditioned on (a) having one ace, or (b) having the ace of spade.

*Solution.* The example shows the subtleness of conditional probabilities. The seemingly indifferent probabilities are in fact different:

$$P(\text{another ace} \mid \text{one ace}) = \frac{P(\text{both aces})}{P(\text{one ace})} = \frac{C_4^2/C_{52}^2}{1 - C_{48}^2/C_{52}^2} = \frac{1}{33};$$
$$P(\text{another ace} \mid \text{ace of spade}) = \frac{P(\text{ace of spade \& another ace})}{P(\text{ace of spade})} = \frac{C_3^1/C_{52}^2}{C_{51}^1/C_{52}^2} = \frac{1}{17}.$$

In the first case, the denominator is interpreted as "at least one ace"; whereas in the second case, it is "ace of space + another card".

**Example 7.2.** A disease has a prevalence rate of 10% (i.e., the probability of being infected is 10%). A diagnostic test for the disease has an accuracy of 98%, meaning it correctly identifies infected individuals as positive 98% of the time. Calculate the probability that an individual is infected given that the test result is positive.

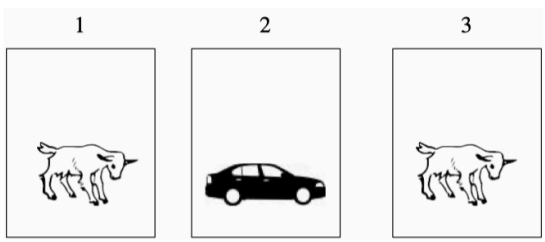
*Solution.* Let $D$ denote being actually infected by the disease; and $T$ denote a positive test. The test accuracy means: $P(T|D) = 98\%$. It also means $P(T|D^C) = 2\%$. We also know that $P(D) = 0.1$. We want to find $P(D|T)$. Note they are two different conditional probabilities, though we mostly confuse the two in everyday life. The two conditional probabilities are associated with Bayes' rule:

$$\begin{aligned} P(D|T) =& \frac{P(T|D)P(D)}{P(T)} \\ =& \frac{P(T|D)P(D)}{P(T|D)P(D) + P(T|D^C)P(D^C)} \\ =& \frac{0.98 \times 0.1}{0.98 \times 0.1 + 0.02 \times 0.9} \approx 84\%. \end{aligned}$$

Note that how $P(T|D)$ is far away from $P(D|T)$!

# 8 Monty Hall problem

**Example 8.1.** (Monty Hall problem) Suppose you are on Monty Hall's TV show. There are three doors. One of them has a car behind it. The other two doors have goats. Monty knows which one has the car. Monty now asks you to pick one door. You will win whatever is behind the door. After you pick one door. Monty opens another door that shows a goat. Monty then asks you if you want to switch. Is it optimal to switch?




We present two solutions to the problem. The first one is using the law of total probability. Let $S$: succeed assuming switch; $D_j$: door $j$ has the car, $j \in 1, 2, 3$. Without loss of generality, assume the initial pick is Door 1. Monty will always open the door with a goat. By the law of total probability,

$$\begin{aligned} P(S) = & \underbrace{P(S|D_1)}_{\text{switch from initial pick}} P(D_1) + \underbrace{P(S|D_2)}_{\text{Monty opens door 3}} P(D_2) + \underbrace{P(S|D_3)}_{\text{Monty opens door 2}} P(D_3) \\ = & 0 + 1 \times \frac{1}{3} + 1 \times \frac{1}{3} = \frac{2}{3}. \end{aligned}$$

The problem can also be solved using the Bayes' rule. Let $D_j$: door $j$ has the car; $M_j$: Monty opens door $j$, $j \in 1, 2, 3$. Assume the initial pick is Door 1. If Monty opens door 3, the probability of winning the car assuming switching is

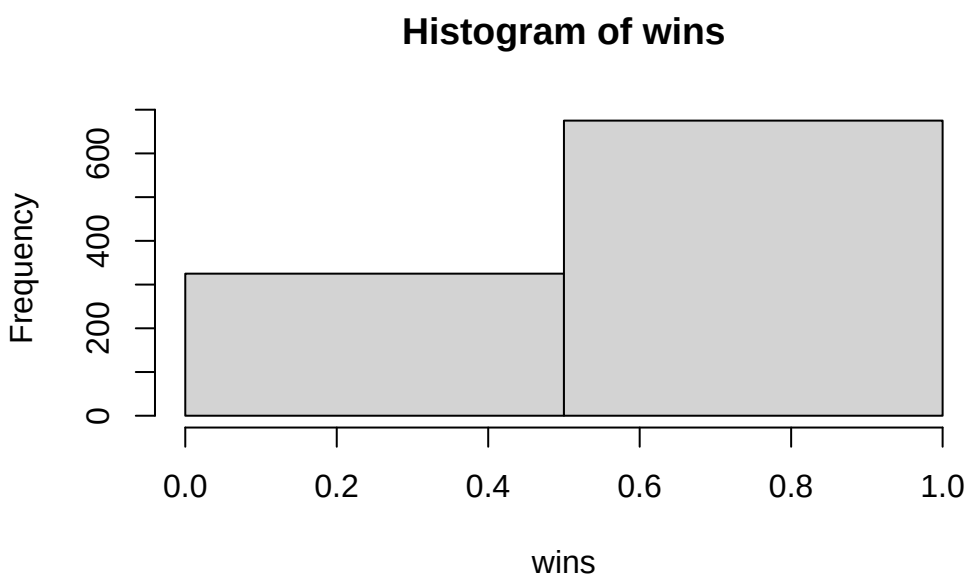
$$\begin{aligned} P(D_2|M_3) = & \frac{P(M_3|D_2)P(D_2)}{P(M_3)} \\ = & \frac{P(M_3|D_2)P(D_2)}{P(M_3|D_1)P(D_1) + P(M_3|D_2)P(D_2) + P(M_3|D_3)P(D_3)} \\ = & \frac{1 \times \frac{1}{3}}{\frac{1}{2} \times \frac{1}{3} + 1 \times \frac{1}{3} + 0} = \frac{2}{3}. \end{aligned}$$

Note that, if door 1 has the car, Monty will open door 2 and 3 with equal probability, thus $P(M_3|D_1) = \frac{1}{2}$. And Monty will never open the door with the car, therefore $P(M_3|D_3) = 0$. Similarly, if Monty opens door 2, we have $P(D_3|M_2) = \frac{2}{3}$. Therefore, the optimal choice is always to switch. Intuitively, because Monty knows which door has the car, the fact that he always opens the door without the car gives additional information regarding the choice of the door.

## R simulation

```
monty_hall <- function() {

  # three doors numbered 1,2,3
  doors <- c(1,2,3)

  # a random door has a car behind it
  car_door <- sample(doors, 1)

  # the guest pick a door randomly
  init_pick <- sample(doors, 1)

  if (car_door == init_pick) {
    # if the guest pick the car door
    # Monty opens either of the other two doors
    monty_open <- sample(setdiff(doors, car_door), 1)
  } else {
    # otherwise, Monty opens the door without the car
    monty_open <- setdiff(doors, c(car_door, init_pick))
  }

  # if the guest switch, he chooses the door other than
  # his initial choice and the opened door
  switch_door <- setdiff(doors, c(init_pick, monty_open))

  # he wins if the switched door is the car door
  win <- switch_door == car_door
  win
}

# repeat the experiment 1000 times and collect the results
wins <- replicate(1000, monty_hall())
```

```r
# convert the results to numbers: 1-win,0-lose
wins <- as.numeric(wins)
# plot the distribution of values
hist(wins, breaks = 2)
```



# 9 Simpson's paradox

**Example 9.1.** (Simpson's paradox). There are two doctors, Dr. Lee and Dr. Wong, performing two types of surgeries — heart surgery (hard) and band-aid removal (easy). Dr. Lee has higher overall surgery success rate. Is Dr. Lee necessarily a better doctor than Dr. Wong?

No. Consider the following example:

| | Dr. Lee | | | Dr. Wong | | |
|---|---|---|---|---|---|---|
| | Heart | Band-Aid | Total | Heart | Band-Aid | Total |
| Success | 2 | 81 | 83 | 70 | 10 | 80 |
| Failure | 8 | 9 | 17 | 20 | 0 | 20 |
| Success rate | 20% | 90% | 83% | 78% | 100% | 80% |

The truth is Dr. Lee has overall higher success rate because he only does easy surgeries (band-aid removal). Dr. Wong does mostly hard surgeries and thus has a lower overall success rate. Yet, he is better at each single type of surgery. To formalize the argument, let $S$: successful surgery; $D$: treated by Dr. Lee, $D^c$: treated by Dr. Wong; $E$: heart surgery, $E^c$: band-aid removal. Dr. Wong is better at each type of surgery,

$$P(S|D,E) < P(S|D^c,E)$$
$$P(S|D,E^c) < P(S|D^c,E^c);$$

But, Dr. Lee has a higher overall successful rate,

$$P(S|D) > P(S|D^c).$$

This is because there is a "confounder" $E$:

$$P(S|D) = \underbrace{P(S|D,E)}_{<P(S|D^c,E)} \underbrace{P(E|D)}_{\text{weight}} + \underbrace{P(S|D,E^c)}_{<P(S|D^c,E^c)} \underbrace{P(E^c|D)}_{\text{weight}}.$$

A **confounder** is a variable that influences with both explanatory variable and the outcome variable, which therefore "confounds" the correlation between the two. In our example, the type of surgery ($E$) is associated with both the doctor and the outcome. Without the confounder being controlled, it is impossible to draw valid conclusions from the statistics.

In general terms, Simpson's paradox refers to the paradox in which a trend that appears across different groups of aggregate data is the reverse of the trend that appears when the aggregate data is broken up into its components. It is one of the most common sources of statistical misuse. Here is another example.

**Example 9.2.** (UC Berkeley gender bias). One of the best-known examples of Simpson's paradox comes from a study of gender bias among graduate school admissions to University of California, Berkeley. The admission figures for the fall of 1973 showed that men applying were more likely than women to be admitted, and the difference was so large that it was unlikely to be due to chance.

| | Male | | Female | |
|---|---|---|---|---|
| | Applicants | Admitted | Applicants | Admitted |
| Total | 8,442 | 44% | 4,321 | 35% |

However, when taking into account the information about departments being applied to, the conclusion turns to the opposite: in most departments, the admission rate for women is higher than men. The lower overall admission rate is caused by the fact that women tended to apply to more competitive departments with lower rates of admission, whereas men tended to apply to less competitive departments with higher rates of admission.

| Department | Male | | Female | |
|---|---|---|---|---|
| | Applicants | Admitted | Applicants | Admitted |
| A | 825 | 62% | 108 | 82% |
| B | 560 | 63% | 25 | 68% |
| C | 325 | 37% | 593 | 34% |
| D | 417 | 33% | 375 | 35% |
| E | 191 | 28% | 393 | 24% |
| F | 373 | 6% | 341 | 7% |
| Total | 2691 | 45% | 1835 | 30% |

See https://setosa.io/simpsons for a really good illustration of the Simpson's paradox.

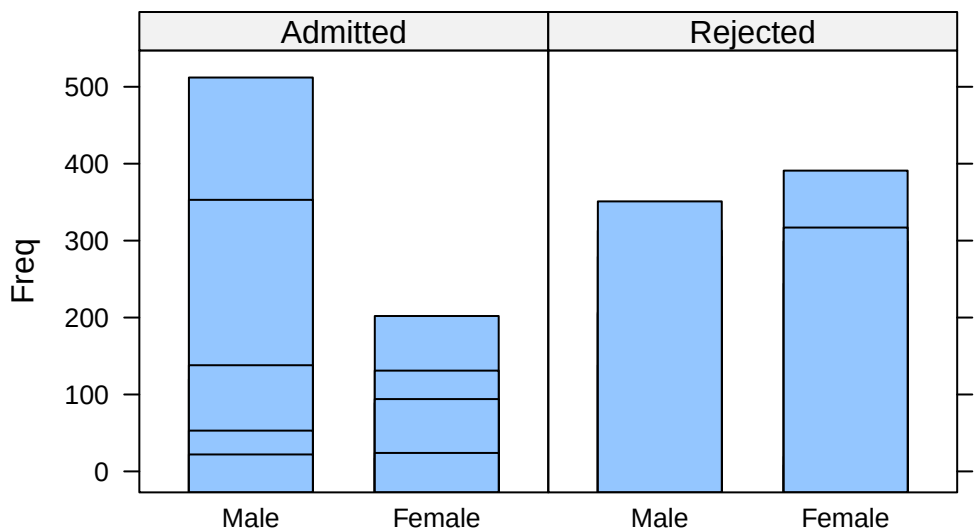
## R demostration

```
library(lattice)
library(knitr)
```

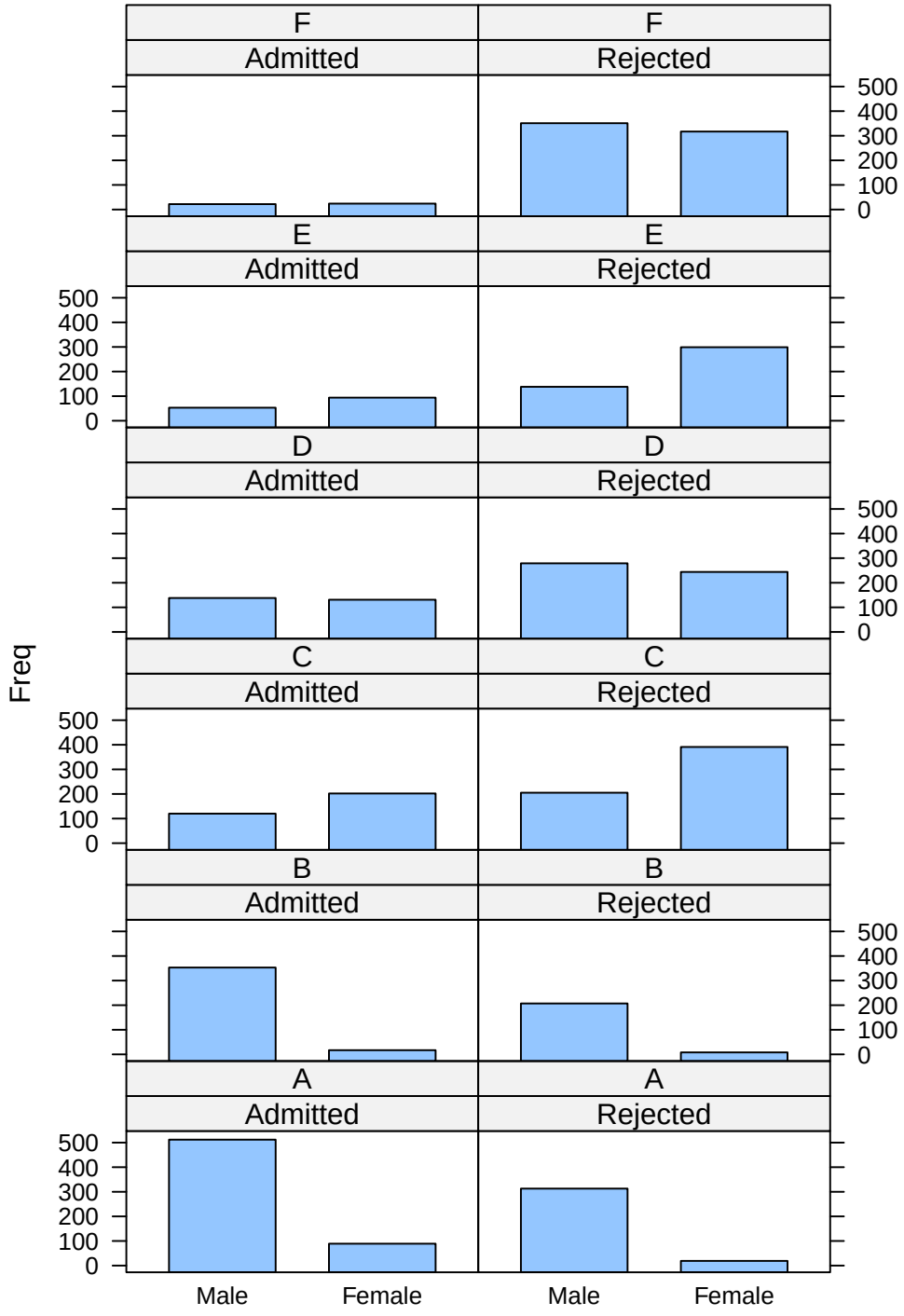
```r
# R has a built-in dataset `UCBAdmissions`
# we convert it to data frame for analysis
UCB <- as.data.frame(UCBAdmissions)

# list the first a few rows
kable(head(UCB, 8))
```

| Admit | Gender | Dept | Freq |
|---|---|---|---|
| Admitted | Male | A | 512 |
| Rejected | Male | A | 313 |
| Admitted | Female | A | 89 |
| Rejected | Female | A | 19 |
| Admitted | Male | B | 353 |
| Rejected | Male | B | 207 |
| Admitted | Female | B | 17 |
| Rejected | Female | B | 8 |

```r
# plot the number of admissions for each gender
# conditioned on admission status
barchart(Freq ~ Gender | Admit, UCB)
```



```r
# plot the number of admissions for each gender
# conditioned on admission status and department
barchart(Freq ~ Gender | Admit + Dept, UCB)
```


F
Admitted
F
Rejected
E
Admitted
E
Rejected
D
Admitted
D
Rejected
C
Admitted
C
Rejected
B
Admitted
B
Rejected
A
Admitted
A
Rejected
Freq
0
100
200
300
400
500
Male
Female

**The importance of conditional thinking**

Whenever we talk about probability or statistics, always remind ourselves what we are the conditioning on. Any statistical reasoning without specifying the conditions can be misleading. We are prone to such fallacies everyday everywhere.

- "10 millions new jobs were added during the term of President X." But it doesn't tell you this was achieved conditioned on that the economy had just had the worst recession.
- "Private schools' graduates earned 50% more than those graduated from public schools." But it doesn't tell you the background of those students who enrolled in private schools.

Be vigilant to these claims when you see them next time.

# 10 Independence

**Definition 10.1** (Independence for two events)**.** If event $B$'s occurrence does not change the probability of $A$, then we say $A$ and $B$ are independent. That is to say $A$ and $B$ are independent if

$$P(A \cap B) = P(A)P(B).$$

Assuming $P(B) > 0$, this is equivalent to

$$P(A|B) = P(A)$$

**Theorem 10.1.** *If events $A$ and $B$ are independent, then*

- *$A$ and $B^c$ are independent;*
- *$A^c$ and $B^c$ are independent.*

$A$ and $B$ are independent means they do not provide information to each other in the sense that conditional probability is not different from the unconditional probability. It is not an intuitive idea as it seems. It will become clearer when we discuss random variables in later chapters.

**Definition 10.2** (Independence for three events)**.** Events $A$, $B$, and $C$ are said to be (mutually) independent if all of the following equations hold:

$$\begin{aligned} P(A \cap B) =& P(A)P(B), \\ P(A \cap C) =& P(A)P(C), \\ P(B \cap C) =& P(B)P(C), \\ P(A \cap B \cap C) =& P(A)P(B)P(C). \end{aligned}$$

**Definition 10.3** (Independence for $n$ events)**.** For $n$ events $A_1, A_2, \dots, A_n$ to be (mutually) independent, we require any pair to satisfy $P(A_i \cap A_j) = P(A_i)P(A_j)$ (for $i \neq j$), any triplet to satisfy $P(A_i \cap A_j \cap A_k) = P(A_i)P(A_j)P(A_k)$ (for $i$, $j$, $k$ distinct), and similarly for all quadruplets, quintuplets, and so on.

**Don't assume independence without justification**

In probability, assuming events are independent without justification can lead to catastrophic errors. A tragic example is the Sally Clark case, where a mother was wrongfully convicted of murdering her two children after both died of sudden infant death syndrome (SIDS). The prosecution argued that the probability of two SIDS deaths in one family was in million, assuming the events were independent. However, this assumption ignored the potential influence of shared genetic, environmental, or health factors, which could increase the likelihood of a second SIDS death if one had already occurred. By treating dependent events as independent, the calculated probability was grossly misleading, contributing to a miscarriage of justice. Always evaluate relationships between events carefully before assuming independence.

## Common confusions

**Independence is not the same as disjointness.**

$A$ and $B$ are disjoint means if $A$ occurs, $B$ cannot occur. But independence means $A$ occurs has nothing to do with $B$.

**Pairwise independence does not imply independence.**

In Definition 10.2, If the first three conditions hold, we say that $A$, $B$, and $C$ are *pairwise independent*. Pairwise independence does not imply independence. Convince yourself with the following example.

**Example 10.1.** Consider two fair, independent coin tosses, and let $A$ be the event that the first is Heads, $B$ the event that the second is Heads, and $C$ the event that both tosses have the same result. Show that $A$, $B$, and $C$ are pairwise independent but not independent.

*Solution.* For each event, $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{2}$. Consider the two events together, there are four possible outcomes: HH, HT, TH, TT. $P(C) = P(HH) + P(TT) = \frac{1}{2}$. Thus,

$$P(A \cap B) = P(HH) = \frac{1}{4} = P(A)P(B)$$
$$P(A \cap C) = P(HH) = \frac{1}{4} = P(A)P(C)$$
$$P(B \cap C) = P(HH) = \frac{1}{4} = P(B)P(C)$$

But $A, B, C$ are not independent, because

$$P(A \cap B \cap C) = P(HH) = \frac{1}{4} \neq P(A)P(B)P(C).$$

# 11 Optimal mating problem*

### The question

How many people should you date before you settle down with someone for marriage? The answer is you should date 37% of your potential options and choose the next one who is better. [1]

> **The 37% rule**
>
> The 37% Rule, also known as the **Optimal Stopping Theory**, provides a strategy to maximize the chances of making the best choice when faced with a sequence of options where decisions are irreversible. It suggests that you should review and reject the first 37% of the total options without selecting any, then choose the next option that is better than all those previously considered.

### Mathematical framework

Let's assume there's a pool of $N$ people out there from which you are choosing. We'll also assume that you have a clear-cut way of rating people. You know who is the best to be your partner. We will call that person Mr/Ms X. The people that you will show up in random order. X can show up anywhere in the sequence. Sadly, a person you have dated and then rejected isn't available to you any longer later on. So you cannot date all of them and pick the best one.

Your dating strategy is to date $M$ of the $N$ people and then settle with the next person who is better. Our task is to find the optimal $M$. If $M$ is too small, you will likely land with someone before X shows up. If $M$ is too large, X will likely pass X and pick someone less optimal. Of course, there is no perfect solution. We want to find the $M$ that maximizes the probability of landing X.

Let $P(M, N)$ be the probability of successfully picking X if you date $M$ people out of $N$ and then go for the next person who is better than the previous ones. The overall probability is:

$$\begin{aligned} P(M, N) = {} & P(\text{Success} \mid \text{X in position 1}) + P(\text{Success} \mid \text{X in position 2}) + \cdots \\ & + P(\text{Success} \mid \text{X in position N}) \end{aligned}$$

[1] See Kissing the frog: A mathematician's guide to mating and Strategic dating: The 37% rule for reference.

For a given value of $M$, if X is among the first $M$ people you date, then you have missed your chance. The probability of settling with X is zero. Therefore, the first $M$ terms are all zero.

If X is in $M+1$, you're in luck: since X is better than all others so far, you will pick X for sure. Therefore,

$$P(\text{Success } |M+1) = P(\text{X is in position } M+1) = \frac{1}{N}$$

Since X is equally likely to be in any position, the probability of X being in $M+1$ out of $N$ people is $1/N$.

If X is in $M+2$, you'll pick him/her up as long as the $M+1$st person didn't have a higher rating than all the previous $M$ people.

$$\begin{aligned} P(\text{Success}|M+2) &= P(\text{Success})P(\text{X is in position } M+2) \\ &= \frac{M}{M+1}\frac{1}{N} \end{aligned}$$

This is because the highest rated person can be anywhere in the sequence $1, \dots, M+1$. We want him/her to be in the first $M$ people. The chance is $M/(M+2)$.

Similarly, if X is the $M+3$ person, you'll pick him/her up to as long as neither the $M+1$ nor the $M+2$ person have a higher rating than all the previous $M$ people. The probability is

$$P(\text{Success}|M+3) = \frac{M}{M+2}\frac{1}{N}$$

Putting them all together, we have

$$\begin{aligned} P(M,N) &= \frac{1}{N} + \frac{M}{N(M+1)} + \frac{M}{N(M+2)} + \cdots + \frac{M}{N(N-1)} \\ &= \frac{M}{N}\left(\frac{1}{M} + \frac{1}{M+1} + \frac{1}{M+2} + \cdots + \frac{1}{N-1}\right) \end{aligned}$$

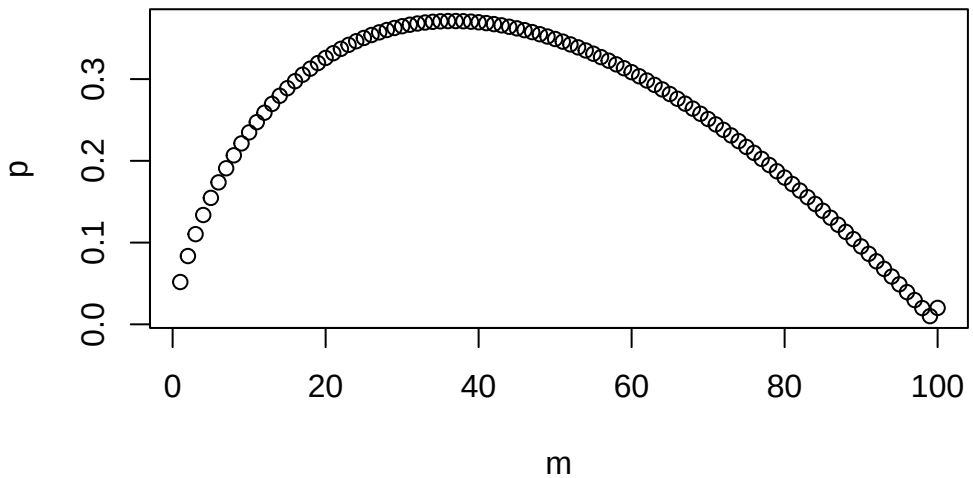
**Maximizing your chance of success**

For a given number of $N$ people you want to choose $M$ such that $P(M,N)$ is the highest. That is you want to find the $M$ such that

$$P(M-1,N) < P(M,N) \text{ and } P(M+1,N) < P(M,N)$$

We can ask the computer to find the solution.

```r
N <- 100
m <- 1:100
p <- sapply(m, function(m) m/N*sum(1/seq(m, N-1)))
plot(m, p)
```




For $N = 100$, the highest probability if achieved when $M = 37$.

**The limiting solution**

We can find the solution analytically if $M, N$ are large. For large $n$, the harmonic sequence can be approximated by the logarithm function:

$$1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} \approx \int_1^{n+1} \frac{1}{x} dx = \ln(n+1) \approx \ln(n) + \gamma$$

where $\gamma$ is a constant.

The optimal condition is

$$\frac{M-1}{N}\left(\frac{1}{M-1} + \cdots + \frac{1}{N-1}\right) < \frac{M}{N}\left(\frac{1}{M} + \cdots + \frac{1}{N-1}\right) < \frac{M+1}{N}\left(\frac{1}{M+1} + \cdots + \frac{1}{N-1}\right)$$

After some manipulation, this is equivalent to

$$\frac{1}{M+1} + \cdots + \frac{1}{N-1} < 1 < \frac{1}{M} + \frac{1}{M+1} + \cdots + \frac{1}{N-1}$$

When $M, N$ are large, this condition is approximated by

$$\ln(N-1) - \ln(M+1) < 1 < \ln(N-1) - \ln(M)$$

Also, for large $M, N$, it makes no difference between $M \pm 1$ and $M$. So the condition effectively becomes

$$\ln(N) - \ln(M) = \ln\left(\frac{N}{M}\right) \approx 1$$

This means $N/M \approx e$ and $M/N \approx 1/e \approx 0.3679$.

# 12 Review of calculus*

Calculus is a prerequisite to work with continuous distributions. The following chapters assume readers are proficient in calculus. We nonetheless review some basic concepts here as a warm-up. This review is not exhaustive, so please refer to a specific textbook if needed for a more comprehensive understanding.

## Differentiation

We define the derivative of a function $f(x)$ to be

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Loosely speaking, a function is continuous if there is no jump in the graph, differentiable if the curve is smooth. Some commonly used derivatives:

$$\begin{aligned}
\frac{d}{dx}(x^n) &= nx^{n-1} \\
\frac{d}{dx}(e^x) &= e^x \\
\frac{d}{dx}(\ln(x)) &= \frac{1}{x} \\
\frac{d}{dx}(\sin(x)) &= \cos(x) \\
\frac{d}{dx}(\cos(x)) &= -\sin(x) \\
(fg)' &= f'g + fg' \\
\left(\frac{f}{g}\right)' &= \frac{f'g - fg'}{g^2} \\
[f(g(x))]' &= f'(g(x))g'(x)
\end{aligned}$$

When dealing with limits of the form "$\frac{0}{0}$" or "$\frac{\infty}{\infty}$", the L'Hospital rule is very handy.

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}.$$

One important application of derivatives is the Taylor's theorem, which gives the approximation of a function around a given point by polynomials. Assume function $f$ is at least $k$ times differentiable, then

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(k)}(a)}{k!}(x-a)^k + \cdots$$

which means we can approximate a function arbitrarily well by higher order polynomials. Some commonly used Taylor series (expanding around $a = 0$):

$$\begin{aligned}
\frac{1}{1-x} &= 1 + x + x^2 + x^3 + \cdots \quad \text{for } |x| < 1 \\
e^x &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \\
\sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots \\
\cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots \\
\ln(1+x) &= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots \quad \text{for } |x| < 1 \\
\arctan(x) &= x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots \quad \text{for } |x| \leq 1
\end{aligned}$$

**Approximating $\pi$ with Taylor series**

Taylor series are one of the most amazing results in calculus. For example, in the last formula, if we let $x = 1$:

$$\frac{\pi}{4} = \arctan(1) = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots$$

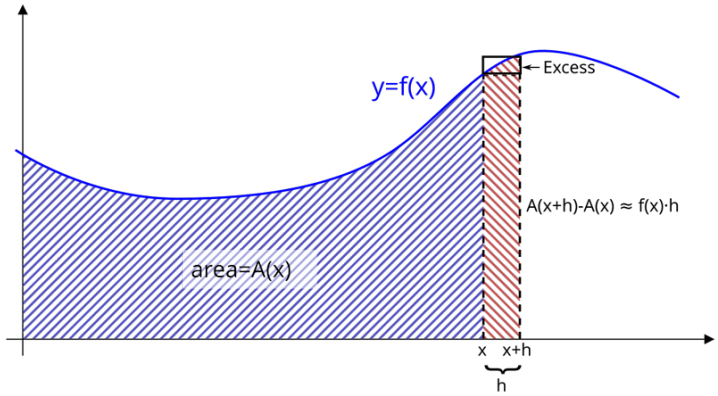
Therefore, we can approximate $\pi$ by summing up a sequence of fractions:

$$\pi = 4\left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots\right).$$

## Integration

Integration is the inverse operation of differentiation. Integral has the geometric interpretation as the area under the curve. Let $A(x)$ be the area under the curve of $y = f(x)$. Thus $A(x) = \int_0^x f(t)dt$. The change of the area resulted from a tiny little change of $x$ is approximated

by $A(x+h)-A(x) \approx f(x)h$. That is $\frac{A(x+h)-A(x)}{h} = f(x)$. If the change is infinitesimal, $h \to 0$, we have $A'(x) = f(x)$.




The Fundamental Theorem of Calculus: if $F$ is the anti-derivative of $f$, then

$$F(x) = \int_a^x f(t)dt$$

$$\int_a^b f(x)dx = F(b) - F(a)$$

One interpretation of the integral is — the integral of a rate of change of a quantity gives the net change in that quantity. Think about speed and distance: $\int_a^b v(t)dt = s(b) - s(a)$.

Because the integral is just a sum over infinitely many approximating rectangles, $\int_a^b f(x)dx = \lim_{n\to\infty} \sum_{i=1}^n f(x_i)\Delta x$. Integrals behave just like sums. For example, $\frac{1}{b-a}\int_a^b f(x)dx$ has the interpretation of the average of $f(x)$ from $a$ to $b$.

Indefinite integrals are the general antiderivatives without specifying the interval of the integration. It always comes with a constant $C$. Some commonly used integrals:

$$
\begin{aligned}
\int dx &= x + C \\
\int x^n dx &= \frac{x^{n+1}}{n+1} + C \\
\int e^x dx &= e^x + C \\
\int \frac{1}{x} dx &= \ln|x| + C \\
\int \cos(x) dx &= \sin(x) + C \\
\int \sin(x) dx &= -\cos(x) + C \\
\int \frac{1}{1+x^2} dx &= \arctan(x) + C
\end{aligned}
$$

Two common integration techniques are *substitution* and *integration by parts*.

**Example 12.1** (Integration by substitution)**.** Find $\int \sqrt{3x+2}dx$.

*Solution.* Let $u = 3x + 2$, then $du = 3dx$. Then

$$\int \sqrt{3x+2}dx = \frac{1}{3}\int \sqrt{u}du = \frac{2}{9}u^{3/2} + C = \frac{2}{9}(3x+2)^{3/2} + C.$$

**Example 12.2** (Integration by parts)**.** Find $\int x \sin x dx$.

*Solution.* Integration by parts follows the formula:

$$\int f(x)g'(x)dx = f(x)g(x) - \int f'(x)g(x)dx$$

Let $f(x) = x$, $g'(x) = \sin x$. Then $g(x) = -\cos x$. Then,

$$\int x \sin x dx = -x\cos x - \int (-\cos x)dx = -x\cos x + \sin x + C.$$

# Part II

# Random Variables

# 13 What is a random variable?

In the previous chapter, we have been working with *events*, which is a conceptualization of real world outcomes occurred with probabilities. In this chapter, we introduce a much more powerful conceptualization that deals with uncertain outcomes — random variables, which is the foundation of all probability and statistical studies.

In high school, all mathematical models come with certainty. For example, the falling time of any object from height $h$ down to the earth is: $t = \sqrt{2h/g}$, where $g$ is the gravity constant. The outcome is *deterministic*. The variables that enter into the equation either have unknown values or known certain values. Errors are possible only due to frictions or measurement errors.

But many real world processes come naturally with uncertainty. Think about the temperature tomorrow, or the stock market returns. We can only make predictions with probabilities. Yes, you may argue this uncertainly is due to incomplete information. If we have all the knowledge regarding the climate, we can predict exactly the temperature. But given the imperfection of the human knowledge, the only feasible option is to build this uncertainly into our mathematical models. Random variable is core concept and the Swiss knife that we use to deal with uncertainties mathematically.

Informally, a random variable differs from a normal variable as it is "random".

> A random variable is a variable whose value is uncertain, but comes with probabilities.

A random variable, say $X$, is never associated with a certain value, such as $X = 1$, or $X = 2$. It could be any of these values, but with different probabilities, say, $X$ takes the value 1 with probability 0.4, and takes the value 2 with probability 0.3. The formal definition of a random variable is as follows.

**Definition 13.1** (Random variable)**.** Given an experiment with sample space $S$, a random variable is a function from the sample space $S$ to the real numbers $\mathbb{R}$.

As an example, flipping a coin twice, let $X$ be the number of heads. Then $X(\cdot)$ is a functions that maps events in $\{HH, HT, TH, TT\}$ into real numbers. In our case, the mapping goes like

$$X(HH) = 2, X(HT) = 1, X(TH) = 1, X(TT) = 0.$$

$X$ is therefore an encoding of events in the sample space into real numbers. We could, of course, have different encodings. Conder the random variable $Y$ as the number of tails. Then we have $Y = 2 - X$.

$$Y(HH) = 0, Y(HT) = 1, Y(TH) = 2, Y(TT) = 2.$$

We could also define $Z$ as the number heads in the 1st toss only. The encoding goes like

$$Z(HH) = 1, Z(HT) = 1, Z(TH) = 0, Z(TT) = 0.$$

We have listed three ways of "encoding" the same experiment as random variables. All of them are valid random variables, but they map the outcomes into different numbers. We can say that, a random variable is a numeric "summary" of an aspect of an experiment.

**Notation for random variables**

We usually use capital letters, such as $X, Y, Z$, to denote random variables. We use small letters, such as $x, y, z$, to denote specific values. $P(X = x)$ means the probability of $X$ taking the value $x$. Don't confuse the random variable $X$ with the number $x$.

**Don't confuse random variables, numbers, and events**

Random variables are never fixed numbers. Functions of random variables, such as $X^2$, $|X|$, $e^X$, are also random variables. Random variables are not events. It does not make sense to write $P(X)$, because $X$ is not an event. But $X = a$ is an event, it makes sense to write $P(X = a)$.

**Definition 13.2** (Distribution)**.** Let $X$ be a random variable. The distribution of $X$ is the collection of all probabilities of the form $P(X \in C)$ for all sets $C$ of real numbers such that $\{X \in C\}$ is an event.

A distribution specifies the probabilities associated with all values of a random variable. In the above example, the distribution of $X$ is given by

$$P(X = 0) = \frac{1}{4}, P(X = 1) = \frac{1}{2}, P(X = 2) = \frac{1}{4}.$$

The distribution of $Y$ is given by

$$P(Y = 0) = \frac{1}{4}, P(Y = 1) = \frac{1}{2}, P(Y = 2) = \frac{1}{4}.$$

The distribution of $Z$ is given by

$$P(Z = 0) = \frac{1}{2}, P(Z = 1) = \frac{1}{2}.$$

You may have noted that the probabilities in a distribution always sums up to 1, as all possible events constitute the entire sample space.

**Specifying the distribution**

Listing all the values is not a smart way to specify a distribution. We like to use a function (if possible), such as $f(x) \stackrel{?}{=} e^{-x}$, to specify the probability of a random variable $X$ taking the value $x$. This is convenient, because once we know the function, we know all the probabilities. But how to specify this function depends on whether a random variable is discrete or continuous.

# 14 Data and random variables

Imagine you're standing in a bustling cafe, sipping your coffee while observing the scene around you. Customers come and go, some ordering their usual drinks, others trying something new. The number of people who walk in during an hour, the time each spends waiting in line, even the total sales for the day—these are all examples of things we can measure, and all are influenced by uncertainty. We understand these seemingly unpredictable happenings through random variables and their distributions.

A random variable is a mathematical abstraction that provides a bridge between theoretical probability and real-world data. Every dataset you encounter—whether it's student grades, daily temperatures, or stock market prices—can be viewed as observations from random variables.

The fact is, we describe something as a random variable not because they are random in nature (like rolling a die), but it is the problem posed does not allow a definite answer (the question itself involves likelihood) or we do not have enough information to give an answer with certainty. In such cases, we would like to give the possible values with their chances of being (the idea of distribution).

Despite the outcome of any one event being uncertain, we can use patterns from past observations to predict the general behavior of these variables. By collecting data, we can figure out how often certain outcomes happen and connect them to theoretical models called **distributions**.

For instance, if you track the heights of 100 people, you might notice that most are close to the average, with fewer at the extremes. This "bell-shaped curve" is the hallmark of something called the **normal distribution**, one of the most common patterns in nature. Heights, test scores, and even measurement errors tend to follow this distribution.

But not all data fits the same shape. If you're counting the number of cars passing through an intersection in a given hour, you might find the **Poisson distribution** is a better fit. This pattern shows up whenever you're dealing with counts of events over time or space—like customer arrivals at a store or typos in a book.

$$\begin{array}{ccccc}
\text{Question with uncertainty} & \rightarrow & \text{Data Collection} & \rightarrow & \text{Patterns} \\
 & & \downarrow & & \downarrow \\
 & & \text{Random Variables} & \rightarrow & \text{Distributions} \quad \rightarrow \quad \text{Predictions}
\end{array}$$

By linking real-world data to these theoretical models, random variables let us make predictions. How many customers will show up next week? What's the chance of a traffic jam during rush hour? Random variables give us the tools to answer these questions with confidence.

Viewing the world through the lens of random variables has several benefits: (1) It helps us deal with uncertainty. Random variables give us a framework to understand situations where outcomes aren't guaranteed. For example, a meteorologist uses random variables to estimate the chance of rain, so you know whether to carry an umbrella. (2) It connects theory to reality. By analyzing data, we can identify which theoretical models describe the randomness we observe. This helps businesses, scientists, and policymakers make better decisions. (3) It allows for better planning.

Suppose you're running an online store. Knowing that the number of daily orders follows a certain distribution can help you manage inventory and staffing. Suppose you're tracking the number of customers visiting your cafe each day. You notice the number fluctuates between 50 and 100, with an average of 75. By treating this as a random variable, you can estimate the likelihood of having fewer than 60 customers tomorrow (useful for planning staff schedules); or the probability of exceeding 90 customers on a holiday (important for stocking supplies).

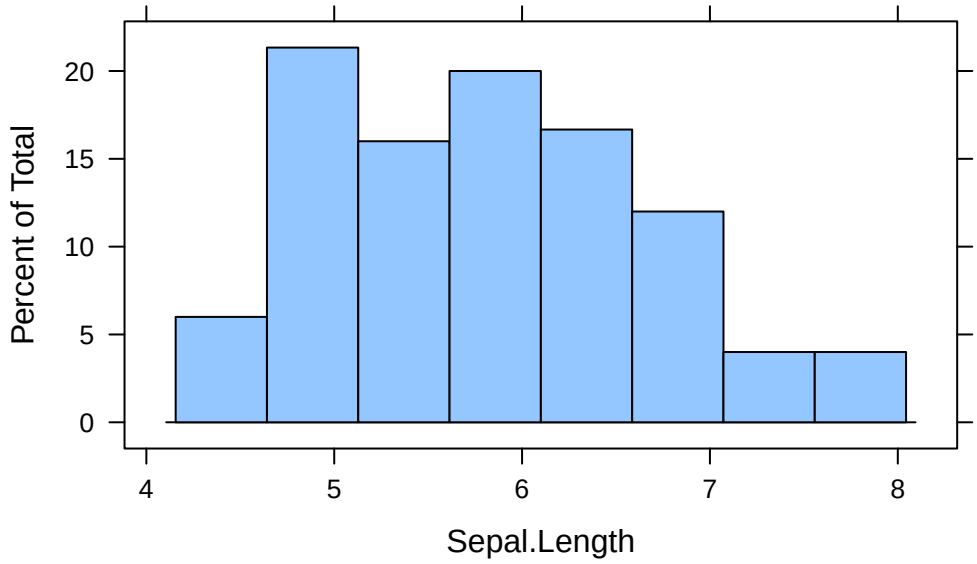
In the grand scheme of things, random variables are more than just mathematical tools—they're a way to make sense of life's unpredictability. So, the next time you see data—whether it's sports stats, sales figures, or even your social media likes—remember: Behind the numbers is a pattern of randomness waiting to be understood. And with random variables, you have the key to unlock it.

## R demonstration

We view every variable in a dataset as random variables. The values we see in the dataset are considered as samples of the random variables whose entire distribution is unknown.

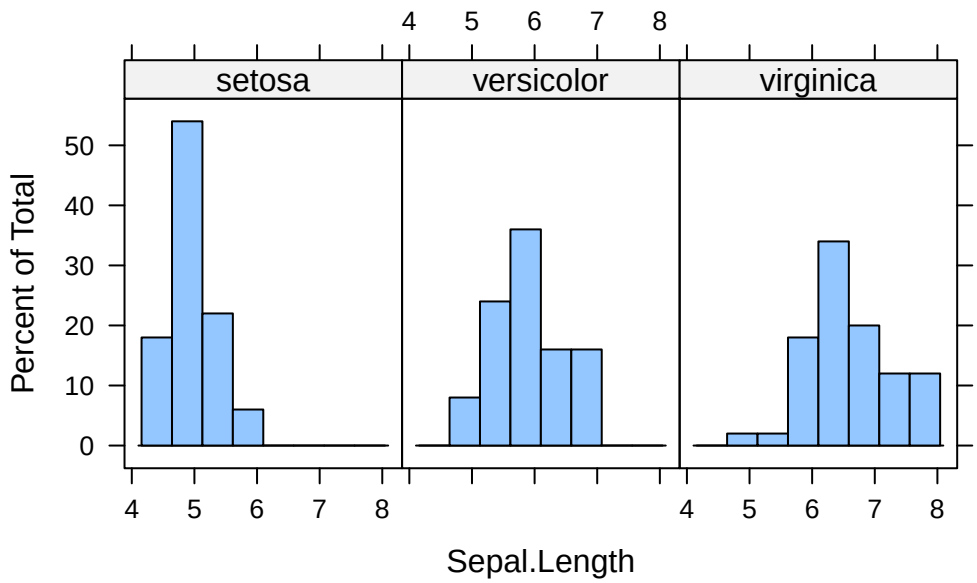
One way to visualize the distribution of a variable is to plot a histogram. A histogram groups data points into intervals, showing how often data values fall within each range. The horizontal axis represents the intervals (or bins), and the vertical axis shows the frequency or count of data points in each bin. Histograms provide insights into the shape of the data distribution, revealing patterns such as skewness, symmetry, and the presence of outliers, making them a fundamental tool in exploratory data analysis.

```r
library(lattice)

histogram(~Sepal.Length, iris)
```



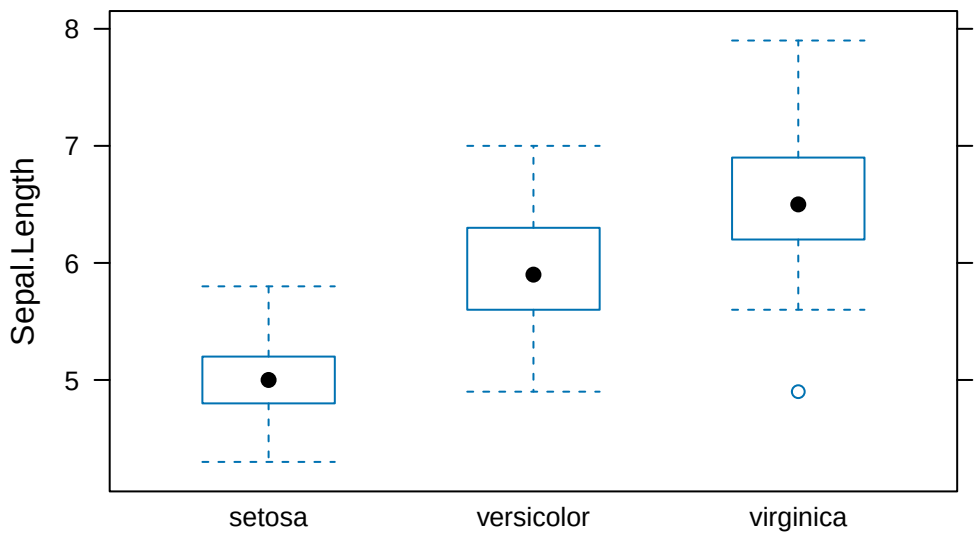

To explore conditional probability, we may plot the histogram conditioned on other variables.

```
histogram(~Sepal.Length|Species, iris)
```



A more parsimonious way to visualize the distribution is to use a boxplot. A boxplot, also known as a box-and-whisker plot, is a visual summary of a dataset that highlights its central tendency, spread, and potential outliers. It displays the median, quartiles, and range of the data. The box represents the interquartile range (IQR), which contains the middle 50% of the data, with the lower and upper edges corresponding to the first (Q1) and third quartiles (Q3). The line inside the box marks the median. Whiskers extend from the box to indicate the range of values within 1.5 times the IQR from Q1 and Q3, while points beyond this range are considered outliers. Boxplots are useful for comparing distributions across different groups and identifying data asymmetry or variability.

```
bwplot(Sepal.Length~Species, iris)
```



# 15 Discrete variables

**Definition 15.1** (Discrete random variable)**.** We say $X$ is a discrete random variable if $X$ can take a finite or countable number of values $x_1, x_2, \dots, x_n$.

**Definition 15.2** (Support)**.** The finite or countably infinite set of values $x$ such that $P(X = x) > 0$ is called the support of $X$.

**Definition 15.3** (Probability mass function)**.** If a random variable $X$ has a discrete distribution, the probability mass function (PMF) of $X$ is defined as the function $f$ such that

$$f(x) = P(X = x).$$

> **Notation for PMF**
>
> PMF is also known as the **probability function**, since $f(x)$ gives the probability. Throughout this lecture notes, we use PMF for abbreviation. But you will find the textbook uses *p.f.* They are different conventions for the same thing.
> Note that how $f(x)$ differs from the probability function $P(\cdot)$. $f(x)$ is a real-valued function, whereas $P(\cdot)$ is the probability operator. Also note that $x$ is not the same as random variable $X$. Sometimes, we add a subscript, $f_X(x)$, to specify this is the PMF for random variable $X$.
> Because we also use $f(x)$ to denote the PDF for continuous random variables. PDF is known as the density function, it is not probability. So we better avoid confusing the two. To distinguish a PMF from a PDF, we may use $p(x)$ for PMF and $f(x)$ for PDF. Mostly in this lecture note, we simply use $P(X = k)$ for PMF.

**Proposition 15.1.** *A valid PMF must satisfy the following criteria:*

- *Non-negative:* $P(X = k) \geq 0$ *for all* $k$*;*
- *Sum up to one:* $\sum_k P(X = k) = 1$.

There are different ways to represent a PMF. We can (1) list all the possible values and their associated probabilities; (2) write a formula for the PMF; or (3) visualize it in a graph.

**Example 15.1.** A random variable $X$ is said to have the **Bernoulli distribution** if $X$ has only two possible values, 0 and 1, and $P(X = 1) = p$, $P(X = 0) = 1 - p$. The PMF of $X$ is

$$P(X = k) = p^k(1 - p)^{1-k}$$

where $k \in \{0, 1\}$.

# 16 Continuous variables

**Definition 16.1** (Continuous random variable)**.** We say a random variable $X$ has a continuous random variable if the possible values of $X$ takes the form of a continuum.

**Definition 16.2** (Probability density function)**.** For a continuous random variable $X$, the probability density function (PDF) of $X$ is a real-valued function $f$ such that

$$P(a \leq X \leq b) = \int_a^b f(x)dx.$$

Continuous random variables are usually measurements. Examples include height, weight, temperature, the amount of money and so on.

**Density is not probability**

PDF differs from the discrete PMF in important ways:

- For a continuous random variable, $P(X = x) = 0$ for all $x$;
- The quantity $f(x)$ is not a probability. To get the probability, we integrate the PDF (probability is the area under the PDF):

$$P(a < X \leq b) = F(b) - F(a) = \int_a^b f(x)dx.$$

- Since any single value has probability 0, including or excluding endpoints does not matter.

$$P(a < X < b) = P(a < X \leq b) = P(a \leq X < b) = P(a \leq X \leq b).$$

**Proposition 16.1.** *The PDF $f$ of a continuous random variable must satisfy the following criteria:*

- *Non-negative:* $f(x) \geq 0$*;*
- *Integrate to one:* $\int_{-\infty}^{\infty} f(x)dx = 1$*.*

**Example 16.1.** A **uniform distribution** is a probability distribution where all values within a specified interval $[a, b]$ are equally likely to occur, and its probability density function (PDF) is given by:

$$f(x) = \frac{1}{b-a} \quad \text{for} \quad a \leq x \leq b$$

and $f(x) = 0$ otherwise.

**Don't confuse a random variable with its distribution**

If random variable $X$ has distribution $f(x)$, the distribution of $X^2$ is not $f^2(x)$. To get the distribution of $X + Y$, you can't just add up $f_X(x) + f_Y(y)$. The right way to do it will be discussed in later chapters (transformation and convolution).

# 17 Cumulative distribution

**Definition 17.1** (Cumulative distribution function)**.** The cumulative distribution function (CDF) of a random variable $X$ is the function $F$ given by $F(x) = P(X \leq x)$.

For discrete random variables, $F(x) = \sum_{k \leq x} p(k)$.

For continuous random variables, $F(x) = \int_{-\infty}^{x} f(t)dt$. We thus have $\frac{dF(x)}{dx} = f(x)$.

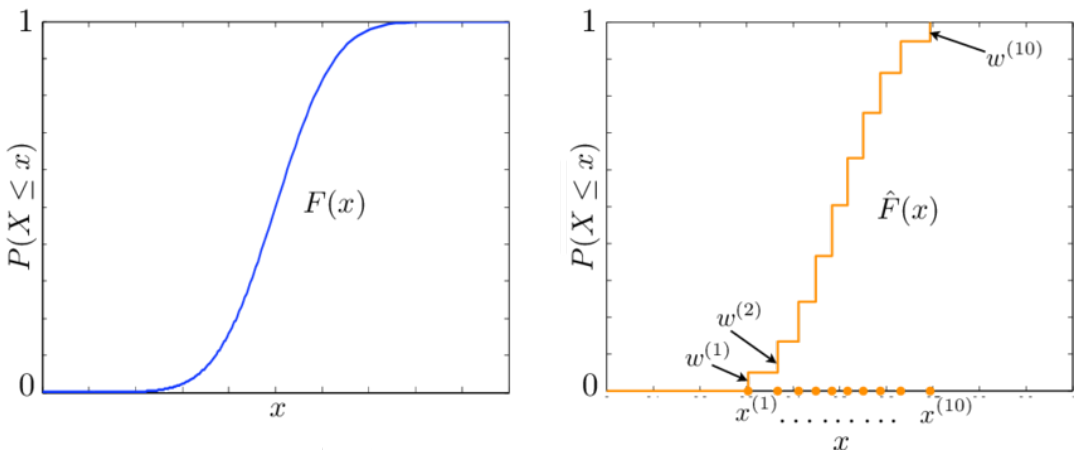
Unlike PMF or PDF, a cumulative distribution function can be defined for both discrete and continuous random variables. CDF gives the full distribution of a random variable. Given the CDF, we can figure out any probability distribution of the random variable:

$$P(x_1 < x \leq x_2) = F(x_2) - F(x_1).$$

**Proposition 17.1.** *Any CDF has the following properties:*

- $P(X > x) = 1 - F(x)$
- $P(x_1 < x \leq x_2) = F(x_2) - F(x_1)$
- *Increasing: if* $x_1 \leq x_2$*, then* $F(x_1) \leq F(x_2)$*.*
- *Right-continuous: for any* $a$*,* $F(a) = \lim_{x \to a+} F(x)$*.*
- $F(x) \to 0$ *as* $x \to -\infty$*;* $F(x) \to 1$ *as* $x \to +\infty$*.*

The CDF for a continuous random variable is <u>differentiable</u>, while the CDF for a discrete random variable consists of jumps and flat regions.


1
0
$P(X \leq x)$
$F(x)$
$x$
1
0
$P(X \leq x)$
$\hat{F}(x)$
$w^{(10)}$
$w^{(2)}$
$w^{(1)}$
$x^{(1)}$
$x^{(10)}$
$x$

# Part III

# Discrete Distributions

# 18 Bernoulli distribution

We introduce some theoretical distributions from now on. These distributions are important because they provide standardized models for common "patterns" of random processes. We develop these distributions from idealized assumptions. In practice, we usually do not know what is the "true" distribution of the question of interest. Typically, we "fit" the question into a theoretical distribution according to their proximity to the assumptions.

**Definition 18.1** (Bernoulli distribution)**.** A random variable $X$ is said to have the Bernoulli distribution, denoted as $X \sim Bern(p)$, if $X$ has only two possible values, 0 and 1, and $P(X = 1) = p$, $P(X = 0) = 1 - p$.

The PMF of a Bernoulli random variable $X$ is given by

$$P(X = k) = \begin{cases} p & \text{if } k = 1, \\ 1 - p & \text{if } k = 0. \end{cases}$$

This can also be expressed as

$$P(X = k) = p^k (1 - p)^{1-k}, \quad k \in \{0, 1\}.$$

**Definition 18.2** (Bernoulli trial)**.** A Bernoulli trial is a random experiment with binary outcomes, often labeled as "success" and "failure," where the probability of success $p$ remains constant across trials.

Bernoulli trials have numerous applications across various fields. For instance, in quality control, a product can either pass or fail inspection; in healthcare, a treatment may either be effective or ineffective; and in marketing, a consumer may either purchase a product or not. As long as the probability of success is constant across trials, we can model such scenarios with Bernoulli distributions.

# 19 Binomial distribution

**Definition 19.1** (Binomial distribution)**.** Suppose $X_1, X_2, \dots, X_n$ are independent and identical Bern($p$) distributions. Let $X$ be the total number of successes of the $n$ independent trials. That is, $X = X_1 + X_2 + \cdots + X_n$. Then $X$ has the Binomial distribution, $X \sim \text{Bin}(n, p)$.

The PMF of $X$ directly follows from the combination theory:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

This is a valid PMF because, by the Binomial theorem, we have

$$\sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k} = (p + (1-p))^n = 1.$$

> **Binomial distribution and Binomial theorem**
>
> You may have noticed the connection between Binomial distribution and Binomial theorem. Consider using the polynomial $px + q$ to represent the outcome of a single Bernoulli trial, where $x$ is the indicator for a success. Then $(px + q)^n$ is the outcome for $n$ independent trials. The coefficient of $x^k$ gives the probability of there being exactly $k$ successes.

**Example 19.1.** In the previous example of tossing two coins, we compute the distribution of $X$ by counting the equally likely outcomes in an event. We can get the same result by realizing it is a Binomial distribution. $X \sim \text{Bin}(2, 1/2)$. Since each coin tossing is an independent Bernoulli trial. The probabilities come directly from the PMF.

$$\begin{aligned}
P(X = 0) &= \binom{2}{0} \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^2 = \frac{1}{4}; \\
P(X = 1) &= \binom{2}{1} \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^1 = \frac{1}{2}; \\
P(X = 2) &= \binom{2}{2} \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^0 = \frac{1}{4}.
\end{aligned}$$

Utilizing the Binomial distribution also allows us to generalize the problem. Suppose we are tossing $n$ coins, we want to find the probability of getting $k$ heads. It is almost impossible to count all the possible outcomes, but the answer immediately follows from the Binomial PMF.

**Theorem 19.1.** *Let $X \sim Bin(n, p)$ and $Y \sim Bin(m, p)$ be two independent Binomial random variables. Then $X + Y \sim Bin(n + m, p)$.*

*Proof.* By the definition of the Binomial distribution, $X = \sum_{i=1}^{n} X_i$ where $X_i \sim \text{Bern}(p)$; $Y = \sum_{j=1}^{m} Y_j$ where $Y_j \sim \text{Bern}(p)$. Therefore,

$$X + Y = \sum_{i=1}^{n} X_i + \sum_{j=1}^{m} Y_j = \sum_{k=1}^{n+m} Z_k$$

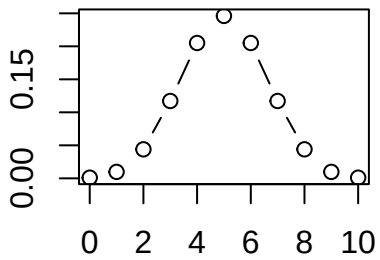
where $Z_k \sim \text{Bern}(p)$. Since $X_i$ and $Y_j$ are identical Bernoulli random variables,

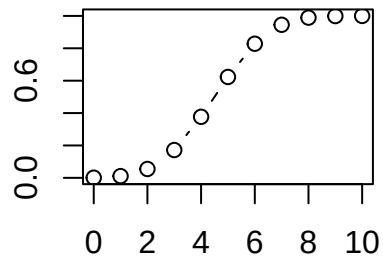
$$Z_k = \begin{cases} X_k, & \text{for } k = 1, \dots, n \\ Y_{k-n}, & \text{for } k = n+1, \dots, n+m \end{cases}$$

By definition, $X + Y = \sum_{k=1}^{n+m} Z_k \sim \text{Bin}(n + m, p)$. □

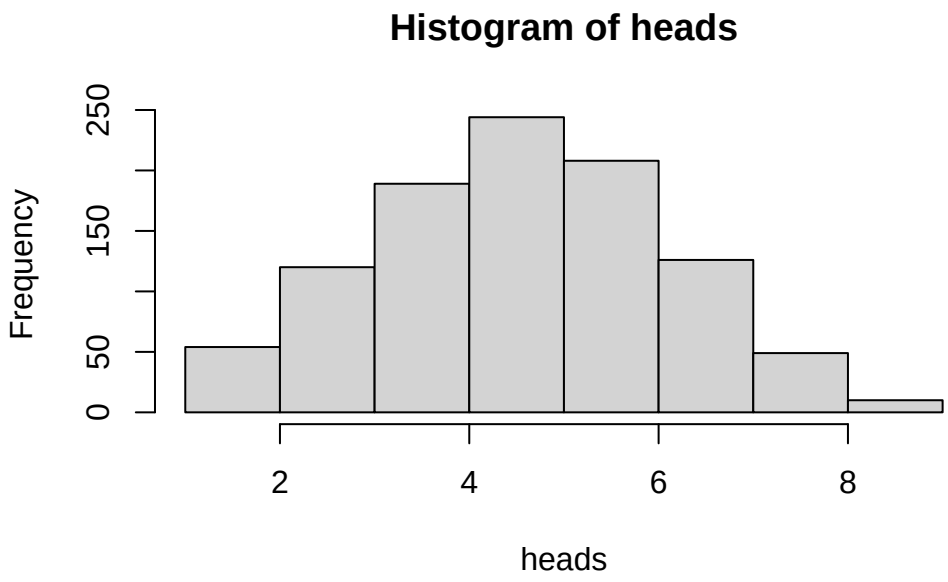
There are built-in functions in R to work with Binomial distributions.

```
# computes P(X=5) for Bin(10,0.5)
p <- dbinom(5, 10, 0.5)

par(mfrow=c(1,2))

# plot the PMF for Bin(10,0.5)
curve(dbinom(x, 10, 0.5), from=0, to=10, n=11, type="b", ann=F)

# `pbinom` computes the CDF
curve(pbinom(x, 10, 0.5), from=0, to=10, n=11, type="b",ann=F)
```




```r
# draw a random value from a given Binomial distribution
# this allows us to simulate a random experiment
# e.g. the number of heads when flipping 10 fair coins
outcome <- rbinom(1, 10, 0.5)

# Repeat the experiment 1000 times
heads <- rbinom(1000, 10, 0.5)

# the histogram will converge to the ideal Binomial distribution
# if the experiment is repeated a large number of times
hist(heads)
```


Histogram of heads
Frequency
0
50
150
250
2
4
6
8
heads

# 20 Dice rolling formula*

The Binomial distribution gives the formula for the probability of observing $k$ heads when flipping $n$ coins. Can we find a formula for the probability of getting a total of $p$ points when rolling $n$ dice?

The probability of obtaining $p$ points on $n$ $s$-sided dice can be computed as the coefficient of $x^p$ in

$$f(x) = (x + x^2 + ... + x^s)^n$$

since each possible arrangement contributes one term.

$$f(x) = x^n(1 + x + \cdots + x^{s-1})^n = x^n \left(\frac{1 - x^s}{1 - x}\right)^n$$

To obtain the coefficient of $x^p$, expand the binomial power:

$$x^n(1 - x^s)^n(1 - x)^{-n} = x^n \sum_{k=0}^{n} (-1)^k \binom{n}{k} x^{sk} \sum_{l=0}^{\infty} \binom{n + l - 1}{l} x^l$$

The coefficient of $x^p$ include all terms with $p = n + sk + l$. Therefore,

$$c_p = \sum_{k=0}^{n} (-1)^k \binom{n}{k} \binom{p - sk - 1}{p - sk - n}$$

But $p - sk - n > 0$ only when $k < (p - n)/s$, so the other terms do not contribute. Furthermore, applying the symmetric property of the binomial formula, we have

$$\binom{p - sk - 1}{p - sk - n} = \binom{p - sk - 1}{n - 1}$$

Therefore, the probability of getting $p$ points when rolling $n$ $s$-sided dice is given by

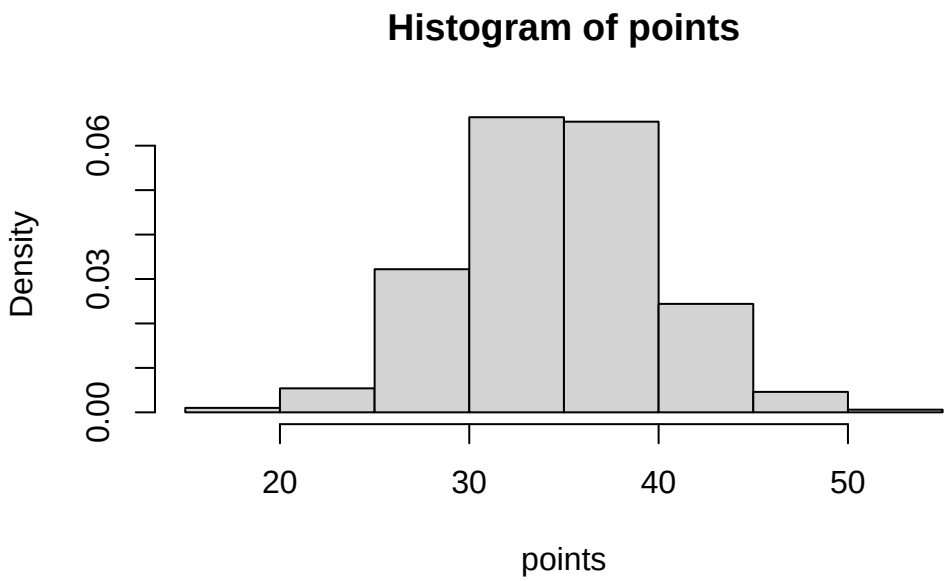
$$f(p,n,s) = \sum_{k=0}^{\lfloor (p-n)/s \rfloor} (-1)^k \binom{n}{k} \binom{p-sk-1}{n-1}.$$

**Binomial formula for negative $n$**

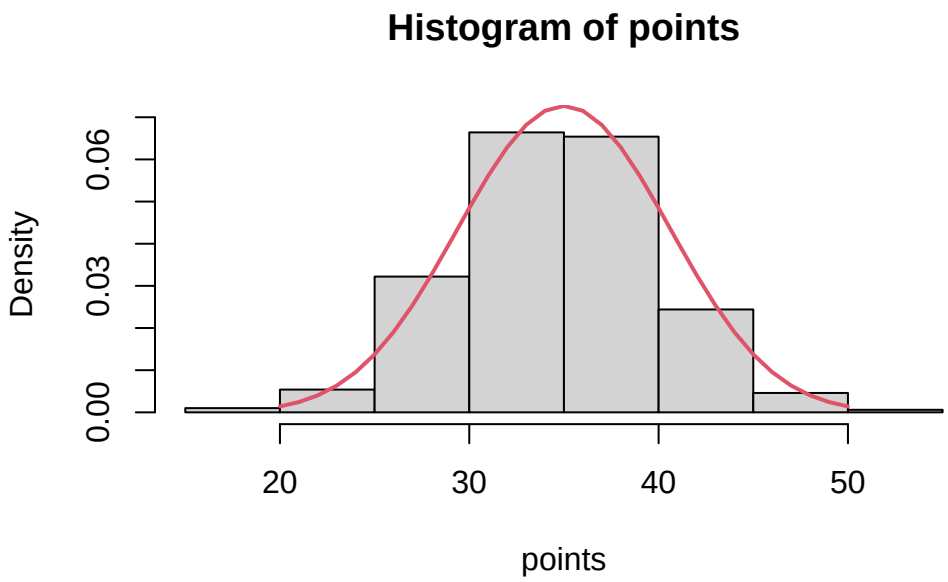
$$\begin{aligned}
\binom{-n}{k} &= \prod_{i=0}^{k-1} \frac{-n-i}{k-i} = (-1)^k \prod_{i=0}^{k-1} \frac{n+i}{k-i} \\
&= (-1)^k \frac{n(n+1)\dots(n+k-1)}{k!} \\
&= (-1)^k \frac{(n+k-1)!}{k!(n-1)!} \\
&= (-1)^k \binom{n+k-1}{k}
\end{aligned}$$

We can verify our formula by simulating the dice rolling game.

```r
set.seed(0)

# simulates rolling n dice and returns the sum
roll_dice <- function(n, s=6) {
  sum(sample(seq(1,s), n, replace = T))
}

# rolling 10 dice 1000 times and collect the results
points <- replicate(1000, roll_dice(10))

# distribution of the sum of points
hist(points, freq = F)
```




```r
# formula for computing probability of dice points
dice_formula <- function(p, n, s=6) {
  prob <- 1/s^n*sum(
    sapply(seq(0, floor((p-n)/s)),
           function(k) (-1)^k*choose(n,k)*choose(p-s*k-1,n-1)))
}

# computing the probability of getting 20-50 when rolling 10 dice
x <- 20:50;
y <- sapply(x, function(p) dice_formula(p,n=10))

# overlay the formula on the histogram
# it turns out the formula does a nice job!
hist(points, ylim = c(0, 0.07), freq = F)
lines(x, y, col = 2, lwd=2)
```


Histogram of points
Density
0.00
0.03
0.06
20
30
40
50
points

# 21 Expectation

**Definition 21.1** (Expectation of a discrete random variable)**.** Let $X$ be a discrete random variable. The expectation of $X$ (or the mean of $X$) is defined as:

$$E(X) = \sum_{\text{all } x} xP(X = x).$$

In other words, the expected value of $X$ is a *weighted average* of the possible values that $X$ can take on, weighted by their probabilities.

> **Note**
>
> The expected value of $X$ is a fixed number, $E(X) \in \mathbb{R}$. It is not a random variable such as $g(X)$.

Sometimes, we would like to omit the parentheses for simplicity and write $EX := E(X)$. We also like to denote expectation by the Greek letter $\mu := E(X)$.

**Example 21.1.** The expectation of a Bernoulli random variable $X \sim Bern(p)$:

$$E(X) = 1 \times P(X = 1) + 0 \times P(X = 0) = p.$$

**Example 21.2.** The expectation of a Binomial random variable $X \sim Bin(n, p)$:

$$\begin{aligned}
E(X) =& \sum_{k=0}^{n} kp(k) \\
=& \sum_{k=0}^{n} k \cdot \binom{n}{k} p^k q^{n-k} \\
=& \sum_{k=1}^{n} n \cdot \binom{n-1}{k-1} p^k q^{n-k} \\
=& np \sum_{k=1}^{n} \binom{n-1}{k-1} p^{k-1} q^{n-k} \\
=& np \underbrace{\sum_{j=0}^{n-1} \binom{n-1}{j} p^j q^{n-1-j}}_{\text{another Binomial PMF}} \\
=& np.
\end{aligned}$$

**Proposition 21.1.** *Expectation has the following properties:*

- $E(X+Y) = E(X) + E(Y)$
- $E(aX+b) = aE(X) + b$

**Example 21.3.** Redo the expectation of $X \sim Bin(n,p)$ with properties of expectation:

$$E(X) = E(X_1 + \cdots + X_n) = nE(X_i) = np$$

where $X_i \sim \text{Bern}(p)$.

## Law of averages

You may wonder what is the difference between $E(X)$ defined in Definition 21.1 and the average of values defined as $\bar{X} = \frac{1}{n}(X_1 + X_2 + \cdots + X_n)$.

The short answer is this: $E(X)$ is a theoretical value, while $\bar{X}$ is an approximation to $E(X)$ with finite observations. They are associated by the following theorem.

> **Law of averages**
>
> The law of averages (or the law of large numbers) states that if you repeat a random experiment, such as tossing a coin or rolling a die, a very large number of times, your individual outcomes, when averaged, should be very close to the theoretical mean (a constant parameter). In mathematical language,
>
> $$\bar{X}_n \to^p \mu \text{ when } n \to \infty.$$
>
> where $\to^p$ reads as "converge in probability".

There is another fundamental difference. In probability theory, we treat $E(X)$ as a fixed number; while $\bar{X}$ is another random variable! Because the sample $\{X_1, X_2, \ldots, X_n\}$ is generated randomly. Consider the coin flipping example, while $E(X) = 0.5$ is a constant, each time you compute the average of, say, 10 flips, you get a different number. We will come back to this point later.

# 22 Hypergeometric

**Example 22.1.** Let's explore an example that appears to be Binomial but is, in fact, not a Binomial distribution. Given a 5-card hand. Find the distribution of the number of aces.

Let $X$ be the number of aces. It is tempting to say $X \sim Bin(5, p)$. But this not correct. Because having one ace is NOT independent from having another ace. We need to use the classical approach:

$$P(X = k) = \frac{\binom{4}{k}\binom{48}{5-k}}{\binom{52}{5}}.$$

This is a Hypergeometric distribution.

Suppose we have a box filled with $w$ white and $b$ black balls. We draw $n$ balls out of the box <u>with replacement</u>. Let $X$ be the number of white balls. Then $X \sim Bin(n, w/(w+b))$. Since the draws are independent Bernoulli trials, each with probability $w/(w+b)$ of success. If we instead sample <u>without replacement</u>, then the number of white balls follows a **Hypergeometric distribution**. We denote this by $X \sim \mathrm{HGeom}(w, b, n)$.

**Theorem 22.1.** *If $X \sim HGeom(w, b, n)$, then the PMF of $X$ is*

$$p_X(k) = \frac{\binom{w}{k}\binom{b}{n-k}}{\binom{w+b}{n}},$$

*for integers $k$ satisfying $0 \le k \le w$ and $0 \le n - k \le b$, and $p_X(k) = 0$ otherwise.*

In Example 22.1, the number of aces in the hand has the $\mathrm{HGeom}(4, 48, 5)$ distribution, which can be seen by thinking of the aces as white balls and the non-aces as black balls. The probability of having exactly three aces is 0.0017%.

**Example 22.2.** Let $X \sim \mathrm{HGeom}(w, b, n)$. Find $E(X)$ the expected number of white balls. Similarly, we can decompose $X$:

$$X = I_1 + \cdots + I_n$$

where $I_j$ equals 1 if the $j$th ball is white and 0 otherwise. We have said that $\{I_j\}$ are not independent, but the linearity of expectation still holds:

$$E(X) = E(I_1 + \cdots + I_n) = E(I_1) + \cdots + E(I_n).$$

Meanwhile we have

$$E(I_j) = P(j\text{-th ball is white}) = \frac{w}{w+b}$$

since unconditionally the $j$th ball is equally likely to be any of the balls. Thus, $E(X) = \frac{nw}{w+b}$.

The Binomial and Hypergeometric distributions are often confused. Both are discrete distributions taking on integer values between 0 and $n$ for some $n$, and both can be interpreted as the number of successes in $n$ Bernoulli trials. However, a crucial part of the Binomial story is that the Bernoulli trials involved are <u>independent</u>. The Bernoulli trials in the Hypergeometric story are <u>dependent</u>, since the sampling is done without replacement.

# 23 Geometric distribution

**Definition 23.1.** Consider a sequence of independent Bernoulli trials, each with the same success probability $p$. Let $X$ be the number of failures before the first successful trial. Then $X$ has a **Geometric distribution**, $X \sim \text{Geom}(p)$.

Let's derive the PMF for the Geometric distribution. By definition,

$$P(X = k) = q^k p$$

where $q = 1 - p$. This is a valid PMF because

$$\sum_{k=0}^{\infty} q^k p = p \sum_{k=0}^{\infty} q^k = \frac{p}{1-q} = 1.$$

The expectation of $X$ is given by

$$E(X) = \sum_{k=0}^{\infty} k \cdot q^k p = p \sum_{k=0}^{\infty} k q^k = p \frac{q}{p^2} = \frac{q}{p}.$$

To see why this holds, taking derivative with respect to $q$ on both sides of $\sum_{k=0}^{\infty} q^k = \frac{1}{1-q}$ yields

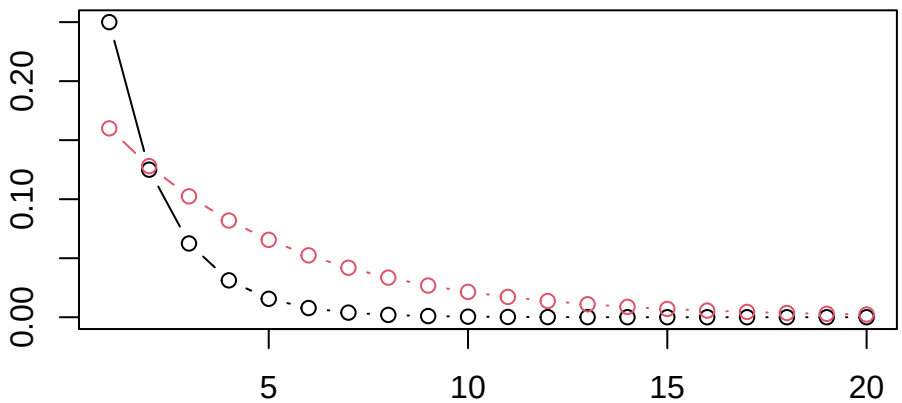
$$\sum_{k=1}^{\infty} k q^{k-1} = \frac{1}{(1-q)^2};$$

Then multiply both sides by $q$:

$$\sum_{k=1}^{\infty} k q^k = \frac{q}{(1-q)^2} = \frac{q}{p^2}.$$

```
x <- 1:20

# Geometric PMF with varying parameters
y1 <- dgeom(x, prob=0.5)
y2 <- dgeom(x, prob=0.2)
```

```r
# Plot the probability mass function
plot(x, y1, type="b", ann=F)
lines(x, y2, type="b", col=2)
```




A generalization of the Geometric distribution is the Negative Binomial distribution.

**Definition 23.2.** In a sequence of independent Bernoulli trials with success probability $p$, if $X$ is the number of failures before the $r$-th success, then $X$ is said to have a **Negative Binomial distribution**, denoted $X \sim \mathrm{NBin}(r, p)$.

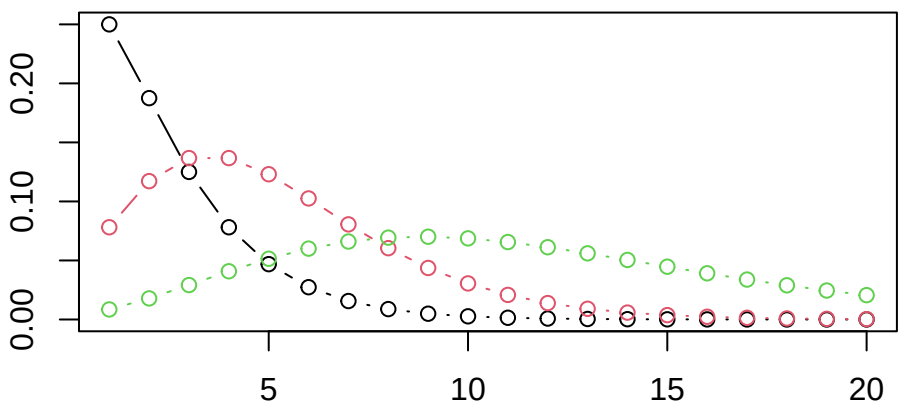
The PMF for Negative Binomial distribution, by definition, is given by

$$P(X = k) = \binom{k + r - 1}{r - 1} q^k p^r.$$

To compute the expectation, let $X = X_1 + \cdots + X_r$ where $X_i$ is the number of failures between the $(i-1)$-th success and the $i$-th success, $1 \leq i \leq r$. Then $X_i \sim \mathrm{Geom}(p)$. By linearity of expectations,

$$E(X) = E(X_1) + \cdots + E(X_r) = r\frac{1-p}{p}.$$

```r
x <- 1:20

# Negative Binomial PMF with varying parameters
y1 <- dnbinom(x, size=2, prob=0.5)
y2 <- dnbinom(x, size=5, prob=0.5)
y3 <- dnbinom(x, size=5, prob=0.3)

# Visualize the probability mass function
plot(x, y1, type="b", ann=F)
lines(x, y2, type="b", col=2)
lines(x, y3, type="b", col=3)
```




**Example 23.1** (Toy collector)**.** There are $n$ types of toys. Assume each time you buy a toy, it is equally likely to be any of the $n$ types. What is the expected number of toys you need to buy until you have a complete set?

*Solution.* Define the following random variables:

$$\begin{aligned}
T =& T_1 + T_2 + \cdots + T_n \\
T_1 =& \text{number of toys until 1st new type} \\
T_2 =& \text{additional number of toys until 2nd new type} \\
T_3 =& \text{additional number of toys until 3rd new type} \\
&\vdots
\end{aligned}$$

We know, $T_1 = 1$, $T_2 - 1 \sim \text{Geom}\left(\frac{n-1}{n}\right)$,..., $T_j - 1 \sim \text{Geom}\left(\frac{n-(j-1)}{n}\right)$. Thus,

$$\begin{aligned}
E(T) =& E(T_1) + E(T_2) + \cdots + E(T_n) \\
=& 1 + \frac{n}{n-1} + \frac{n}{n-2} + \cdots + \frac{1}{n} \\
=& n(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}) \\
\to& n(\log n + 0.577).
\end{aligned}$$

If $n = 5$, $E(T) \approx 11$; if $n = 10$, $E(T) \approx 29$.

# 24 Dependence

We need a tool to study collections of variables. Knowledge of each individual PMF is of little help. Because variables can be dependent on each each other (they are not necessarily independent). We need to know their inter-relationship. Joint distribution gives the probability that two or more random variables simultaneously takes particular values.

**Definition 24.1** (Joint distribution)**.** The joint PMF of random variables $(X, Y)$ is given by

$$f(x, y) = P(X = x, Y = y).$$

The joint CDF of random variables $(X, Y)$ is given by

$$F(x, y) = P(X \leq x, Y \leq y).$$

**Theorem 24.1.** *The discrete random variables $X$ and $Y$ are **independent** if and only if*

$$P(X = x, Y = y) = P(X = x)P(Y = y)$$

*for all possible values of $x, y$.*

*Equivalently, the condition can be stated with CDF: the random variables $X$ and $Y$ are **independent** if and only if*

$$P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y)$$

*for all possible values of $x, y$.*

*Proof.* $X$ and $Y$ are independent implies the event $\{X = x\}$ and $\{Y = y\}$ are independent for any $x, y$. By Definition 10.1, we have

$$P(X = x, Y = y) = P(\{X = x\} \cap \{Y = y\}) = P(X = x)P(Y = y).$$

□

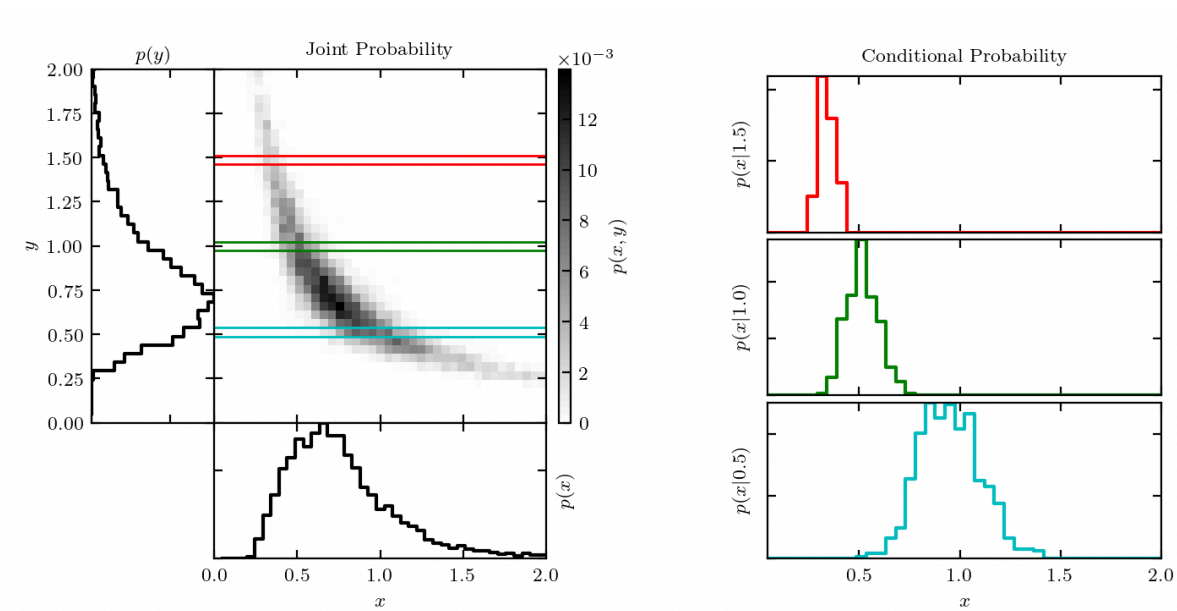
> **Note**
>
> If $X$ and $Y$ are independent, then any function of $X$ is independent of any function of $Y$.

**Definition 24.2** (Marginal distribution)**.** The marginal distribution gives the distribution of a subset of variables in a joint distribution without reference to the values of the other variables.

The marginal PMF of $X$ given the joint PMF of $(X, Y)$ is given by

$$f_X(x) = \sum_y P(X = x, Y = y) = \sum_y f_{X,Y}(x, y).$$

> **Note**
>
> It is easy to compute the marginal distribution given the joint distribution. However, in general, we cannot deduce the joint distribution from the marginal distribution. Unless the random variables are independent, the joint distribution is **not** the product of marginal distributions.




**Example 24.1.** Let $X$ be an indicator of an individual being a current smoker. Let $Y$ be the indicator of his developing lung cancer at some point in his life. The joint PMF of $X$ and $Y$ is as specified in the table below.

| | $Y = 1$ | $Y = 0$ | **Total** |
|---|---|---|---|
| $X = 1$ | 0.05 | 0.20 | **0.25** |
| $X = 0$ | 0.03 | 0.72 | **0.75** |

| | | | |
|---|---|---|---|
| **Total** | **0.08** | **0.92** | **1** |

The marginal PMF for having lung cancer is

$$\begin{aligned} P(Y=1) =& P(Y=1, X=0) + P(Y=1, X=1) = 0.08, \\ P(Y=0) =& P(Y=0, X=0) + P(Y=0, X=1) = 0.92. \end{aligned}$$

In this example, $X, Y$ are not independent, because

$$P(X=1, Y=1) \neq P(X=1)P(Y=1).$$

**Definition 24.3.** If a given number of random variables are independent and have the same distribution, we call them **independent and identically distributed**, or **i.i.d** for short.

- Independent and identically distributed ($X, Y$ independent die rolls)
- Independent and not identically distributed ($X$: die roll; $Y$: coin flip)
- Dependent and identically distributed ($X$: number of Heads; $Y$: number of Tails)
- Dependent and not identically distributed ($X$: economic growth; $Y$: presidential election)

> **Note**
>
> We view random sample as a collection of i.i.d random variables from the same population distribution. For example, let $X_i$ be the test score of student $i$. We say $X_1, X_2, \dots, X_n \overset{iid}{\sim} G$ where $G$ is the (unknown) population distribution for test scores.
> The **independent** assumption means that one observation does not influence another, while the **identically distributed** assumption ensures all observations follow the same probability law. This perspective simplifies statistical analysis and is foundational for many statistical inference.

# 25 Conditional distribution

**Definition 25.1** (Conditional distribution)**.** The conditional PMF of $Y$ given $X = x$ is defined as

$$f_{Y|X}(y|x) = P(Y = y|X = x) = \frac{P(Y = y, X = x)}{P(X = x)}$$

for any $x$ such that $P(X = x) > 0$.

**Definition 25.2** (Conditional expectation)**.** The conditional expectation of $Y$ given $X = x$ is defined as

$$\psi(x) = E(Y|X = x) = \sum_{\text{all } y} y \; f_{Y|X}(y|x).$$

$\psi(x)$ depends on the value of $x$ taken by $X$, so it can be thought of as a function $\psi(X)$ of $X$ itself.

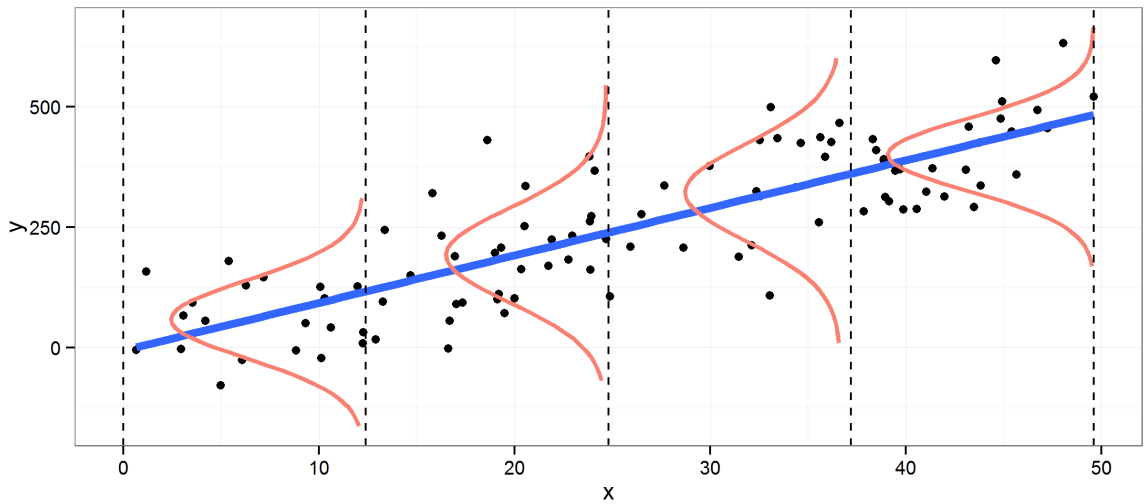
$$\psi(X) = E(Y|X)$$

is called the conditional expectation of $Y$ given $X$.

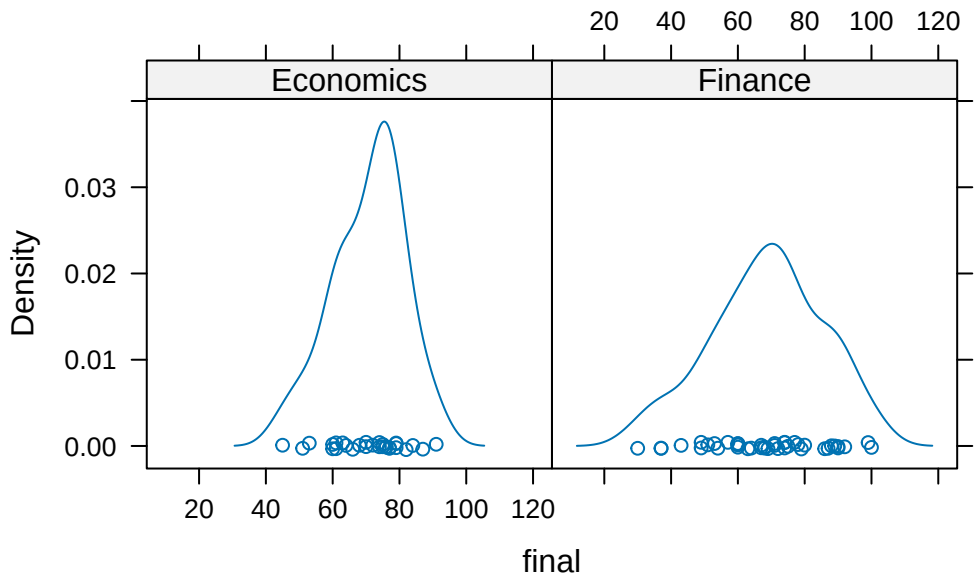
> **Note**
>
> Although $E(X)$ is a number, $E(Y|X)$ is a random variable. It is a function of random variable $X$, and therefore it is a random variable itself.

Conditional distribution is a key concept in probability, describing how the distribution of one random variable depends on the values of other variables—an idea central to many practical applications. For instance, we might be interested in how income distributions vary by education level or how the probability of a disease changes with age.

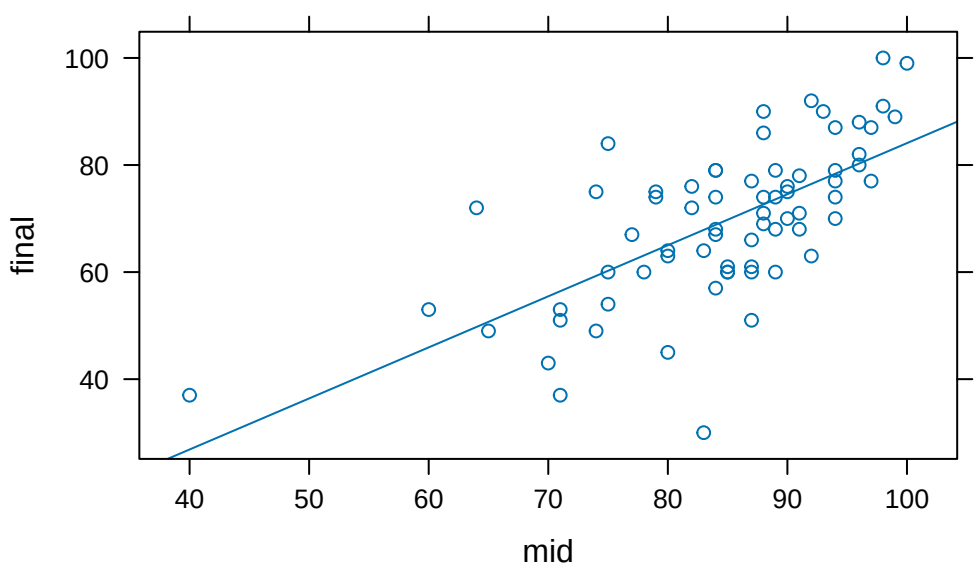
Conditional expectation gives the expected value of one variable given the value of another. It is frequently used for making predictions, such as predicting your earnings given that you graduate from a this college.




```r
library(lattice)

# conditional distribution of exam scores
exam <- read.csv("../dataset/exam.csv")

# distribution of exam scores conditioned on major
densityplot(~ final | major, data = exam)
```




```
# final exam scores conditioned on midterm exam scores
xyplot(final ~ mid, type=c("p", "r"), data = exam)
```



# 26 Poisson distribution

Now we introduce arguably the most popular discrete distribution—Poisson distribution. Poisson distribution is used to model independent events occurring at a constant mean rate. It is like the Binomial distribution in the sense that they both model the number of occurrence of events, but it is parametrized on the "rate" of the event (how many times an event occurs in a unit of time on average) rather than the total number of events and the probability of each event. It is therefore more practical in real-world modeling since we mostly observe the rate rather than the totality. We introduce the Poisson distribution by showing that it is a limiting case of the Binomial distribution.

**Question:** Suppose we are studying the distribution of the number of visitors to a certain website. Every day, a million people independently decide whether to visit the site, with probability $p = 2\times 10^{-6}$ of visiting. What is the probability of getting $k$ visitors on a particular day?

We can model the problem with a Binomial distribution. Let $X \sim \text{Bin}(n, p)$ be the number of visitors, where $n = 10^6$ and $p = 2 \times 10^{-6}$. But it is easy to run into computational difficulties with such a large $n$ and small $p$. This is not uncommon, if we want to model the number of emails one receives per day, or the number of phone calls in a service center. In such cases, we could reasonably assume $n \to \infty$ and $p \to 0$ while $np = \lambda$ is a constant. We may call $\lambda$ — the "rate", as it can be interpreted as the average visitors per day.

Take limit of the Binomial distribution:

$$\begin{aligned}
P(X = k) &= \lim_{n\to\infty} \binom{n}{k} p^k (1-p)^{n-k} \\
&= \lim_{n\to\infty} \binom{n}{k} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} \\
&= \lim_{n\to\infty} \frac{n!}{(n-k)!k!} \cdot \frac{\lambda^k}{n^k} \underbrace{\left(1 - \frac{\lambda}{n}\right)^n}_{\to e^{-\lambda}} \underbrace{\left(1 - \frac{\lambda}{n}\right)^{-k}}_{\to 1} \\
&= \lim_{n\to\infty} \underbrace{\frac{n!}{n^k(n-k)!}}_{\to 1} \frac{\lambda^k}{k!} e^{-\lambda} \\
&= \frac{\lambda^k}{k!} e^{-\lambda}.
\end{aligned}$$

This is the PMF of the Poisson distribution.

> **The limiting definition of exponential function**
>
> $$e^x = \lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n$$

**Definition 26.1** (Poisson distribution)**.** A random variable $X$ has the Poisson distribution with parameter $\lambda$ if the PMF of $X$ is

$$P(X=k) = \frac{e^{-\lambda}\lambda^k}{k!}, \quad k = 0, 1, 2, ...$$

We denote this as $X \sim \text{Pois}(\lambda)$.

We can easily verify this is a valid PMF because $\sum_{k=0}^{\infty}\frac{\lambda^k}{k!} = e^{\lambda}$.

**Theorem 26.1.** *If $X \sim Bin(n,p)$ and we let $n \to \infty$ and $p \to 0$ such that $\lambda = np$ remains fixed, then the PMF of $X$ converges to the PMF of $Pois(\lambda)$.*

The expectation of the Poisson distribution is

$$\begin{aligned}
E(X) =& \sum_{k=0}^{\infty} k \cdot \frac{e^{-\lambda}\lambda^k}{k!} \\
=& e^{-\lambda}\sum_{k=1}^{\infty}\frac{\lambda^k}{(k-1)!} \\
=& \lambda e^{-\lambda}\sum_{k=1}^{\infty}\frac{\lambda^{k-1}}{(k-1)!} \\
=& \lambda e^{-\lambda}e^{\lambda} = \lambda.
\end{aligned}$$

To get the variance, we first compute $E(X^2)$. By LOTUS,

$$E(X^2) = \sum_{k=0}^{\infty} k^2 \cdot \frac{e^{-\lambda}\lambda^k}{k!} = e^{-\lambda}\sum_{k=1}^{\infty} k^2\frac{\lambda^k}{k!}$$

Differentiate $\sum_{k=0}^{\infty}\frac{\lambda^k}{k!} = e^{\lambda}$ on both sides with respect to $\lambda$ and multiply (replenish) again by $\lambda$:

$$\sum_{k-1}^{\infty} k \frac{\lambda^k}{k!} = \lambda e^{\lambda}$$

Repeat:

$$\sum_{k-1}^{\infty} k^2 \frac{\lambda^k}{k!} = \lambda(e^{\lambda} + \lambda e^{\lambda})$$

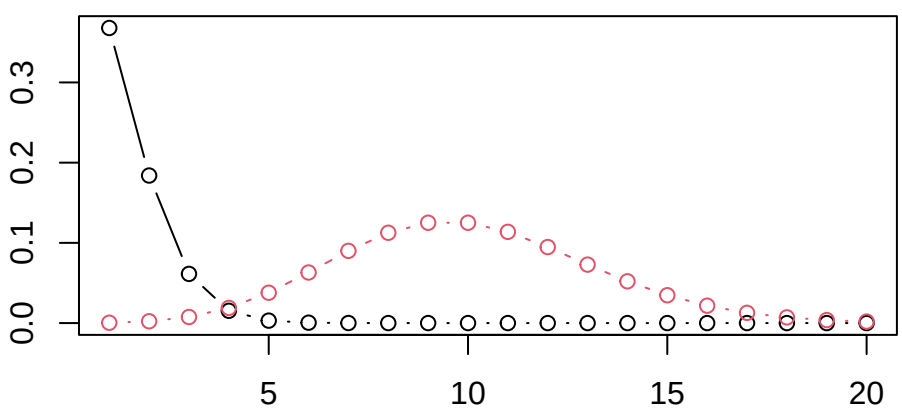
Therefore, we have

$$E(X^2) = e^{-\lambda}(\lambda + \lambda^2)e^{\lambda} = \lambda + \lambda^2$$

Finally,

$$Var(X) = E(X^2) - (E(X))^2 = \lambda + \lambda^2 - \lambda^2 = \lambda.$$

```
x <- 1:20
y1 <- dpois(x, lambda=1)
y2 <- dpois(x, lambda=10)
plot(x, y1, type="b", ann=F)
lines(x, y2, type="b", col=2)
```



**Example 26.1.** Continued with the website visiting example, there are one million people visiting the site every day, each with probability $p = 2 \times 10^{-6}$. Give an approximation for the probability of getting at least three visitors on a particular day.

Let $X$ be the number of visitors. Since $n$ is large, $p$ is small, $np = 2$ is fixed, $X$ is well approximated by Pois(2). Therefore,

$$\begin{aligned} P(X \geq 3) = 1 - P(X < 3) &= 1 - P(X = 0) - P(X = 1) - P(X = 2) \\ &= 1 - e^{-2} - 2e^{-2} - \frac{2^2}{2!}e^{-2} \\ &= 1 - 5e^{-2} \approx 0.32. \end{aligned}$$

The Poisson distribution is often used in situations where we are counting the number of successes in a particular region or interval of time, where there are a large number of trials, each with a small probability of success. The Poisson paradigm says in situations like this, we can approximate the number of successes by a Poisson distribution. It is more general than Theorem 26.1, as we relax the assumption of independence and identical events.

**Proposition 26.1** (Poisson paradigm)**.** *Let $A_1, \dots, A_n$ be events with $p_j = P(A_j)$, where $n$ is large, the $p_j$ are small, and the $A_j$ are independent or weakly dependent. Then $X = \sum_{j=1}^n I(A_j)$, that is how many of the $A_j$ occur, is approximately distributed as Pois($\lambda$) with $\lambda = \sum_{j=1}^n p_j$.*

The Poisson paradigm is also called the *law of rare events*. The interpretation of "rare" is that the $p_j$ are small, but $\lambda$ is relatively stable. The number of events that occur may not be exactly Poisson, but the Poisson distribution often gives good approximations. Note that the conditions for the Poisson paradigm to hold are fairly flexible: the $n$ trials can have different success probabilities, and the trials don't have to be independent, though they should not be very dependent. So there are a wide variety of situations that can be cast in terms of the Poisson paradigm. This makes the Poisson a very popular model.

Poisson distribution is also used to model the number of **events occurring randomly over time** with **constant rate**, such as the number of customers visiting a store, the number of phone calls to a call center, and so on.

Why the random occurrence of events has anything to do with the Poisson distribution? Consider in this way: one can divide the time line into infinitely small intervals (e.g. milliseconds). In each interval, an event either happens or not. The chance that an event occurs in a millisecond is very small. While there are infinitely many trials. So counting events occurring randomly at a fixed average rate over time is mathematically equivalent to counting rare events in many trials.

**Definition 26.2** (Poisson process)**.** A sequence of arrivals in continuous time is a Poisson process with rate $\lambda$ per unit of time if

- The number of arrivals in an interval of length $t$ is distributed $\text{Pois}(\lambda t)$;
- The numbers of arrivals in disjoint time intervals are independent.

# 27 Earthquake simulation

**Example 27.1.** What is the probability of $k$ earthquakes occurred in a random year in Sichuan?

This is the case of rare events with approximately constant rate of occurrence. We intend to model this with the Poisson distribution. The key is to figure out the parameter $\lambda$.

Historical records[1] show that, from 26 BCE to 2021 CE, there were 309 earthquakes with magnitude of 5.0 or greater. Let $X$ be the number of earthquakes with magnitude 5.0 or greater. The annual rate $\lambda$ of earthquakes is therefore $\frac{309}{2048} = 0.15$.

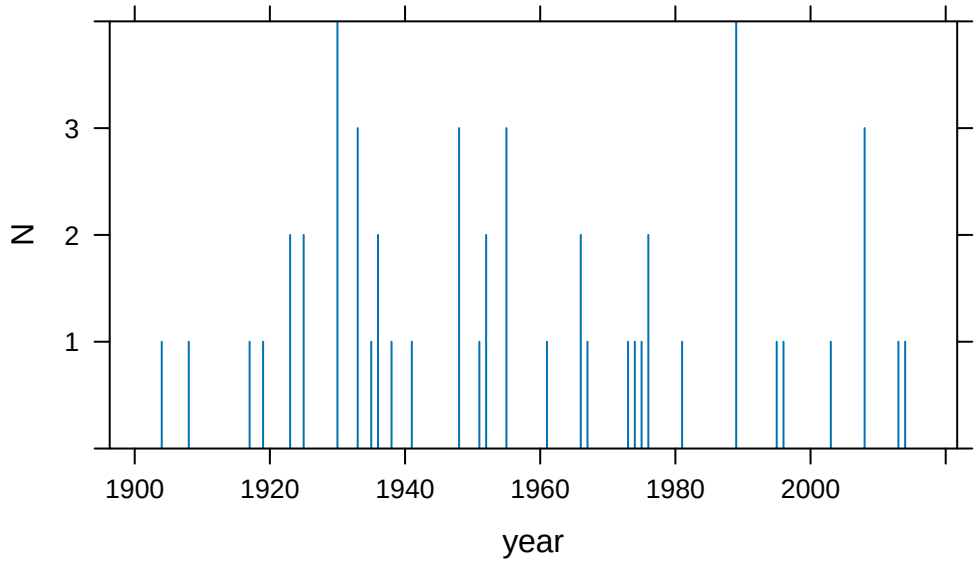
Assume earthquakes are independent events (not always the case). Then $X \sim \text{Pois}(0.15)$. By the distribution of the Poisson distribution,

$$P(X = k) = \begin{cases} 0.86 & k = 0 \\ 0.13 & k = 1 \\ 0.01 & k = 2 \end{cases}.$$

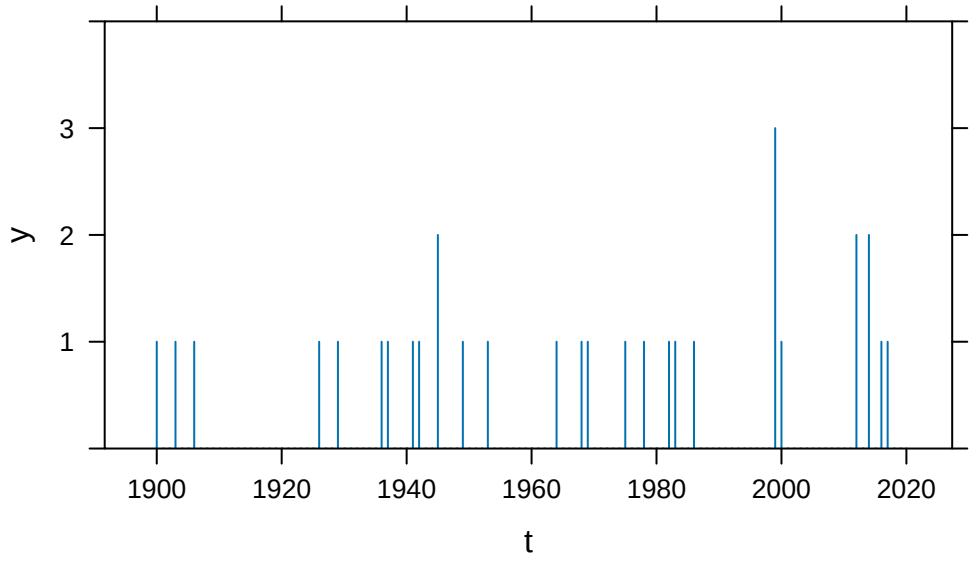
This is of course a rough approximation. Let's see how close the occurrences of earthquakes simulated by a Poisson process to the actual patterns of events.

```
library(lattice)
library(data.table)

# Earthquakes of Sichuan 1900-2019. source: USGS
eq <- fread("../dataset/earthq.csv")

# only consider strong earthquakes with magnitude >= 6
# count the number of earthquakes by year
eq6 <- eq[mag >=6, .N, by = year]

# plot the occurrence of earthquakes by year
xyplot(N~year, data=eq6, type="h", ylim=c(0,4))
```

[1] See this article from the Sichuan Earthquake Administration.




```r
# earthquakes are likely to be followed by earthquakes
# the rate of occurrence, however, looks constant
t <- seq(1900, 2019)
rate <- 0.3

# initialize the random generator
set.seed(3000)

# model the occurrence with Poisson distribution
y <- rpois(length(t), rate)

# plot the simulated occurrence of earthquakes
xyplot(y ~ t, type="h", ylim = c(0,4))
```




The simulated result looks somewhat similar to the history. However, there are a few notable difference. The simulated ones are distributed more evenly, whereas in reality, earthquakes are not independent. Moreover, the Poisson process seems to generate less rare events (more than 4 earthquakes a year) than the reality.

# 28 Birthday problem revisited

The beauty if approximating discrete problems by continuous function is that it makes calculation easier. Now we revisit the birthday problem with Poisson distribution.

**Example 28.1.** If we have $m$ people and $\binom{m}{2}$ pairs. Each pair of people has probability $p = 1/365$ of having the same birthday. Find the probability of at least one match.

*Solution.* The probability of match is small, and the number of pairs is large. We consider using the Poisson paradigm to approximate the number $X$ of birthday matches. $X \approx Pois(\lambda)$ where $\lambda = \binom{m}{2}\frac{1}{365}$. Then the probability of at least one match is

$$P(X \geq 1) = 1 - P(X = 0) \approx 1 - e^{-\lambda}.$$

For $m = 23$, $\lambda = 253/365$ and $1 - e^{-\lambda} \approx 0.5$, which agrees with our previous finding that we need 23 people to have 50% chance of a birthday match.

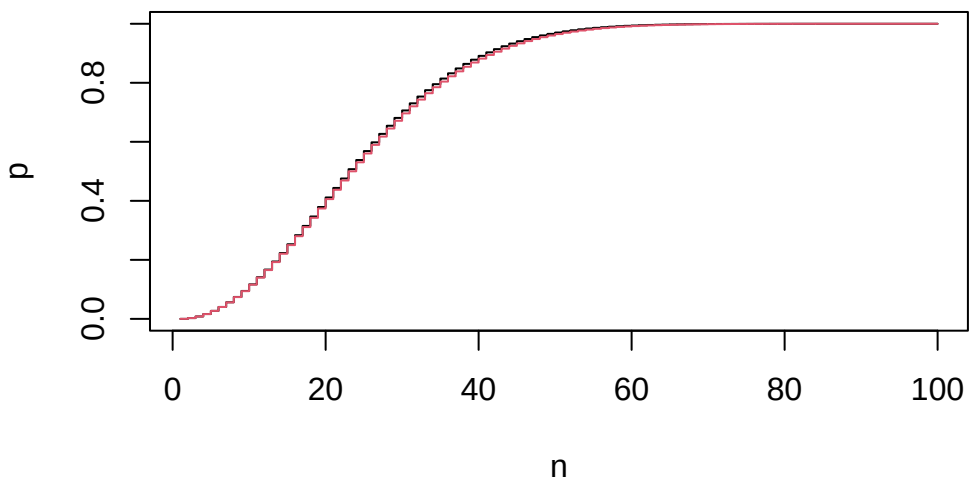
**Example 28.2.** Continued with the assumption above. What's the probability of two people who were born not only on the same day, but also at the same hour and the same minute?

*Solution.* This is the birthday problem with $c = 365 \cdot 24 \cdot 60 = 525600$ categories rather than 365 categories. By Poisson approximation, the probability of at least one match is approximately $1 - e^{-\lambda_1}$ where $\lambda_1 = \binom{m}{2}\frac{1}{525600}$. This would require $m = 854$ to reach the break even point, 50% chance of getting a match.

You may wonder how good the Poisson approximation is. We can compare it with the true values.

```
# compute the probability of coincidences for 1,2...100 people
n <- 1:100
p <- sapply(n, pbirthday)

# approximate the probability by Poisson paradigm
lambda <- choose(n, 2)/365
q <- 1 - exp(-lambda)

# black line is the true probability
```

```r
# red line is the Poisson approximation
plot(n, p, type = "s")
lines(n, q, col = 2, type="s")
```



# 29 Convolution

A convolution is a sum of independent random variables. The main task in this section is to determine the distribution of $T = X + Y$, where $X$ and $Y$ are independent random variables whose distributions are known.

**Theorem 29.1** (Convolution)**.** *If $X$ and $Y$ are independent discrete random variables, then the PMF of their sum $T = X + Y$ is*

$$\begin{aligned} P(T = t) &= \sum_x P(Y = t - x)P(X = x) \\ &= \sum_y P(X = t - y)P(Y = y) \end{aligned}.$$

*If $X$ and $Y$ are independent continuous random variables, then the PDF of their sum $T = X+Y$ is*

$$\begin{aligned} f_T(t) &= \int_{-\infty}^{\infty} f_Y(t - x)f_X(x)dx \\ &= \int_{-\infty}^{\infty} f_X(t - y)f_Y(y)dy. \end{aligned}$$

**Theorem 29.2** (Sum of Binomial random variables)**.** *Let $X \sim Bin(n, p)$ and $Y \sim Bin(m, p)$ be two independent Binomial random variables. Then $X + Y \sim Bin(n + m, p)$.*

*Proof.* We have proved the theorem in Theorem 19.1. Here is another way to prove it using convolution.

$$
\begin{aligned}
P(X+Y=k) &= \sum_{i=0}^{k} P(X=i)P(Y=k-i) \\
&= \sum_{i=0}^{k} \binom{n}{i} p^i (1-p)^{n-i} \binom{m}{k-i} p^{k-i} (1-p)^{m-k+i} \\
&= \sum_{i=0}^{k} \binom{n}{i} \binom{m}{k-i} p^k (1-p)^{m+n-k} \\
&= p^k (1-p)^{m+n-k} \sum_{i=0}^{k} \binom{n}{i} \binom{m}{k-i} \\
&= p^k (1-p)^{m+n-k} \binom{n+m}{k}.
\end{aligned}
$$

The last step: $\binom{n+m}{k} = \sum_{i=0}^{k} \binom{n}{i}\binom{m}{k-i}$

is known as the Vandermonde's identity. □

**Example 29.1** (Sum of Poisson random variables)**.** If $X \sim \text{Pois}(\lambda_1)$, $Y \sim \text{Pois}(\lambda_2)$, and $X, Y$ are independent, then $X + Y \sim \text{Pois}(\lambda_1 + \lambda_2)$.

*Proof.* Intuitively, $X$ is the number of events occurring at rate $\lambda_1$; $Y$ is the number of events occurring at rate $\lambda_2$. Therefore, $X + Y$ should be events occurring at rate $\lambda_1 + \lambda_2$.

To get the PMF of $X + Y$, condition on $X$ and use the law of total probability:

$$
\begin{aligned}
P(X+Y=k) &= \sum_{j=0}^{k} P(Y=k-j)P(X=j) \\
&= \sum_{j=0}^{k} \frac{e^{-\lambda_2}\lambda_2^{k-j}}{(k-j)!} \cdot \frac{e^{-\lambda_1}\lambda_1^{j}}{j!} \\
&= \frac{e^{-(\lambda_1+\lambda_2)}}{k!} \sum_{j=0}^{k} \binom{k}{j} \lambda_1^j \lambda_2^{k-j} \\
&= \frac{e^{-(\lambda_1+\lambda_2)}}{k!} (\lambda_1+\lambda_2)^k.
\end{aligned}
$$

We thus arrive at the PMF for $\text{Pois}(\lambda_1 + \lambda_2)$. Intuitively, if there are two different types of events occurring at rates $\lambda_1$ and $\lambda_2$, independently, then the overall event rate is $\lambda_1 + \lambda_2$. □

# 30 Application: seller ratings*

This example involves multiple types of discrete distributions. The technique used to solve this problem aligns with Bayesian inference, which is beyond the scope of this course. However, it remains an interesting case. The procedure illustrates the process of statistical modeling: we begin with an assumption and a proposed statistical model, then update it with new data. Finally, we draw inferences based on the model, typically addressing the question we aim to answer. You are not required to understand everything in this example. Nonetheless, it helps to develop a mindset of statistical inference early in the study.

Suppose you are shopping a product online. There are three sellers with the following ratings:

- Seller 1: 100% positive out of 10 reviews
- Seller 2: 96% positive out of 50 reviews
- Seller 3: 93% positive out of 200 reviews

Which seller is likely to give the best service?

The problem is intriguing because it is obvious that higher ratings do not necessarily means higher satisfaction. We have to weight in the number of reviews. The more reviews, the more trustworthy the ratings are. Let $X_j^{(i)}$ be a random variable that means consumer $j$ is satisfied with seller $i$, where $i \in \{1, 2, 3\}$. Assume $X_j^{(i)}$ follows a Bernoulli distribution:

$$X_j^{(i)} = \begin{cases} 1 & \text{satisfied with probability } \theta_i \\ 0 & \text{otherwise} \end{cases}$$

where $\theta_i$ is an unknown parameter of seller $i$ that captures their "genuine" satisfaction rate. We assume the consumers independently write their ratings. The overall positive rate of seller $i$ is therefore $R_i = \frac{1}{n_i} \sum_j X_j^{(i)}$ where $n_i$ is the total number of reviews. We want to infer the value of $\theta_i$ from their observed positive rate $R_i$. From now on we drop the seller index $i$ to simply the notation since it is symmetric for all sellers.

Because we have no prior knowledge about $\theta$. We assume that $\theta$ takes any value from $[0, 1]$ equally likely, i.e. $\theta \sim \text{Unif}(0, 1)$. Assuming each $X_j$ is independent and identical, then

$$S = X_1 + X_2 + \cdots + X_n$$

follows the Binomial distribution with PMF:

$$p(k|\theta) = \binom{n}{k}\theta^k(1-\theta)^{n-k}$$

Our goal is to find: $p(\theta|k)$. Recall that the Bayes' rule allows us to invert the conditional probability:

$$p(\theta|k) = \frac{p(k|\theta)p(\theta)}{p(k)} = \frac{p(k|\theta)p(\theta)}{\int_{-\infty}^{\infty} p(k|\theta)p(\theta)d\theta}$$

Since $\theta \sim \text{Unif}(0,1)$, we have

$$p(\theta) = \begin{cases} 1 & \text{if } \theta \in [0,1] \\ 0 & \text{otherwise} \end{cases}$$

We now focus on $\theta \in [0,1]$, since the probability is 0 otherwise. Substitute in the PMF of the Binomial distribution,

$$p(\theta|k) = \frac{\binom{n}{k}\theta^k(1-\theta)^{n-k}}{\int_0^1 \binom{n}{k}\theta^k(1-\theta)^{n-k}d\theta}$$

The hard part is to evaluate the integral. We state without proof (this is known as the Beta function, which we will prove in later chapters):

$$\int_0^1 \theta^k(1-\theta)^{n-k} = \frac{k!(n-k)!}{(n+1)!}$$

Therefore,

$$p(\theta|k) = \frac{(n+1)!}{k!(n-k)!}\theta^k(1-\theta)^{n-k}$$

Now suppose you are the next customer. The probability that you would be satisfied is

$$
\begin{aligned}
P(X_{n+1} = 1|S = k) &= \int_0^1 P(x_{n+1} = 1|\theta)p(\theta|k)d\theta \\
&= \int_0^1 \theta \times \frac{(n+1)!}{k!(n-k)!}\theta^k(1-\theta)^{n-k}d\theta \\
&= \frac{(n+1)!}{k!(n-k)!}\int_0^1 \theta^{k+1}(1-\theta)^{(n+1)-(k+1)}d\theta \\
&= \frac{(n+1)!}{k!(n-k)!} \times \frac{(k+1)!(n-k)!}{(n+2)!} \\
&= \frac{k+1}{n+2}.
\end{aligned}
$$

Now we substitute the ratings for the three sellers:

- Seller 1: $n = 10, k = 10$
- Seller 2: $n = 50, k = 48$
- Seller 3: $n = 200, k = 186$

The probabilities that you would be satisfied with each seller are: 92%, 94%, 93%. The result is known as the **Laplace's rule of succession**. The rule of thumb is, pretending we have too more reviews: one is positive, the other is negative. Compute the satisfaction rate as $\frac{k+1}{n+2}$.

# Part IV

# Expectation and Variance

# 31 Expectation revisited

**Definition 31.1.** For discrete random variable $X$, the expectation of $X$ is defined as

$$E(X) = \sum_{\text{all } x} xP(X = x);$$

For continuous random variable $X$ with density function $f(x)$, the expectation is defined as

$$E(X) = \int_{-\infty}^{\infty} x\ f(x)\ dx.$$

**Proposition 31.1** (Linearity)**.** *For random variables $X_1, X_2, \dots, X_n$, regardless of their dependencies, it holds that*

$$E(X_1 + \cdots + X_n) = E(X_1) + \cdots + E(X_n).$$

*Proof.* We prove the simplest case $E(X + Y) = E(X) + E(Y)$.

$$\begin{aligned}
E(X+Y) &= \sum_{z=x+y} zP(X+Y=z) \\
&= \sum_x \sum_y (x+y)P(X=x, Y=y) \\
&= \sum_x \sum_y xP(X=x, Y=y) + \sum_x \sum_y yP(X=x, Y=y) \\
&= \sum_x x \sum_y P(X=x, Y=y) + \sum_y y \sum_x P(X=x, Y=y) \\
&= \sum_x xP((X=x) \cap \bigcup_{\text{all } y} (Y=y)) + \sum_y yP(\bigcup_{\text{all } x} (X=x) \cap (Y=y)) \\
&= \sum_x xP(X=x) + \sum_y yP(Y=y) \\
&= E(X) + E(Y).
\end{aligned}$$

□

**Proposition 31.2.** *Further properties on the linearity of expectations:*

- *If* $Y = aX + b$*, then* $E(Y) = aE(X) + b$*.*
- $E(a_1X_1 + \cdots + a_nX_n + b) = a_1E(X_1) + \cdots + a_nE(X_n) + b$

**Proposition 31.3** (Multiplication)**.** *If* $X$ *and* $Y$ *are independent, we have*

$$E(XY) = E(X)E(Y).$$

*In general, if* $X_1, \ldots, X_n$ *are independent, we have*

$$E(X_1X_2 \cdots X_n) = E(X_1)E(X_2) \cdots E(X_n).$$

*Proof.* For discrete and independent $X, Y$,

$$\begin{aligned} E(XY) &= \sum_x \sum_y xyP(X = x, Y = y) \\ &= \sum_x \sum_y xyP(X = x)P(Y = y) \quad \text{if independent} \\ &= \sum_x xP(X = x) \sum_y yP(Y = y) \\ &= E(X)E(Y). \end{aligned}$$

□

**Multiplication does not hold without independence**

It is misleadingly natural to extend the generality of the addition rule to multiplication. But the multiplication rule of expectation is very restrictive. Always remember to check independence before applying the multiplication rule.

**Sufficient but not necessary condition**

If $X, Y$ are independent, it follows that $E(XY) = E(X)E(Y)$. However, the latter does not imply independence. Consider a counter-example,

$$X = \begin{cases} 1 & \text{with prob. } 1/2 \\ 0 & \text{with prob. } 1/2 \end{cases}, \quad Z = \begin{cases} 1 & \text{with prob. } 1/2 \\ -1 & \text{with prob. } 1/2 \end{cases};$$

Then

$$Y = XZ = \begin{cases} -1 & \text{with prob. } 1/4 \\ 0 & \text{with prob. } 1/2 \\ 1 & \text{with prob. } 1/4 \end{cases}.$$

We have $E(X) = 1/2$, $E(Y) = 0$, $E(XY) = 0$. So $E(XY) = E(X)E(Y)$. But clearly $X, Y$ are not independent.

**Theorem 31.1** (Law of the unconscious statistician (LOTUS))**.** *Let $X$ be a random variable, and $g$ be a real-valued function of a real variable. If $X$ has a discrete distribution, then*

$$E[g(X)] = \sum_{all\ x} g(x)P(X = x).$$

LOTUS says we can compute the expectation of $g(X)$ without knowing the PMF of $g(X)$.

**Example 31.1.** Compute $E(X)$ and $E(X^2)$ given the following distribution.

| $X$ | 0 | 1 | 2 |
|---|---|---|---|
| $X^2$ | 0 | 1 | 4 |
| $P$ | 1/4 | 1/2 | 1/4 |

*Solution.* According to the distribution table, we compute the expectations as

$$E(X) = 0 \times 1/4 + 1 \times 1/2 + 2 \times 1/4 = 1;$$
$$E(X^2) = 0 \times 1/4 + 1 \times 1/2 + 4 \times 1/4 = 3/2.$$

Note that $E(X^2) \neq [E(X)]^2$.

**Don't pull non-linear functions out of expectation**

In general, $E[g(X)] \neq g(E(X))$. Linearity implies $E[g(X)] = g(E(X))$ if $g$ is a linear function. For a nonlinear function $g$, you can't pull function $g$ out of expectation $E$. The right way to find $E[g(X)]$ is with LOTUS.

**Example 31.2** (St. Petersburg Paradox)**.** Flip a fair coin over and over again until the head lands the first time. You will win $2^k$ dollars if the head lands in the $k$-th trial (including the successful trial). What is the expected payoff of this game?

*Solution.* Let $X = 2^k$. We want to find $E(X)$. The probability of the first head showing up in the $k$-th trial is $\frac{1}{2^k}$. Therefore,

$$E(X) = \sum_{k=1}^{\infty} 2^k \cdot \frac{1}{2^k} = \sum_{k=1}^{\infty} 1 = \infty$$

The expected payoff is infinitely high! This is against most people's intuition, which is likely to be a small number. This is because we mistakenly go through the calculation $E(X) = E(2^k) = 2^{E(k)}$ in our mind. $E(k)$ the expected number of flips before a head is 2. Thus, $2^{E(k)} = 4$.

Another way to resolve the paradox is that we don't typically reason about infinity. No one would play this game infinitely many times. For finite number of plays, the probability of getting very large payoff, say $2^{100}$, is none. We can demonstrate this with a simulation.

```r
# Function to simulate the St. Petersburg Paradox
st_petersburg_game <- function(n_simulations) {
  # Initialize a vector to store the outcomes
  outcomes <- numeric(n_simulations)

  for (i in 1:n_simulations) {
    # Start with the initial reward
    reward <- 2
    # Flip a coin until it lands tails
    while (runif(1) < 0.5) {
      reward <- reward * 2
    }
    # Store the reward for this simulation
    outcomes[i] <- reward
  }

  # Return the outcomes
  return(outcomes)
}

# Set the number of simulations
n_simulations <- 1000

# Run the simulation
results <- st_petersburg_game(n_simulations)

# Calculate the average outcome (expected value)
expected_value <- mean(results)

# Print the results
cat("Simulated Expected Value:", expected_value)
```

```
Simulated Expected Value: 18.012
```

```r
# However, as the number of simulations increases
# We would see higher and higher maximum reward
cat("Maximum Reward Observed:", max(results))
```

```
Maximum Reward Observed: 4096
```

# 32 Life expectancy

Life expectancy is the average number of years a person is expected to live. It is a crucial indicator of the quality of living and one of the three components of the Human Development Index (HDI) (the other two components are education and per capita GDP). Here is a toy example to compute life expectancy with hypothetical data.[1]

| (1) Age | (2) Population | (3) Mortality rates | (4) # Survive | | (5) # Died at age | (6) P(Age) |
|---|---|---|---|---|---|---|
| 0 | 200 | 1% | 1000 | | 10 | 1% |
| 20 | 300 | 2% | 990 | =1000(1-1%) | 20 | 2% |
| 40 | 250 | 10% | 970 | =990(1-2%) | 97 | 10% |
| 60 | 150 | 20% | 873 | =970(1-10%) | 175 | 17% |
| 80 | 100 | 100% | 699 | =873(1-20%) | 699 | 70% |
| Total | 1000 | | | | | |

To simplify our analysis, we will assume there are only five possible ages: 0, 20, 40, 60, and 80. A baby is born at age 0, and can either die at that age or survive to age 20. We intentionally exclude intermediate ages such as 5 and 10 for the sake of computational simplicity.

It's important to note that life expectancy is not the same as the average age of the population. For instance, based on the hypothetical data presented, the average age can be calculated as:

$$\overline{\text{Age}} = (0 \times 200 + 20 \times 300 + 40 \times 250 + 60 \times 150 + 80 \times 100)/1000 = 33.$$

However, the expected age, denoted as $E(\text{Age})$, is defined as:

$$E(\text{Age}) = \sum \text{Age} \times P(\text{Age}).$$

To compute this expected value, we need to determine $P(\text{Age})$, the probability of living to a specific age or dying at that age. This requires consideration of the mortality rate at each age, which is given in Column 3.

[1] This is an overly simplified example that only serves to clarify the definition of expectation. See this tutorial from MEASURE Evaluation for the actual computation of life expectancy.

Assuming 1000 babies are born at age 0, with a mortality rate of 1% at that age, we find that 99% of the babies survive to age 20. Thus, the number of babies that survive to age 20 is: $1000 \times (1-1\%) = 990$. We can apply similar calculations to determine the number of survivors at each subsequent age.

The number of individuals who die at a specific age (Column 5) is the difference between the number of survivors at that age and the next (Column 4). To find the probability of living to a specific age, we compute: $P(\text{Age}) = \text{Column } 4/1000$.

Finally, we compute the expected value of age (or life expectancy) as follows:

$$E(Age) = 0 \times 1\% + 20 \times 2\% + 40 \times 10\% + 60 \times 17\% + 80 \times 70\% = 70.6.$$

This figure differs from the average age. Since the mortality rate is low at younger ages, the probabilities $P(\text{Age})$ for these ages are also low, while they are higher for older ages. This example illustrates the distinction between average and expected values. In everyday conversation, we may use these terms interchangeably, but in certain contexts, expected values can significantly differ from averages.

# 33 Two envelope paradox

**Example 33.1** (Two-envelope paradox)**.** Imagine you are given two identical envelopes, each containing money. One contains twice as much as the other. You may pick one envelope and keep the money it contains. Having chosen an envelope at will, but before inspecting it, you are given the chance to switch envelopes. Should you switch?

The paradox arises when you try to solve the expectation. Let $A$ denote the amount of money in the envelope you have chosen, and $B$ denote the amount of money in the other envelope.

We know $B$ is either twice as much as $A$, or half as much as $A$. Each with probability 1/2. So

$$E(B) = \frac{1}{2}(2A) + \frac{1}{2}(A/2) = \frac{5}{4}A$$

Since $E(B) > A$, you should always switch! However, after you switch to $B$, by the same argument, you should switch back to $A$. You you switch back and forth indefinitely!

**Where do things go wrong?** The error in this calculation lies in a subtle misunderstanding: the two $A$s in the calculation actually represent different values, that are incorrectly equated. In particular, the $2A$ represents the expected value in the other envelope given that it is the larger one, and the $A/2$ represents the expected value in the other envelope given that it is the smaller one.

$$E(B) = E(B|B < A)P(B < A) + E(B|B > A)P(B > A)$$

Suppose the amount of money in the two envelopes are $a$ and $2a$ respectively. $E(B|B < A) = a$ and $E(B|B > A) = 2a$. Therefore,

$$E(B) = \frac{1}{2}a + \frac{1}{2}2a = \frac{3}{2}a.$$

The same calculation applies to $E(A)$. Thus, $E(A) = E(B)$.

**Example 33.2** (HH vs HT)**.** Flip a coin indefinite times. Let $X$ denote the number of flips until you see HH. Let $Y$ denote the number of flips until you see HT. Find $E(X)$ and $E(Y)$.

It is tempting to think they are the same, since either H or T happens with probability 1/2. But the answer is extremely counter-intuitive: $E(X) > E(Y)$!

**HH case.** Let $E_0$ = E(X|No H observed), and $E_1$ = E(X|One H observed). Then

$$E_0 = 1 + \frac{1}{2}E_1 + \frac{1}{2}E_0$$

The first term is we need to flip once. If the first flip is H, the additional expected number of flips is $E_1$. If the first flip is T, we have to start over again ($E_0$).

$$E_1 = 1 + \frac{1}{2}(0) + \frac{1}{2}E_0$$

Once we have observed an H, we do another flip. If it is another H, we are done. If it is a T, we have to start over again ($E_0$).

Solve the two equations, we have $E_0 = 6$, $E_1 = 4$. Thus, $E(X) = 6$.

**HT case.** Let $E_0$ = E(Y|No H observed), and $E_1$ = E(Y|One H observed). Then

$$E_0 = 1 + \frac{1}{2}E_1 + \frac{1}{2}E_0$$

If the first flip is H, we need $E_1$. If the first flip is T, we have wasted the flip, so it is $E_0$ again.

$$E_1 = 1 + \frac{1}{2}(0) + \frac{1}{2}E_1$$

If we have a T by 1/2 chance, we are done (the first term). If it is an H, we get another $E_1$.

In this case, we have $E_0 = 4$, $E_1 = 2$. Thus, $E(Y) = 4$.

```
# Function to simulate the number of flips until a specific pattern appears
simulate_pattern <- function(pattern, n_simulations) {
  results <- numeric(n_simulations)

  for (i in 1:n_simulations) {
    flips <- character(0)  # Initialize an empty vector to store flips
    count <- 0  # Initialize the flip counter

    while (TRUE) {
      # Simulate a coin flip (H or T)
      flip <- sample(c("H", "T"), 1)
```

```r
      flips <- c(flips, flip)  # Append the flip to the sequence
      count <- count + 1  # Increment the flip counter

      # Check if the last two flips match the pattern
      if (length(flips) >= 2 && all(tail(flips, 2) == pattern)) {
        results[i] <- count  # Record the number of flips
        break  # Exit the loop
      }
    }
  }

  return(results)
}

# Set the number of simulations
n_simulations <- 2000

# Simulate for HH
results_HH <- simulate_pattern(c("H", "H"), n_simulations)
expected_HH <- mean(results_HH)

# Simulate for HT
results_HT <- simulate_pattern(c("H", "T"), n_simulations)
expected_HT <- mean(results_HT)

# Create a table of results
results_table <- data.frame(
  Pattern = c("HH", "HT"),
  Expected_Flips = c(expected_HH, expected_HT)
)

# Print the table
knitr::kable(results_table)
```

| Pattern | Expected_Flips |
|---|---:|
| HH | 5.936 |
| HT | 4.000 |

# 34 Linearity and indicators

**Definition 34.1** (Indicator variable)**.** An indicator variable $\mathbb{I}_A$ for an event $A$ is a random variable defined as:

$$\mathbb{I}_A = \begin{cases} 1 & \text{if event } A \text{ occurs,} \\ 0 & \text{if event } A \text{ does not occur.} \end{cases}$$

The indicator variable $\mathbb{I}_A$ "indicates" whether the event $A$ happens (1) or not (0).

The expected value of an indicator variable is equal to the probability of the event $A$:

$$E[\mathbb{I}_A] = 1 \cdot P(A) + 0 \cdot P(A^c) = P(A)$$

This is known as the **fundamental bridge**, as it allows us to convert between probability and expectation.

Indicator variables are often used in linearity of expectation calculations. This allows us to break down a problem into easy-to-solve small problems. For example, if $X = \sum_{i=1}^{n} \mathbb{I}_{A_i}$, then:

$$E[X] = \sum_{i=1}^{n} E[\mathbb{I}_{A_i}] = \sum_{i=1}^{n} P(A_i)$$

**Example 34.1.** In a group of $n$ people, what is the expected number of distinct birthdays among the $n$ people (the expected number of days on which at least one of the people was born)? What is the expected number of people sharing a birthday (any day)?

*Solution.* Let $X$ be the number of distinct birthdays, and write $X = I_1 + \cdots + I_{365}$, where

$$I_j = \begin{cases} 1 & \text{if someone was born on day } j \\ 0 & \text{otherwise} \end{cases}.$$

Then

$$\begin{aligned} E(I_j) &= P(\text{someone was born on day } j) \\ &= 1 - P(\text{no one was born on day } j) \\ &= 1 - \left(\frac{364}{365}\right)^n. \end{aligned}$$

Then by linearity,

$$E(X) = 365\left(1 - \left(\frac{364}{365}\right)^n\right).$$

Let $Y$ be the number of people sharing a birthday, and $Y = J_1 + \cdots + J_n$ where $J_k$ is an indicator that the $j$-th person shares his birthday with somebody else.

$$\begin{aligned} E(J_k) &= P(\text{someone shares birthday with } k) \\ &= 1 - P(\text{no one shares birthday with } k) \\ &= 1 - \left(\frac{364}{365}\right)^{n-1}. \end{aligned}$$

Therefore,

$$E(Y) = \sum_{k=1}^{n} E(J_k) = n\left(1 - \left(\frac{364}{365}\right)^{n-1}\right).$$

For some numeric values, $E(Y) = 2.3$ if $n = 30$; $E(Y) = 6.3$ if $n = 50$.

**Example 34.2.** Suppose that there are $n$ people sitting in a classroom with exactly $n$ seats. At some point, everyone got up, ran around the room, and sat back down randomly (i.e., all seating arrangements are equally likely). What is the expected value of the number of people sitting in their original seat?

*Solution.* Number the people from 1 to $n$. Let $X_i$ be the Bernoulli random variable with value 1 if person $i$ returns to their original seat and value 0 otherwise. Since person $i$ is equally likely to sit back down in any of the $n$ seats, the probability that person $i$ returns to their original seat is $1/n$. Therefore $E[X_i] = 1/n$. Now, let $X$ be the number of people sitting in their original seat following the rearrangement. Then $X = X_1 + X_2 + \cdots + X_n$. By linearity of expected values, we have $E[X] = \sum E[X_i] = \sum 1/n = 1$.

**Example 34.3.** Let $\Pi$ be a permutation over $\{1, 2, \ldots, n\}$. That is a reordering of the numbers. A fixed point of a permutation are the points not moved by the permutation. For example, in the permutation below

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $\Pi$ | 2 | 4 | 3 | 1 |

The fixed point is 3. Find the expected number of fixed points of a random permutation.

*Solution.* Let $X$ be the number of fixed points of a random permutation. Then $X = \sum_{k=1}^{n} 1_{\Pi(k)=k}$ where $1_{\Pi(k)=k}$ indicates the $k$-th number stays the same after the permutation. By linearity,

$$E(X) = E\left(\sum_{k=1}^{n} 1_{\Pi(k)=k}\right) = \sum_{k=1}^{n} E\left(1_{\Pi(k)=k}\right) = \sum_{k=1}^{n} \frac{1}{n} = 1.$$

**Example 34.4.** (Buffon's needle). Rule a surface with parallel lines a distance $d$ apart. What is the probability that a randomly dropped needle of length $l \leq d$ crosses a line?

*Solution.* Consider dropping any (continuous) curve of length $l$ onto the surface. Imagine dividing up the curve into $N$ straight line segments, each of length $\frac{l}{N}$. Let $X_i$ be the indicator for the $i$-th segment crossing a line. Let $X$ be the total number of times the curve crosses a line. Then,

$$E(X) = E(\sum X_i) = \sum E(X_i) = N \cdot E(X_i).$$

There could be infinitely many segments. It is hard to compute this expectation directly. But here we arrive an important Lemma: the expected number of crossings is proportional to the length of the curve, regardless of the shape of the curve. If we can compute $E(X)$ for some curve, the we can compute $E(X)$ for any length by scaling the value proportional to the length.

Consider a circle of diameter $d$. The circle always crosses the lines twice for sure. That is, $E(X_{\text{circle}}) = 2$. The length of the circle is $\pi d$. Therefore, the value of $E(X)$ for any curve of length $l$ is given by

$$E(X) = \frac{2l}{\pi d}.$$

Now a needle can cross a line either 1 or 0 times. Thus, $E(X) = 1 \cdot P(X = 1) + 0 \cdot P(X = 0)$ is exactly the probability of a needle crossing a line.

This amazing example can be used to approximate the value of $\pi$. Let $q$ be the probability of a needle crossing a line. $q$ can be approximated by large number of simulations. Then $\pi \approx \frac{2l}{qd}$.

```r
# Buffon's Needle Simulation
buffon_needle <- function(N, L, D) {
  # Initialize the number of crossings
  crossings <- 0

  for (i in 1:N) {
    # Randomly generate the position of the needle's midpoint
    y <- runif(1, min = 0, max = D / 2)  # Distance from the nearest line
    # Randomly generate the angle of the needle (in radians)
    theta <- runif(1, min = 0, max = pi / 2)  # Angle with respect to the lines

    # Check if the needle crosses a line
    if (y <= (L / 2) * sin(theta)) {
      crossings <- crossings + 1
    }
  }

  # Estimate pi
```

```r
  pi_estimate <- (2 * L * N) / (D * crossings)

  return(pi_estimate)
}

# Parameters
N <- 100000  # Number of trials
L <- 1       # Length of the needle
D <- 2       # Distance between the lines (D >= L)

# Run the simulation
pi_estimate <- buffon_needle(N, L, D)

# Print the result
cat("Estimated value of pi:", pi_estimate)
```

```
Estimated value of pi: 3.170778
```

# 35 Median and mode

The mean is called a measure of *central tendency* because it tells us something about the center of a distribution, specifically its center of mass. Other measures of central tendency that are commonly used in statistics are the median and the mode, which we now define.

**Definition 35.1** (Median)**.** We say that $c$ is a median of a random variable $X$ if

$$P(X \leq c) \geq 1/2 \text{ and } P(X \geq c) \geq 1/2.$$

Intuitively, the median is a value $c$ such that half the mass of the distribution falls on either side of $c$ (or as close to half as possible, for discrete random variables). Note that the condition given above is more general than

$$P(X \leq c) = P(X \geq c) = \frac{1}{2}$$

Consider a discrete distribution as follows:

$$P(X = k) = \begin{cases} \frac{1}{3}, & k = 1 \\ \frac{1}{2}, & k = 2 \\ \frac{1}{6}, & k = 3 \end{cases}$$

In this case, 2 is a median since $P(X \leq 2) = 5/6 \geq 1/2$ and $P(X \geq 2) = 2/3 \geq 1/2$. However, $P(X \leq 2) \neq P(X \geq 2)$. For strictly continuous random variable $X$, Definition 35.1 does imply

$$P(X \leq c) = P(X \geq c) = \frac{1}{2}$$

Since the CDF of $X$ satisfies $F(c) \geq 1/2$ and $1 - F(c) \geq 1/2$, which implies $F(c) = 1/2$. Moreover, if the CDF of $X$ is strictly increasing, $F^{-1}(1/2)$ is the unique median.

**Definition 35.2** (Mode)**.** For a discrete random variable $X$, we say that $c$ is a mode of $X$ if it maximizes the PMF:

$$P(X = c) \geq P(X = x) \quad \text{for all } x.$$

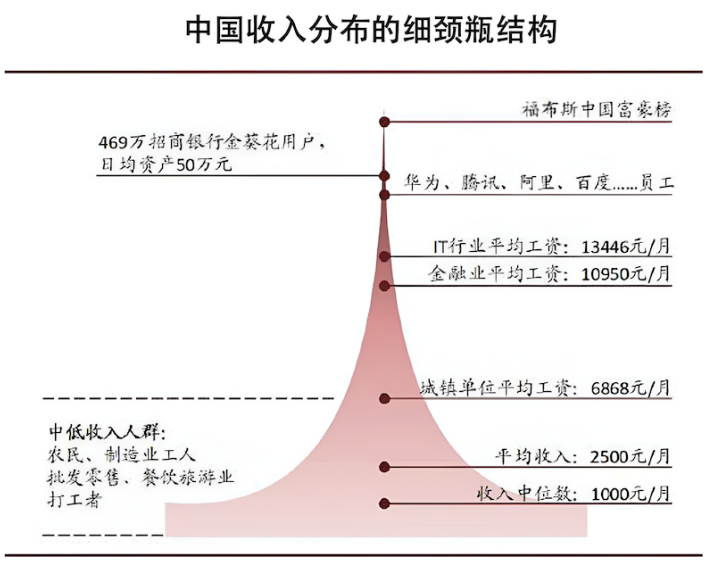
For a continuous random variable $X$ with PDF $f$, we say that $c$ is a mode if it maximizes the PDF:

$$f(c) \geq f(x) \quad \text{for all } x.$$

Intuitively, the mode is a value that has the greatest mass or density out of all values in the support of $X$.

> **Note**
>
> A distribution can have multiple medians and multiple modes. Medians have to occur side by side; modes can occur all over the distribution.

**Example 35.1.** The main reason why the median is sometimes preferred over the mean is that the median is more robust to extreme values. A typical example is the income distribution. Higher incomes are rare, but their absolute values are high. Thus, the mean income tends be higher than what the mass of the population would earn. But the median is more robust to extreme values and is closer to the earnings of an "average" person. For example, the average monthly income in China is ¥$2,500$ in 2019, but the median is only ¥$1,000$.




注：城镇单位平均工资为2018年数据，其他指标为2019年数据。

**Theorem 35.1.** *Let $X$ be an random variable with mean $\mu$ , and let $m$ be a median of $X$.*

- *A value of $c$ that minimizes the mean squared error $E\left(X-c\right)^2$ is $c=\mu$.*
- *A value of $c$ that minimizes the mean absolute error $E\left|X-c\right|$ is $c=m$.*

*Proof.* Part 1 - Minimizing the mean squared error $E[(X-c)^2]$. Expand the mean squared error:

$$E[(X-c)^2] = E[X^2 - 2cX + c^2] = E[X^2] - 2cE[X] + c^2.$$

To find the value of $c$ that minimizes this expression, take the derivative with respect to $c$ and set it to zero:

$$\frac{d}{dc}E[(X-c)^2] = -2E[X] + 2c = 0$$

This implies $c = \mu$. We can confirm with second-order condition that $c = \mu$ is indeed a minimizer.

Part 2 - Minimizing the mean absolute error $E\,|X-c|$.

$$E|X-c| = \int_{-\infty}^{c}(c-x)f(x)dx + \int_{c}^{\infty}(x-c)f(x)dx$$

Take derivative with respect to $c$, applying the Leibniz's rule:

$$(c-x)f(x)\frac{d}{dc}c + \int_{-\infty}^{c} f(x)dx - (x-c)f(x)\frac{d}{dc}c + \int_{c}^{\infty}(-f(x))dx = 0$$

The first-order condition resolves to

$$\int_{-\infty}^{c} f(x)dx = \int_{c}^{\infty} f(x)dx$$

which is exactly the definition of a median. $\square$

# 36 Variance

Expectation is the most commonly used summary of a distribution, as it indicates where values are likely centered. However, it provides limited insight into the distribution's overall shape. For example, two random variables might have the same mean, yet one could have values spread far from the mean while the other has values tightly clustered around it. Variance, on the other hand, describes how far values in a distribution typically deviate from the mean, offering a measure of the distribution's dispersion.

**Definition 36.1.** The **variance** of a random variable $X$ is defined as

$$Var(X) = E\left[X - E(X)\right]^2.$$

By convention, variance is also denoted by Greek letter $\sigma^2$, where $\sigma = \sqrt{Var(X)}$ is called the **standard deviation**.

Variance measures how far $X$ typically deviates from its mean, but instead of averaging the differences, we average the squared differences to ensure both positive and negative deviations contribute. The expected deviation, $E(X - E(X))$, is always zero, so squaring avoids this cancellation. Since variance is in squared units, we take the square root to get the standard deviation, restoring the original units.

**Why squared deviation?**

We can measure the dispersion of a distribution in different ways. For example, $E(|X - E(X)|)$ is also a possible choice. But it is less common because the absolute value function isn't differentiable. Besides, squaring connects to geometric concepts like the distance formula and Pythagorean theorem, which have useful statistical meanings.

**Sample variance**

Definition 36.1 gives the theoretical variance of a distribution. With finite sample from the distribution, we estimate the variance with sample observations:

$$\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

Why dividing by $n-1$? Because we want an unbiased estimator. We will discuss this in

later chapters.

**Theorem 36.1.** *For any random variable $X$,*

$$Var(X) = E(X^2) - (EX)^2.$$

*Proof.* Let $\mu = E(X)$. By definition,

$$\begin{aligned} Var(X) &= E(X-\mu)^2 = E(X^2 - 2\mu X + \mu^2) \\ &= E(X^2) - 2\mu E(X) + \mu^2 = E(X^2) - \mu^2. \end{aligned}$$

□

**Example 36.1.** Find the variance for $X \sim \text{Bern}(p)$.

$$Var(X) = E(X^2) - E^2(X) = p - p^2 = p(1-p).$$

**Proposition 36.1.** *Variance has the following properties:*

- $Var(X) \geq 0$
- $Var(X + c) = Var(X)$
- $Var(cX) = c^2 Var(X)$
- *If $X, Y$ are independent, $Var(X + Y) = Var(X) + Var(Y)$.*
- *If $X_1, X_2, \dots, X_n$ are independent, $Var(\sum_{i=1}^{n} X_i) = \sum_{i=1}^{n} Var(X_i)$.*

**Example 36.2.** Find the variance for $X \sim \text{Bin}(n, p)$. $X = X_1 + \cdots + X_n$ where $X_i$ are $i.i.d$ Bernoulli distributions

$$Var(X) \overset{iid}{=} \sum_{i=1}^{n} Var(X_i) = np(1-p).$$

# 37 Covariance

For more than one random variable, it is also of interest to know the relationship between them. Are they dependent? How strong is the dependence? Covariance and correlation are intended to measure that dependence. But they only capture a particular type of dependence, namely linear dependence.

**Definition 37.1.** The **covariance** between random variables $X$ and $Y$ is defined as

$$Cov(X, Y) = E[(X - EX)(Y - EY)].$$

The covariance between $X$ and $Y$ reflects how much $X$ and $Y$ simultaneously deviate from their respective means.

- If $X > EX$ & $Y > EY$ or $X < EX$ & $Y < EY$, then $Cov(X, Y)$ is positive.
- If $X > EX$ & $Y < EY$ or $X < EX$ & $Y > EY$, then $Cov(X, Y)$ is negative.

**Theorem 37.1.** *For any random variables $X$ and $Y$,*

$$Cov(X, Y) = E(XY) - E(X)E(Y).$$

*Proof.* Let $\mu_X = E(X)$ and $\mu_Y = E(Y)$. By definition,

$$\begin{aligned} Cov(X, Y) &= E(XY - \mu_X Y - \mu_Y X + \mu_X \mu_Y) \\ &= E(XY) - \mu_X E(Y) - \mu_Y E(X) + \mu_X \mu_Y \\ &= E(XY) - E(X)E(Y). \end{aligned}$$

□

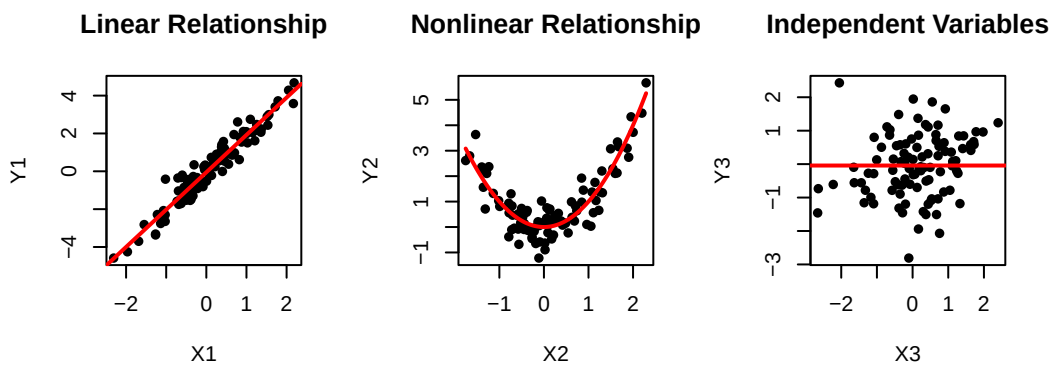
**Theorem 37.2.** *If $X, Y$ are independent, they are uncorrelated. But the converse is false.*

*Proof.* $Cov(X, Y) = E(XY) - E(X)E(Y)$. Independence implies $E(XY) = E(X)E(Y)$. Thus, $Cov(X, Y) = 0$. But $Cov(X, Y) = 0$ does not necessarily imply independence. Consider the following counter example. Let $X$ be a random variable that takes three values -1, 0, 1 with equal probability. And $Y = X^2$. $X$ and $Y$ are clearly dependent. But they their correlation is 0. Since $E(X) = 0$, $E(Y) = 2/3$, $E(XY) = E(X^3) = 0$, $Cov(X, Y) = 0$. □

**Linear dependency**

Covariance and correlation provide measures of the extend to which two random variables are linearly related. It is possible that the covariance is 0 even when $X$ and $Y$ are dependent but the relationship is nonlinear.

```r
# Set seed for reproducibility
set.seed(123)

# Number of observations
n <- 100

# Linear relationship
X1 <- rnorm(n)
Y1 <- 2 * X1 + rnorm(n, sd = 0.5)

# Nonlinear relationship
X2 <- rnorm(n)
Y2 <- X2^2 + rnorm(n, sd = 0.5)

# Independent variables
X3 <- rnorm(n)
Y3 <- rnorm(n)

# Visualize the relationship
# Set up a 1x3 grid for plots
par(mfrow = c(1, 3))

# Plot 1: Linear Relationship
plot(X1, Y1, main = "Linear Relationship", pch = 16)
abline(lm(Y1 ~ X1), col = "red", lwd = 2)

# Plot 2: Nonlinear Relationship
plot(X2, Y2, main = "Nonlinear Relationship", pch = 16)
curve(x^2, add = TRUE, col = "red", lwd = 2)

# Plot 3: Independent Variables
plot(X3, Y3, main = "Independent Variables", pch = 16)
abline(h = mean(Y3), col = "red", lwd = 2)
```




```r
# Compute covariances
cov_linear <- cov(X1, Y1)
cov_nonlinear <- cov(X2, Y2)
cov_independent <- cov(X3, Y3)

# Print the results
knitr::kable(
  data.frame(
    Relationship = c("Linear", "Nonlinear", "Independent"),
    Covariance = c(cov_linear, cov_nonlinear, cov_independent)
  ), digits = 3)
```

| Relationship | Covariance |
|---|---|
| Linear | 1.645 |
| Nonlinear | 0.473 |
| Independent | 0.192 |

**Proposition 37.1.** *Covariance has the following properties:*

- $Cov(X, X) = Var(X)$
- $Cov(X, Y) = Cov(Y, X)$
- $Cov(cX, Y) = Cov(X, cY) = c\,[Cov(X, Y)]$
- $Cov(X + Y, Z) = Cov(X, Z) + Cov(Y, Z)$
- $Var(X + Y) = Var(X) + Var(Y) + 2Cov(X, Y)$

*Proof.* We only prove the variance-covariance property:

$$\begin{aligned} Var(X+Y) &= E[(X+Y-\mu_X-\mu_Y)^2] \\ &= E[(X-\mu_X)^2+(Y-\mu_Y)^2+2(X-\mu_X)(Y-\mu_Y)] \\ &= Var(X)+Var(Y)+2Cov(X,Y). \end{aligned}$$

□

**Theorem 37.3.** *For random variables $X_1, X_2, \dots, X_n$, it holds that*

$$Var\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} Var(X_i) + 2\sum_{i<j} Cov(X_i, X_j).$$

*If $X_1, X_2, \dots, X_n$ are identically distributed and have the same covariance relationships (symmetric), then*

$$Var\left(\sum_{i=1}^{n} X_i\right) = nVar(X_1) + 2\binom{n}{2} Cov(X_1, X_2).$$

While $\mathrm{Cov}(X,Y)$ quantifies how $X$ and $Y$ vary together, its magnitude also depends on the absolute scales of $X$ and $Y$ (multiply $X$ by a constant $c$, the covariance will be different). To establish a measure of association between $X$ and $Y$ that is unaffected by arbitrary changes in the scales of either variable, we introduce a "standardized covariance" called correlation.

**Definition 37.2.** The **correlation** between random variables $X$ and $Y$ is defined as

$$Corr(X,Y) = \frac{Cov(X,Y)}{\sqrt{Var(X)Var(Y)}}.$$

We also denote correlation by $\rho \equiv Corr(X,Y)$.

Unlike covariance, scaling $X$ or $Y$ has no effect on the correlation. We can verify this:

$$Corr(cX,Y) = \frac{Cov(cX,Y)}{\sqrt{Var(cX)Var(Y)}} = \frac{cCov(X,Y)}{c\sqrt{Var(X)Var(Y)}} = Corr(X,Y).$$

**Theorem 37.4.** *For any random variable $X$ and $Y$,*

$$-1 \le Corr(X,Y) \le 1.$$

*Proof.* Without loss of generality, assume $X, Y$ both have variance 1, since scaling does not change the correlation. Let $\rho = Corr(X,Y) = Cov(X,Y)$. Then

$$\begin{aligned} Var(X+Y) &= Var(X)+Var(Y)+2Cov(X,Y) = 2+2\rho \ge 0, \\ Var(X-Y) &= Var(X)+Var(Y)-2Cov(X,Y) = 2-2\rho \ge 0. \end{aligned}$$

Thus $-1 \le \rho \le 1$. □

- $X$ and $Y$ are **positively correlated** if $\rho_{XY} > 0$;
- $X$ and $Y$ are **negatively correlated** if $\rho_{XY} < 0$;
- $X$ and $Y$ are **uncorrelated** if $\rho_{XY} = 0$.

**Theorem 37.5.** *Suppose that $X$ is a random variable and $Y = aX + b$ for some constants $a, b$, where $a \neq 0$. If $a > 0$, then $\rho_{XY} = 1$. If $a < 0$, then $\rho_{XY} = -1$.*

*Proof.* If $Y = aX + b$, then $E(Y) = aE(X) + b$. Thus, $Y - E(Y) = a(X - E(X))$. Therefore,

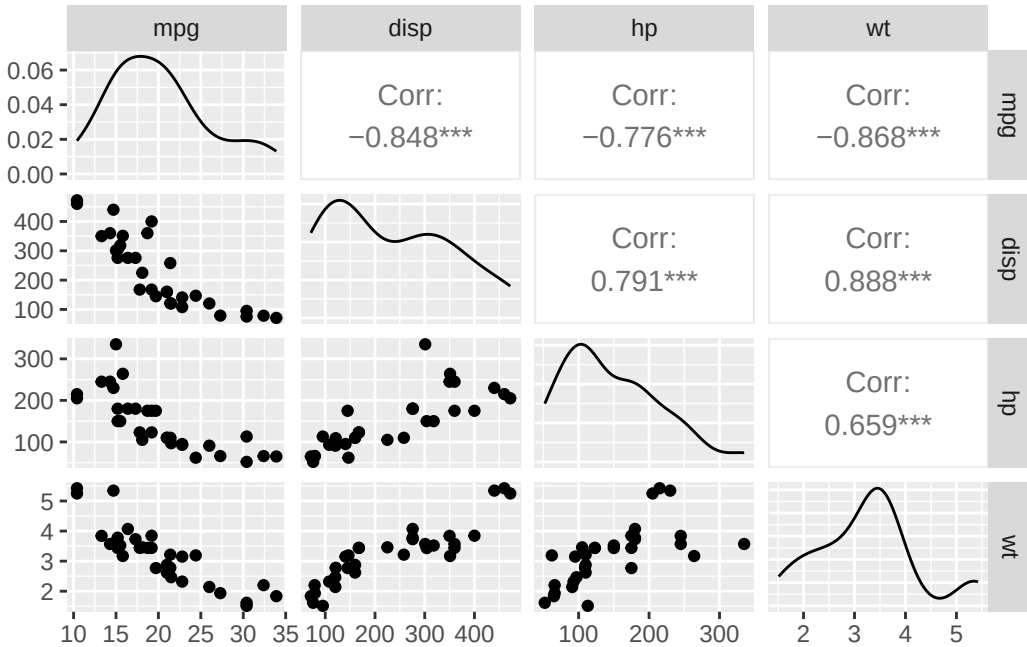
$$Cov(X, Y) = aE[(X - EX)^2] = aVar(X).$$

Since $Var(Y) = a^2 Var(X)$, $\rho_{XY} = \frac{a}{|a|}$. The theorem thus follows. □

**Correlation analysis**

Correlation analysis is a very commonly used method in statistics to measure the strength of the linear relationship between two variables and compute their association. Simply put, correlation analysis tells us how one variable changes with other variables.

```r
# Select variables for covariance analysis
variables <- mtcars[, c("mpg", "disp", "hp", "wt")]

# Compute the covariance matrix
cov_matrix <- cov(variables)

# Compute the correlation matrix
cor_matrix <- cor(variables)

# Plot the correlation pairs
GGally::ggpairs(variables)
```




**Example 37.1.** Let $X \sim \text{HGeom}(w, b, n)$. Find $Var(X)$.

*Solution.* Interpret $X$ as the number of white balls in a sample of size $n$ from an box with $w$ white and $b$ black balls. We can represent $X$ as the sum of indicator variables, $X = I_1 + \cdots + I_n$ , where $I_j$ is the indicator of the $j$-th ball in the sample being white. Each $I_j$ has mean $p = w/(w+b)$ and variance $p(1-p)$, but because the $I_j$ are dependent, we cannot simply add their variances. Instead,

$$\begin{aligned} Var(X) &= Var\left(\sum_{j=1}^{n} I_j\right) \\ &= Var(I_1) + \cdots + Var(I_n) + 2\sum_{i<j} Cov(I_i, I_j) \\ &= np(1-p) + 2\binom{n}{2} Cov(I_i, I_j) \end{aligned}$$

In the last step, because of symmetry, for every pair $i$ and $j$, $Cov(I_i, I_j)$ are the same.

$$\begin{aligned}
Cov(I_i, I_j) &= E(I_i I_j) - E(I_i)E(I_j) \\
&= P(i \text{ and } j \text{ both white}) - P(i \text{ is white})P(j \text{ is white}) \\
&= \frac{w}{w+b} \cdot \frac{w-1}{w+b-1} - p^2 \\
&= p\frac{Np-1}{N-1} - p^2 \\
&= \frac{p(p-1)}{N-1}
\end{aligned}$$

where $N = w + b$. Plugging this into the above formula and simplifying, we eventually obtain

$$Var(X) = np(1-p) + n(n-1)\frac{p(p-1)}{N-1} = \frac{N-n}{N-1}np(1-p).$$

This differs from the Binomial variance of $np(1-p)$ by a factor of $\frac{N-n}{N-1}$. This discrepancy arises because the Hypergeometric story involves sampling without replacement. As $N \to \infty$, it becomes extremely unlikely that we would draw the same ball more than once, so sampling with or without replacement essentially become the same.

**Example 37.2.** (PG exam). Put $k$ balls into $n$ boxes. Let $X$ be the number of empty boxes. Find $E(X)$ and $Var(X)$.

*Solution.* Define an indicator variable

$$I_j = \begin{cases} 1 & j\text{-th box is empty} \\ 0 & \text{otherwise} \end{cases}$$

Then $X = \sum_{j=1}^{n} I_j$. Unconditionally, the probability of one box being empty is $\left(\frac{n-1}{n}\right)^k$. Therefore,

$$E(I_j) = P(j\text{-th box is empty}) = \left(\frac{n-1}{n}\right)^k$$

for $j = 1, 2, \ldots, n$. It follows that

$$E(X) = \sum_{j=1}^{n} I_j = nE(I_j) = n\left(\frac{n-1}{n}\right)^k.$$

To compute the variance,

$$\begin{aligned}
Var(X) = Var(I_1 + \cdots + I_n) &= \sum_{j=1}^{n} Var(I_j) + 2\sum_{i<j} Cov(I_i, I_j) \\
&= nVar(I_j) + 2\binom{n}{2}Cov(I_i, I_j),
\end{aligned}$$

since by symmetry, $Var(I_j)$ is the same for all $j$ and $Cov(I_i, I_j)$ is the same for all $i \neq j$. It suffices to compute $Var(I_j)$ and $Cov(I_i, I_j)$ for any $j$ and $i \neq j$. Since $I_j$ only takes number 0 and 1,

$$E(I_j^2) = \left(\frac{n-1}{n}\right)^k,$$

$$Var(I_j) = E(I_j^2) - (E(I_j))^2 = \left(\frac{n-1}{n}\right)^k - \left(\frac{n-1}{n}\right)^{2k}.$$

For the covariance term,

$$E(I_i I_j) = P(i, j \text{ are both empty}) = \left(\frac{n-2}{n}\right)^k,$$

$$Cov(I_i, I_j) = E(I_i I_j) - E(I_i)E(I_j) = \left(\frac{n-2}{n}\right)^k - \left(\frac{n-1}{n}\right)^{2k}.$$

Therefore,

$$Var(X) = n\left[\left(\frac{n-1}{n}\right)^k - \left(\frac{n-1}{n}\right)^{2k}\right] + 2\binom{n}{2}\left[\left(\frac{n-2}{n}\right)^k - \left(\frac{n-1}{n}\right)^{2k}\right].$$

# 38 Portfolio allocation*

In the world of finance, one of the most well-established principles is the idea of **diversification**. By combining assets with varying levels of risk and return, investors can reduce the overall risk of their portfolio. One of the most powerful tools for achieving this is **correlation**, a statistical measure that quantifies how two assets move in relation to each other. Understanding and applying the concept of correlation can be the key to constructing portfolios that not only aim for higher returns but also minimize risk. This section demonstrates how a lower correlation between assets can significantly reduce portfolio risk while maintaining the same expected return.

**The Role of Correlation in Portfolio Risk**

A core principle in portfolio management is that the overall portfolio risk (or volatility) is not simply the weighted average of the individual asset risks. Instead, it also depends on how the assets in the portfolio are correlated with each other. When assets have lower correlation, the overall risk of the portfolio is reduced. This occurs because the assets will not all move in the same direction at the same time, thus smoothing out large fluctuations in the portfolio value.

To quantify this relationship, we use a formula that calculates the portfolio risk:

$$\sigma_P = \sqrt{w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_A w_B \sigma_A \sigma_B \rho_{AB}}$$

Where: $\sigma_P$ is the overall portfolio risk. $w_A$ and $w_B$ are the weights of the assets in the portfolio. $\sigma_A$ and $\sigma_B$ are the individual standard deviations (risks) of assets A and B. $\rho_{AB}$ is the correlation between the two assets.

Consider a portfolio of two assets, Asset A and Asset B. Both assets have the same expected return of 10% and individual risks (standard deviations) of 15%. The key difference between the two cases is the correlation between the assets.

Case 1: High Correlation

Let's assume that the correlation between the two assets is 0.8, indicating that the two assets tend to move in the same direction. In this scenario, we will calculate the portfolio risk when we allocate 50% of the portfolio to Asset A and 50% to Asset B.

$$\sigma_P = \sqrt{(0.5)^2(0.15)^2 + (0.5)^2(0.15)^2 + 2(0.5)(0.5)(0.15)(0.15)(0.8)} \approx 20.12\%$$

Case 2: Low Correlation

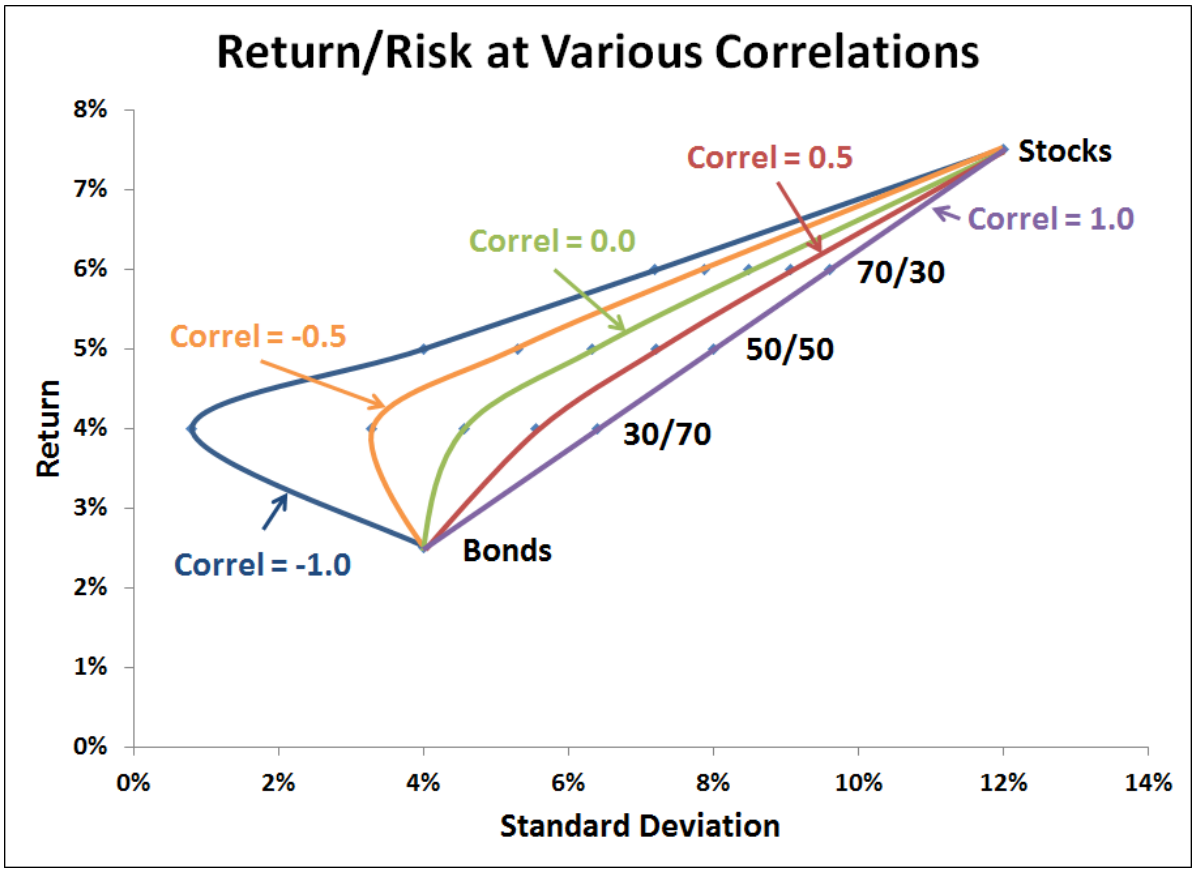
Now, let's assume that the correlation between the two assets is 0.2, indicating a much weaker relationship between the two assets. Again, we allocate 50% to Asset A and 50% to Asset B. Using the same formula, we can compute the portfolio risk:

$$\sigma_P = \sqrt{(0.5)^2(0.15)^2 + (0.5)^2(0.15)^2 + 2(0.5)(0.5)(0.15)(0.15)(0.2)} \approx 16.1\%$$

In this example, by reducing the correlation between the two assets, we reduced the portfolio risk by approximately 4%, even though the expected return remained the same (since the return of both assets was identical).

**The principle of diversification**

Diversification is often referred to as a "free lunch" in finance because it allows investors to reduce portfolio risk without sacrificing expected returns. The reason lower correlation reduces risk is that when assets are highly correlated, they tend to move together, amplifying the portfolio's overall volatility. In contrast, when assets are less correlated, they do not move in lockstep, and the portfolio experiences less overall volatility because the negative fluctuations of one asset can be offset by the positive fluctuations of the other. This is a key advantage of diversification: it allows you to spread risk across assets that behave differently.


Return/Risk at Various Correlations
8%
7%
6%
5%
4%
3%
2%
1%
0%
Return
0%
2%
4%
6%
8%
10%
12%
14%
Standard Deviation
Stocks
Correl = 0.5
Correl = 1.0
Correl = 0.0
70/30
Correl = -0.5
50/50
30/70
Bonds
Correl = -1.0

# 39 Conditional expectation

We have introduced conditional expectation in Definition 25.2. Here we reiterate the definition with continuous random variables.

**Definition 39.1** (Conditional expectation)**.** Let $X$ and $Y$ be continuous random variables with joint density $f_{X,Y}(x,y)$, $X$'s density $f_X(x)$, and conditional density $f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)}$. The conditional expectation of $Y$ given $X = x$ is

$$\begin{aligned} E(Y|X=x) &= \int_{-\infty}^{\infty} y \; f_{Y|X}(y|x)dy \\ &= \int_{-\infty}^{\infty} y \; \frac{f_{X,Y}(x,y)}{f_X(x)} dy \end{aligned}$$

When the denominator is zero, the expression is undefined.

Note that conditioning on a continuous random variable is not the same as conditioning on the event $\{X = x\}$ as it was in the discrete case. The probability of the event is zero, but we define the conditional expectation in terms of the density function.

**Theorem 39.1** (Law of iterated expectation)**.** *For any random variable $X$ and $Y$, it holds that*

$$E(E(Y|X)) = E(Y).$$

*Proof.* Note that $E(Y|X) = g(X)$ is a function of $X$. Apply LOTUS:

$$\begin{aligned} E(E(Y|X)) &= \int g(x)f(x)dx \\ &= \int \left( \int yf(y|x)dy \right) f(x)dx \\ &= \int \int yf(y|x)f(x)dydx \\ &= \int y \int f(y,x)dx\, dy \\ &= \int_{-\infty}^{\infty} yf(y)dy \\ &= E(Y). \end{aligned}$$

□

**Theorem 39.2.** *For any random variable $X$ and $Y$, and any function $g$, we have*

$$E(g(X)Y|X) = g(X)E(Y|X).$$

*Proof.* For any specific value of $X = x$, $g(x)$ is a constant. Thus, $E(g(x)Y|X = x) = g(x)E(Y|X = x)$. This is true for all values of $x$. □

**Theorem 39.3** (Best predictor)**.** *Conditional expectation $E(Y|X)$ is the best predictor for $Y$ using $X$ (minimized the square loss function).*

*Proof.* Let $g(X)$ be a predictor for $Y$ using $X$. We want to find the $g$ such that minimizes $E(Y - g(X))^2$.

$$\begin{aligned}
E(Y - g(X))^2 &= E(Y - E(Y|X) + E(Y|X) - g(X))^2 \\
&= E(Y - E(Y|X))^2 + 2\underbrace{E(Y - E(Y|X)}_{E(Y)=E(E(Y|X))}((E(Y|X) - g(X)) \\
&\quad + E(E(Y|X) - g(X))^2 \\
&= E(Y - E(Y|X))^2 + E(E(Y|X) - g(X))^2 \\
&\geq E(Y - E(Y|X))^2.
\end{aligned}$$

Therefore, $E(Y - g(X))^2$ is minimized when $g(X) = E(Y|X)$. □

**Definition 39.2** (Linear conditional expectation model)**.** An extremely widely used method for data analysis in statistics is linear regression. In its most basic form, we want to predict the mean of $Y$ using a single explanatory variable $X$. A linear conditional expectation model assumes that $E(Y|X)$ is linear in $X$:

$$E(Y|X) = a + bX,$$

or equivalently,

$$Y = a + bX + \epsilon,$$

with $E(\epsilon|X) = 0$. The intercept and the slope is given by

$$b = \frac{Cov(X, Y)}{Var(X)}, a = E(Y) - bE(X).$$

We first show the equivalence of the two expressions of the model. Let $Y = a + bX + \epsilon$, with $E(\epsilon|X) = 0$. Then by linearity,

$$E(Y|X) = E(a|X) + E(bX|X) + E(\epsilon|X) = a + bX.$$

Conversely, suppose that $E(Y|X) = a + bX$, and define

$$\epsilon = Y - (a + bX).$$

Then $Y = a + bX + \epsilon$, with

$$E(\epsilon|X) = E(Y|X) - E(a + bX|X) = E(Y|X) - (a + bX) = 0.$$

To derive the expression for $a$ and $b$, take covariance between $X$ and $Y$,

$$\begin{aligned} Cov(X, Y) &= Cov(X, a + bX + \epsilon) \\ &= Cov(X, a) + bCov(X, X) + Cov(X, \epsilon) \\ &= bVar(X) + Cov(X, \epsilon) \end{aligned}$$

Note that $Cov(X, \epsilon) = 0$ because

$$\begin{aligned} Cov(X, \epsilon) &= E(X\epsilon) - E(X)E(\epsilon) \\ &= E(E(X\epsilon|X)) - E(X)E(E(\epsilon|X)) \\ &= E(XE(\epsilon|X)) - E(X)E(E(\epsilon|X)) \\ &= 0 \end{aligned}$$

Therefore,

$$Cov(X, Y) = bVar(X)$$

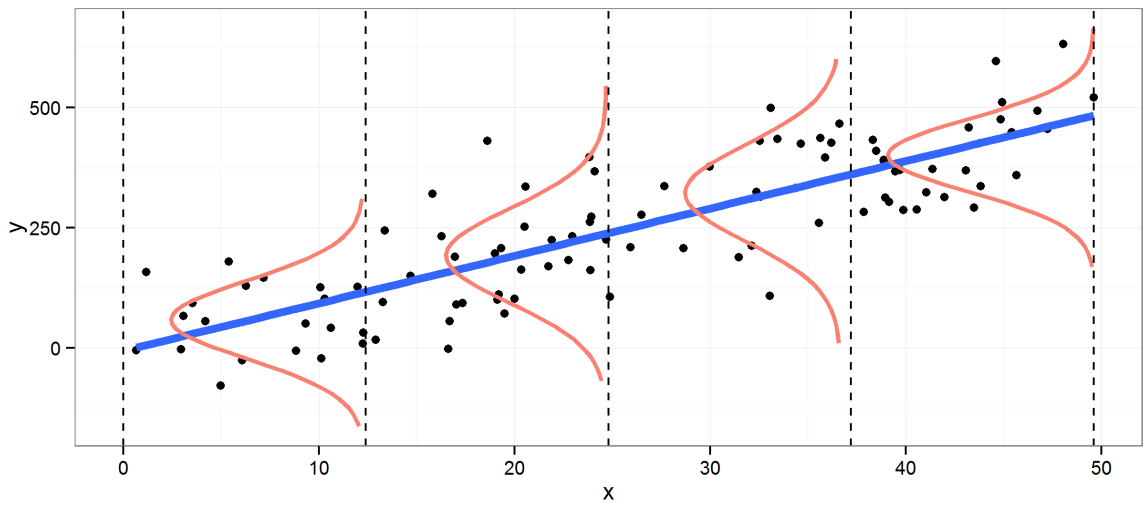
Thus,

$$\begin{aligned} b &= \frac{Cov(X, Y)}{Var(X)}, \\ a &= E(Y) - bE(X) = E(Y) - \frac{Cov(X, Y)}{Var(X)}E(X). \end{aligned}$$

In practice, we don't know the true value of $Cov(X, Y)$ or $Var(X)$. We have to estimate it with sample observations. Thus, we compute $\hat{b} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$. By definition, $b$ gives the marginal change of $E(Y|X)$ with respect to $X$.

| Characteristic | Beta | 95% CI | p-value |
|---|---|---|---|
| hp | -0.07 | -0.09, -0.05 | <0.001 |

Abbreviation: CI = Confidence Interval

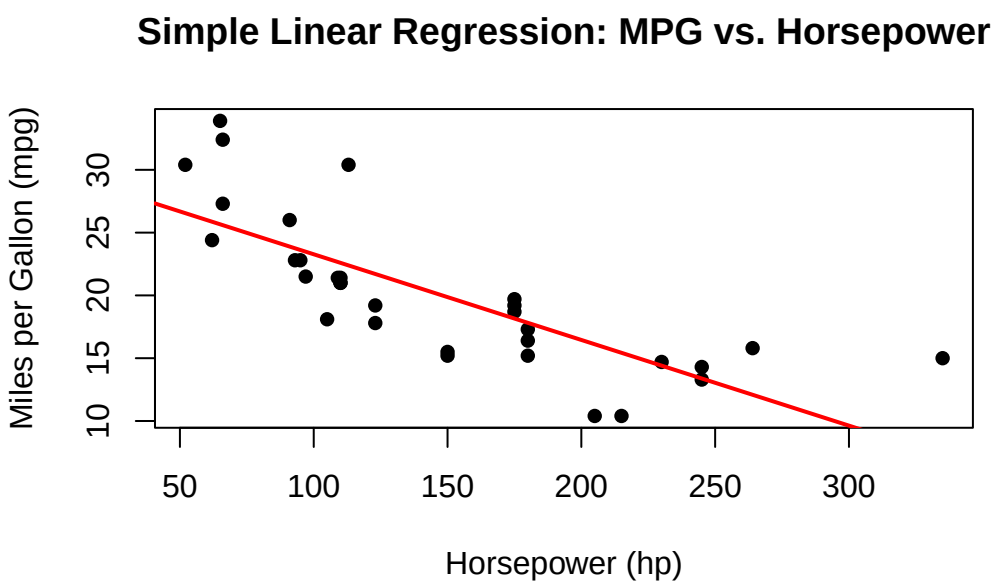



Linear regression is the simple yet powerful modeling tool in statistics. It is useful whenever we want to predict one variable with another. When the assumptions are met (though this is rare), the model gives the conditional expectation — the best predictor. If the assumptions are not met, regression does not give the

```
# Predict the miles/gallon of a car using its horsepower
# Use the built-in dataset `mtcars`
data(mtcars)

# Fit the linear regression model
model <- lm(mpg ~ hp, data = mtcars)

# Display the model summary
gtsummary::tbl_regression(model)
```

```
# Scatter plot of the data
plot(mtcars$hp, mtcars$mpg,
     main = "Simple Linear Regression: MPG vs. Horsepower",
     xlab = "Horsepower (hp)",
```

```r
     ylab = "Miles per Gallon (mpg)",
     pch = 16)  # Use solid circles for points

# Add the regression line
abline(model, col = "red", lwd = 2)
```



# 40 Moments and MGF

**Definition 40.1.** Let $X$ be a random variable with mean $\mu$ and variance $\sigma^2$ . For any positive integer $n$, the $n$-th **moment** of $X$ is $E(X^n)$, the $n$-th **central moment** is $E(X-\mu)^n$, and the $n$-th **standardized moment** is $E\left(\frac{X-\mu}{\sigma}\right)^n$.

In accordance with this terminology, $E(X)$ is the first moment of $X$, $Var(X)$ is the second central moment of $X$. It is natural to ask if there are higher order moments. The answer is yes.

**Definition 40.2.** Let $X$ be a random variable with mean $\mu$, standard deviation $\sigma$, and finite third moment. The **skewness** of $X$ is defined as

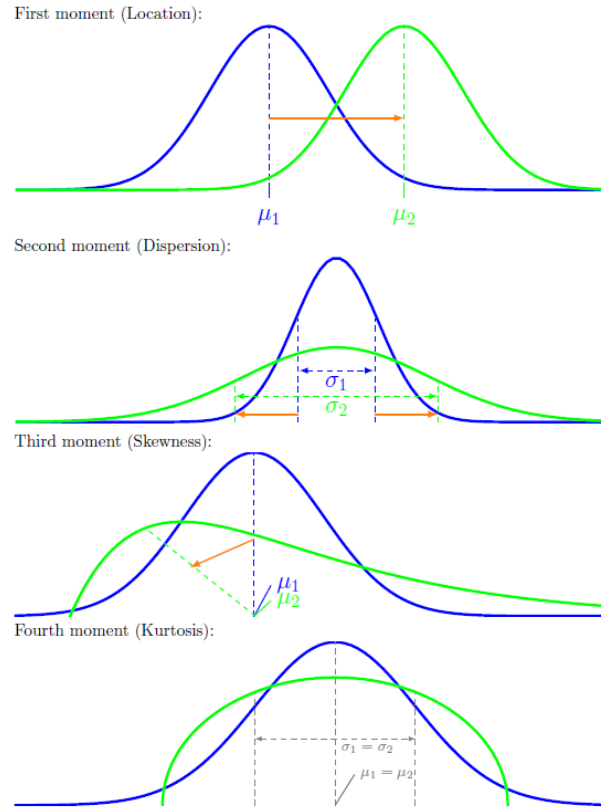
$$\text{Skew}(X) = E\left[\left(\frac{X-\mu}{\sigma}\right)^3\right].$$

**Definition 40.3.** The **Kurtosis** of $X$ is defined as

$$\text{Kurt}(X) = \left[\left(\frac{X-\mu}{\sigma}\right)^4\right].$$

Skewness is the measure of the lopsidedness of the distribution; any symmetric distribution will have a third central moment, if defined, of zero. A distribution that is skewed to the left (the tail of the distribution is longer on the left) will have a negative skewness. A distribution that is skewed to the right (the tail of the distribution is longer on the right), will have a positive skewness.

Kurtosis is a measure of the heaviness of the tail of the distribution. If a distribution has heavy tails, the kurtosis will be high; conversely, light-tailed distributions have low kurtosis.




We see that moments give information about the shape of a distribution. Different orders of moments captures different aspects of the distribution. As higher and higher moments are calculated, they reveal more and more aspects of the distribution. Loosely speaking, it is somewhat like the Taylor theorem in the probability theory. We can approximate a distribution by "expectation of polynomials": $E(X), E(X^2), E(X^3), ...$

**Definition 40.4.** Let $X$ be a random variable. For each real number $t$, define the **moment generating function** (MGF) as

$$M_X(t) = E\left(e^{tX}\right).$$

To see why it is "generating" moments, take the Taylor expansion of the exponential function:

$$e^{tX} = 1 + tX + \frac{t^2X^2}{2!} + \frac{t^3X^3}{3!} + \cdots$$

Hence,

$$M_X(t) = E\left(e^{tX}\right) = 1 + E(X)t + E(X^2)\frac{t^2}{2!} + \cdots$$

A natural question at this point is: What is the interpretation of $t$? The answer is that $t$ has no interpretation in particular; it's just a bookkeeping device that we introduce in order to *encode* the sequence of moments in a differentiable function.

**Theorem 40.1.** *Let $M_X(t)$ be the MGF of $X$. Then the $n$-th moment of $X$ is given by $E(X^n) = M_X^{(n)}(0)$, where $M_X^{(n)}$ denotes the $n$-th derivative of the MGF.*

**Theorem 40.2.** *The MGF (if it exists) uniquely determines the distribution. That is, if two random variables have the same MGF, then they must have the same distribution.*

**Theorem 40.3.** *If $X$ and $Y$ are independent, then the MGF of $X+Y$ is the product of the individual MGFs:*

$$M_{X+Y}(t) = M_X(t)M_Y(t).$$

**Example 40.1.** For $X \sim Bern(p)$, $e^{tX}$ takes on the value $e^t$ with probability $p$ and the value 1 with probability $q$, so $M(t) = E\left(e^{tX}\right) = pe^t + q$. Since this is finite for all values of $t$, the MGF is defined on the entire real line.

**Example 40.2.** The MGF of a $Bin(n,p)$ random variable is $M(t) = (pe^t + q)^n$, since it is the product of $n$ independent Bernoulli MGFs.

# 41 Inequalities

This section introduces some of the most popular inequality in statistics and general mathematics. Interestingly, our probability theories can shed light on these inequalities that are otherwise hard to explain. We don't show formal proofs here, but just point out how these inequalities can be useful in statistics.

**Theorem 41.1.** *(Cauchy-Schwarz inequality).*

$$\left|\sum x_i y_i\right| \leq \sqrt{\sum x_i^2}\sqrt{\sum y_i^2}$$

*Proof.* If $X, Y$ have zero means, their correlation can be written as

$$\rho_{XY} = \frac{E(XY)}{\sqrt{E(X^2)E(Y^2)}}$$

Since $|\rho_{XY}| \leq 1$, we always have

$$|E(XY)| \leq \sqrt{E(X^2)E(Y^2)}.$$

Consider $\{x_i\}$ and $\{y_i\}$ as realizations of $X$ and $Y$ with equal probabilities, such that $E(X) = \frac{1}{n}\sum x_i$. The original inequality is thus proved. □

**Theorem 41.2.** *(Jensen's inequality). For a convex function $f$, we have*

$$\frac{1}{n}\sum f(x_i) \geq f\left(\frac{1}{n}\sum x_i\right);$$

*If $f$ is concave, then*

$$\frac{1}{n}\sum f(x_i) \leq f\left(\frac{1}{n}\sum x_i\right).$$

We do not intend to prove it, but offer a special case in statistics that helps to understand Jensen's inequality. Consider

$$Var(X) = E(X^2) - (E(X))^2 \geq 0$$

We have

$$E(X^2) \geq (E(X))^2.$$

Note that $f(X) = X^2$ is a convex function, and $E(*) = \frac{1}{n}\sum *$, we have shown the first inequality. The concave case is the opposite.

In general, if $g$ is a convex function, then $E(g(X)) \geq g(E(X))$. If $g$ is a concave function, then $E(g(X)) \leq g(E(X))$. In both cases, the only way that equality can hold is if there are constants $a$ and $b$ such that $g(X) = a + bX$ with probability 1.

**Theorem 41.3.** *(Markov inequality). Let $X$ be a random variable, then*

$$P(|X| \geq a) \leq \frac{E|X|}{a}$$

*That is, the probability of $|X|$ deviating from its mean by a multiple of $a$ must be less than $1/a$.*

*Proof.* Define a random variable

$$I_{|X|\geq a} = \begin{cases} 1 & \text{if } |X| \geq a \\ 0 & \text{if } |X| < a \end{cases}$$

Note that $P(|X| \geq a) = E(I_{|X|\geq a})$. It always holds that

$$a \cdot I_{|X|\geq a} \leq |X|$$

Therefore,

$$E\left[a \cdot I_{|X|\geq a}\right] \leq E|X|$$

Hence,

$$P(|X| \geq a) \leq \frac{E|X|}{a}.$$

□

For an intuitive interpretation, let $X$ be the income of a randomly selected individual from a population. Taking $a = 2E(X)$, Markov's inequality says that $P(X \geq 2E(X)) \leq 1/2$, i.e., it is impossible for more than half the population to make at least twice the average income. This is clearly true, since if over half the population were earning at least twice the average income, the average income would be higher. Similarly, $P(X \geq 3E(X)) \leq 1/3$: you can't have more than 1/3 of the population making at least three times the average income, since those people would already drive the average above what it is.

**Theorem 41.4.** *(Chebyshev inequality). Let $X$ be a random variable with mean $\mu$ and standard deviation $\sigma$, then*

$$P\left(|X - \mu| > c\sigma\right) \leq \frac{1}{c^2}$$

*That is, the probability of $X$ deviating from its mean by $a$ times the standard deviation must be less than $1/a^2$.*

*Proof.* We first show

$$P(|X - \mu| > a) \leq \frac{\sigma^2}{a^2}$$

This is true by taking squares and applying the Markov inequality,

$$P(|X - \mu| > a) = P((X - \mu)^2 > a^2) \leq \frac{E(X - \mu)^2}{a^2} = \frac{\sigma^2}{a^2}.$$

Substitute $c\sigma$ for $a$, we have the original inequality. □

This gives us an upper bound on the probability of a random variable being more than $c$ standard deviations away from its mean, e.g., there can't be more than a 25% chance of being 2 or more standard deviations from the mean. Given the mean and standard deviation of a random variable $X$, we know that $\mu \pm 2\sigma$ captures 75% of its possible values; $\mu \pm 3\sigma$ captures 90% of the possible values.

# Part V

# Continuous Distributions

# 42 Continuous vs Discrete

Continuous random variables, in many ways, are more versatile and useful than discrete distributions. One key reason is that many quantities in the physical world, such as temperature, height, weight, and time, are inherently continuous in nature. These variables can take on any value within a range, providing a more accurate representation of real-world phenomena compared to discrete variables, which are limited to distinct values. Additionally, the probability density functions (PDFs) of continuous distributions are often defined by smooth, differentiable functions. This mathematical structure allows us to apply calculus for analysis, enabling precise calculations of probabilities, expected values, and other statistical measures. The ability to integrate and differentiate these functions not only simplifies manipulation but also makes continuous distributions a powerful tool for solving complex problems in physics, engineering, and data analysis.

**Definition 42.1.** A random variable has a continuous distribution if its CDF is *differentiable*. A continuous random variable is a random variable with a continuous distribution.

**Definition 42.2.** For a continuous random variable $X$ with CDF $F$, the probability density function (PDF) of $X$ is the derivative of the CDF, given by $f(x) = F'(x)$. The support of $X$ is the set of all $x$ where $f(x) > 0$.

*Remark.* By the fundamental theorem of calculus, we integrate a PDF to get the CDF:

$$F(x) = \int_{-\infty}^{x} f(t)dt.$$

PDF differs from the discrete PMF in important ways:

- For a continuous random variable, $P(X = x) = 0$ for all $x$;
- The quantity $f(x)$ is not a probability. To get the probability, we integrate the PDF (probability is the area under the PDF):

$$P(a < X \leq b) = F(b) - F(a) = \int_{a}^{b} f(x)dx.$$

- Since any single value has probability 0, including or excluding endpoints does not matter.

$$P(a < X < b) = P(a < X \leq b) = P(a \leq X < b) = P(a \leq X \leq b).$$

**Theorem 42.1.** *The PDF $f$ of a continuous random variable must satisfy the following criteria:*

- *Nonnegative:* $f(x) \geq 0$*;*
- *Integrates to 1:* $\int_{-\infty}^{\infty} f(x)dx = 1$.

**Definition 42.3.** The expectation of a continuous random variable $X$ with PDF $f$ is

$$E(X) = \int_{-\infty}^{\infty} xf(x)dx.$$

**Theorem 42.2.** *If $X$ is a continuous random variable with PDF $f$ and $g : \mathbb{R} \to \mathbb{R}$. The LOTUS applies*

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x)dx.$$

| | Discrete | Continuous |
|---|---|---|
| PMF/PDF | $P(X = x) = p(x)$ | $P(a \leq X \leq b) = \int_a^b f(x)dx$ |
| CDF | $F(x) = P(X \leq x) = \sum_{k \leq x} p(k)$ | $F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)dt$ |
| Expectation | $E(x) = \sum_x xP(X = x)$ | $E(X) = \int_{-\infty}^{+\infty} xf(x)dx$ |
| LOTUS | $E[g(x)] = \sum_x g(x)P(X = x)$ | $E[g(x)] = \int_{-\infty}^{+\infty} g(x)f(x)dx$ |

# 43 Uniform distribution

**Definition 43.1.** Let $a$ and $b$ be two given real numbers such that $a < b$. Let $X$ be a random variable such that it is known that $a \leq X \leq b$ and, for every subinterval of $[a, b]$, the probability that $X$ will belong to that subinterval is proportional to the length of that subinterval. We then say that the random variable $X$ has the Uniform distribution on the interval $[a, b]$. The PDF of $X$ is

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

This is a valid PDF since

$$\int_{-\infty}^{+\infty} f(x)dx = \int_a^b \frac{1}{b-a}dx = \frac{1}{b-a}\int_a^b dx = 1.$$

The CDF of $X$ is

$$F(x) = \int_{-\infty}^{x} f(t)dt = \int_a^x f(t)dt = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & a \leq x \leq b \\ 1 & x > b \end{cases}.$$

The expectation of $X$:

$$E(X) = \int_a^b x\frac{1}{b-a}dx = \frac{1}{b-a}\left[\frac{x^2}{2}\right]_a^b = \frac{a+b}{2}.$$

To figure out the variance, first compute

$$E(X^2) = \int_a^b x^2\frac{1}{b-a}dx = \frac{1}{b-a}\left[\frac{x^3}{3}\right]_a^b = \frac{a^2+ab+b^2}{3}$$

Thus,

$$Var(X) = E(X^2) - E^2(X) = \frac{a^2+ab+b^2}{3} - \frac{(a+b)^2}{4} = \frac{(b-a)^2}{12}.$$

**Example 43.1.** A stick of unit length is broken at a random point X. What is the expected length of the longer piece?

*Solution.* The lengths of the two pieces are $X$ and $1 - X$, with $X \sim Unif(0, 1)$. The longer piece is $\max(X, 1 - X)$. For $X < 0.5$, the longer piece is $1 - X$, and for $X \geq 0.5$, it is $X$. The expected value is:

$$E[\max(X, 1 - X)] = \int_0^{0.5} (1 - X)\, dx + \int_{0.5}^1 X\, dx = \frac{3}{4}.$$

Intuition might suggest that since the stick is broken at a random point, the longer piece should be "somewhat larger" than the shorter piece, but not as large as 3/4. However, the uniform distribution of the break point means that the longer piece can sometimes be much larger than the shorter piece, especially when the break point is close to one end.

```r
# Set seed for reproducibility
set.seed(123)

# Number of simulations
n_simulations <- 1000

# Simulate breaking the stick
X <- runif(n_simulations, min = 0, max = 1)  # Random break points
longer_piece <- pmax(X, 1 - X)  # Length of the longer piece

# Compute the expected length
expected_length <- mean(longer_piece)

# Print the result
cat("Simulated Expected Length of Longer Piece:", expected_length)
```

```
Simulated Expected Length of Longer Piece: 0.7488728
```

# 44 Special integrals

There are many reasons to learn integrals. But the most compelling reason is that math is no longer the same with integrals. We can have many amazing results with integrals that were otherwise not imaginable. This section introduces two integrals that are of special importance to continuous distributions.

**Example 44.1.** Show that $\int_{-\infty}^{+\infty} e^{-x^2} dx = \sqrt{\pi}$.

*Proof.* This is known as Gaussian integral, which is the kernel of the PDF of the normal distribution. It also amazingly relates two of the most famous constants in mathematics. It is not integrable by normal integration techniques. But it can be solved by switching to the polar coordinate.

$$\begin{aligned}
\left(\int_{-\infty}^{+\infty} e^{-x^2} dx\right)^2 &= \int_{-\infty}^{+\infty} e^{-x^2} dx \int_{-\infty}^{+\infty} e^{-y^2} dy \\
&= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} e^{-(x^2+y^2)} dx dy \\
&= \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r dr d\theta && dA = dxdy = rdrd\theta \\
&= \int_0^{2\pi} \int_0^{\infty} \frac{1}{2} e^{-u} du d\theta && \text{let } u = r^2 \\
&= \frac{1}{2} \int_0^{2\pi} d\theta = \pi.
\end{aligned}$$

□

**Example 44.2.** Show that $\int_0^{\infty} t^n e^{-t} dt = n!$

*Proof.* $\Gamma(z) = \int_0^{\infty} t^{z-1} e^{-t} dt$ is known as the Gamma function, which is definitely one of the most interesting functions in mathematics. It is the extension of factorials to real numbers or even complex numbers. It also has many interesting properties, such as $\Gamma(n) = (n-1)!$, $\Gamma(1/2) = \sqrt{\pi}$, $\Gamma(3/2) = \sqrt{\pi}/2$, $\Gamma'(1) = -\gamma$ and so on. The $(n-1)$ in the Gamma function is

due to historical reasons and does not matter in our case. We will prove the integral with $n$ instead of $(n-1)$.

There are many ways to prove this. One is to discover the recursive relationship $\Gamma(n+1) = n\Gamma(n)$. But it does not give a clue why we need this integral to approximate the factorial. We start with an elementary integral

$$\int_0^\infty e^{at} dt = -\frac{1}{a}$$

where $a < 0$. Differentiate both sides $n$ times with respect to $a$:

$$\begin{aligned}
\int_0^\infty e^{at} t dt &= -(-1)a^{-2} \\
\int_0^\infty e^{at} t^2 dt &= -(-1)(-2)a^{-3} \\
\int_0^\infty e^{at} t^3 dt &= -(-1)(-2)(-3)a^{-4} \\
&\vdots \\
\int_0^\infty e^{at} t^n dt &= (-1)^{n+1} n! a^{-(n+1)}
\end{aligned}$$

Let $a = -1$, we have

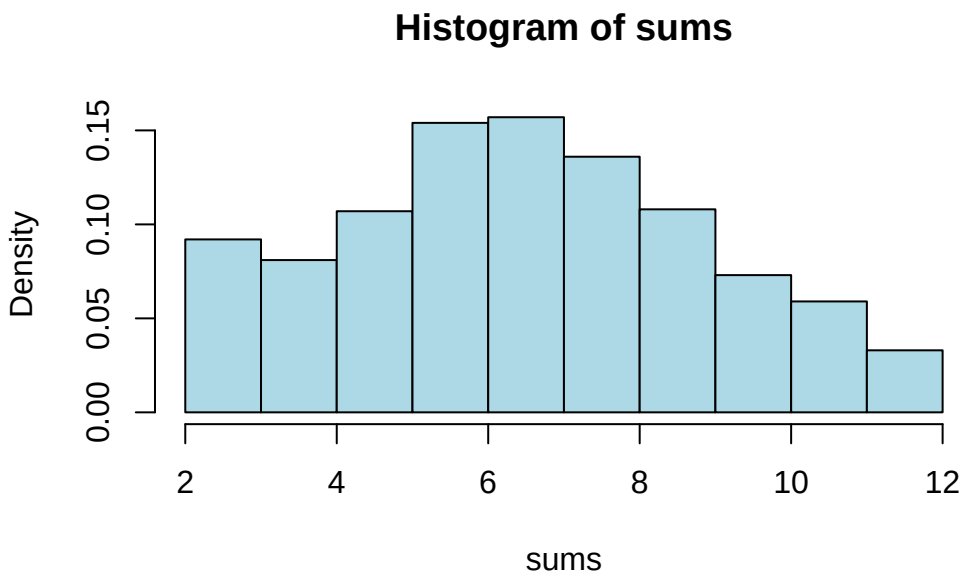
$$\int_0^\infty e^t t^n = n!$$

$\square$

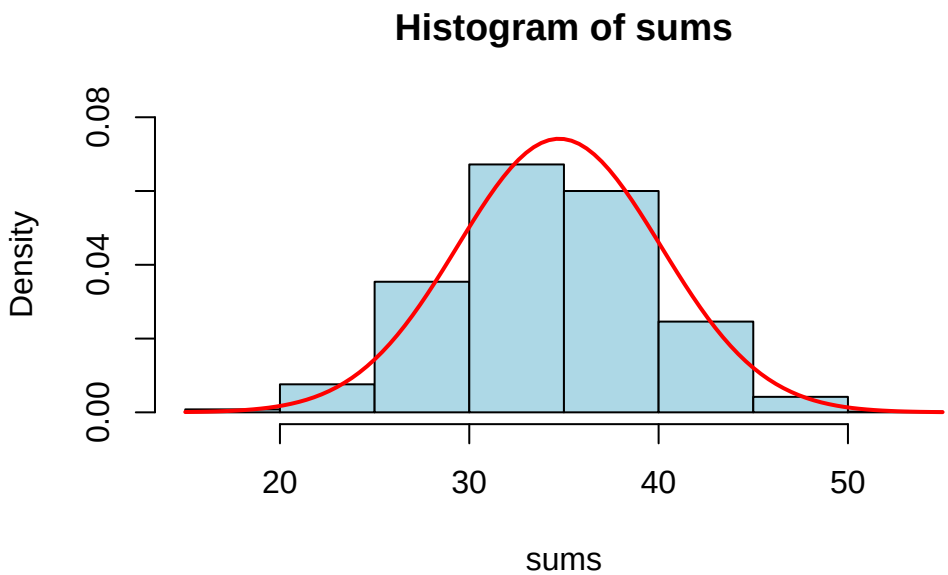
# 45 Sum of random variables

We have seen the sum of coin heads and the sum of dice points follow a "bell-shaped" distribution. This is not a coincidence. The pattern does not exist when the number of coins or dice are small, but becomes apparent when the numbers get large.

```
# Function to simulate the sum of rolling n fair dice
simulate_dice_sum <- function(n, n_simulations) {

  # Simulate rolling n dice n_simulations times
  dice_rolls <- matrix(sample(1:6, n * n_simulations, replace = TRUE),
                       nrow = n_simulations, ncol = n)

  # Compute the sum of each roll
  sums <- rowSums(dice_rolls)

  return(sums)
}

# Set seed for reproducibility
set.seed(123)

# Rolling 2 dice
sums <- simulate_dice_sum(2, 1000)

# Plot the distribution of sums
hist(sums, prob = TRUE, col = "lightblue")
```

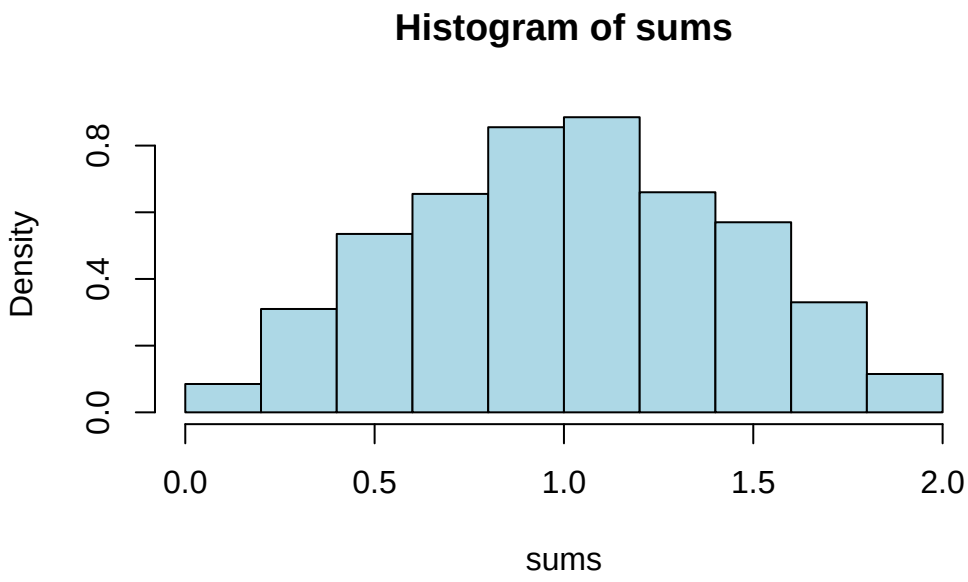



```r
# Rolling 10 dice
sums <- simulate_dice_sum(10, 1000)

# Plot the distribution of sums
hist(sums, prob = TRUE, ylim = c(0,.08), col = "lightblue")

# Overlay the normal curve
curve(dnorm(x, mean = mean(sums), sd = sd(sums)),
      col = "red", lwd = 2, add = TRUE)
```

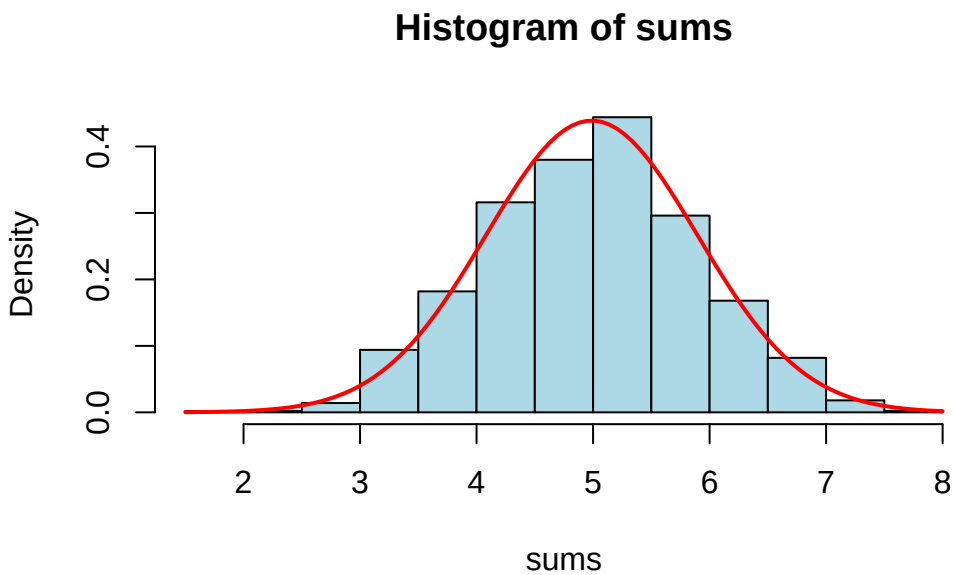



This does not only hold for dice points. In fact, the sum of random variables from any distribution would reveal a similar pattern.

```
# Sum of uniform random variables
simulate_uniform_sum <- function(n, n_simulations) {
  sums <- numeric(n_simulations)

  # Simulate the experiment n_simulations times
  for (i in 1:n_simulations) {
    # Sum of n Uniform random variables
    sums[i] <- sum(runif(n))
  }

  return(sums)
}

# Sum of 2 Uniform random variables
sums <- simulate_uniform_sum(2, 1000)

# Plot the histogram
hist(sums, prob = TRUE, col = "lightblue")
```




```r
# Sum of 10 Uniform random variables
sums <- simulate_uniform_sum(10, 1000)

# Plot the histogram
hist(sums, prob = TRUE, col = "lightblue")

# Overlay the normal curve
curve(dnorm(x, mean = mean(sums), sd = sd(sums)),
      col = "red", lwd = 2, add = TRUE)
```




**Sum of random variables approaches Normal distribution**

Let $X_1, X_2, \dots, X_n$ be a sequence of **i.i.d** random variables with mean $\mu = E(X_i)$ and variance $\sigma^2 = Var(X_i)$. Let

$$S_n = X_1 + X_2 + \cdots + X_n$$

Then, as $n \to \infty$, $S_n$ **converge in distribution** to a normal distribution. That is

$$S_n \to^d N(n\mu, n\sigma^2).$$

# 46 Normal distribution

The most widely used model for random variables with continuous distributions is the family of normal distributions. One reason is that many real world samples appears to be normally distributed (the mass centered around the mean). The other reason is because of the Central Limit Theorem (will be discussed in later chapters), which essentially says the sum (or mean) or any random samples are approximately normal.

**Definition 46.1.** A random variable $Z$ has the standard Normal distribution with mean 0 and variance 1, denoted as $Z \sim N(0,1)$, if $Z$ has a PDF that follows

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}.$$

This is a valid PDF because $\int_{-\infty}^{\infty} f(z)dz = 1$, which directly follows from Example 44.1. We further verify its mean and variance:

$$E(Z) = \int_{-\infty}^{+\infty} z \cdot \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = 0 \quad \text{by symmetry.}$$

$$\begin{aligned}
Var(Z) &= E(Z^2) - (EZ)^2 = E(Z^2) \\
&= \int_{-\infty}^{+\infty} z^2 \cdot \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz \\
&= \frac{2}{\sqrt{2\pi}} \int_0^{\infty} \underbrace{z}_{u} \cdot \underbrace{z e^{-z^2/2} dz}_{dv} \\
&= \frac{2}{\sqrt{2\pi}} \left\{ \left[ z(-e^{-z^2/2}) \right]_0^{\infty} + \underbrace{\int_0^{\infty} e^{-z^2/2} dz}_{\sqrt{2\pi}/2} \right\} \\
&= 1.
\end{aligned}$$

**Definition 46.2.** The CDF of standard normal distribution is usually denoted by $\Phi$. Therefore,

$$\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-t^2/2} dt.$$

By symmetry, we have $\Phi(-z) = 1 - \Phi(z)$.

**Definition 46.3.** Let $X = \mu + \sigma Z$ where $Z \sim N(0,1)$. Then we say $X$ has the Normal distribution with mean $\mu$ and variance $\sigma^2$, denoted as $X \sim N(\mu, \sigma^2)$. The PDF of $X$ is given by

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right].$$

The mean and variance of $X$ can be easily verified by the properties of expectation and variance.

$$E(X) = E(\mu + \sigma Z) = \mu + \sigma E(Z) = \mu,$$
$$Var(X) = Var(\mu + \sigma Z) = \sigma^2 Var(Z) = \sigma^2.$$

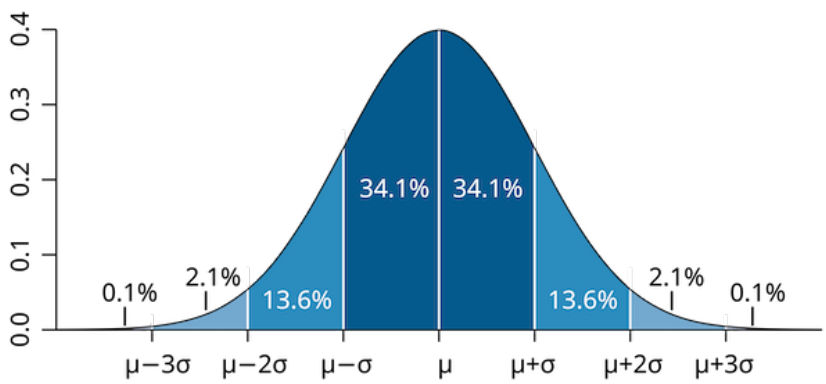
To verify the PDF, we utilize the standard normal CDF:

$$P(X \leq x) = P\left(\frac{X-\mu}{\sigma} \leq \frac{x-\mu}{\sigma}\right) = \Phi\left(\frac{x-\mu}{\sigma}\right)$$

The PDF is the derivative of the CDF,

$$f(x) = \frac{1}{\sigma}\Phi'\left(\frac{x-\mu}{\sigma}\right) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right].$$

The shape of the normal distribution is the famous bell-shaped curve.



**Three-sigma rule**

The normal distribution has the "three-sigma rule":

$$P(|X-\mu| \leq \sigma) \approx 0.68$$
$$P(|X-\mu| \leq 2\sigma) \approx 0.95$$
$$P(|X-\mu| \leq 3\sigma) \approx 0.997$$

Critical values: $\Phi(-1) \approx 0.16, \Phi(-2) \approx 0.025, \Phi(-3) \approx 0.0015$.

**Theorem 46.1.** *Let $X$ have the Normal distribution with mean $\mu$ and variance $\sigma^2$. Let $F$ be the CDF of $X$. Then the* ***standardization*** *of $X$*

$$Z = \frac{X-\mu}{\sigma}$$

*has the standard normal distribution, and, for all $x$:*

$$F(x) = \Phi\left(\frac{x-\mu}{\sigma}\right).$$

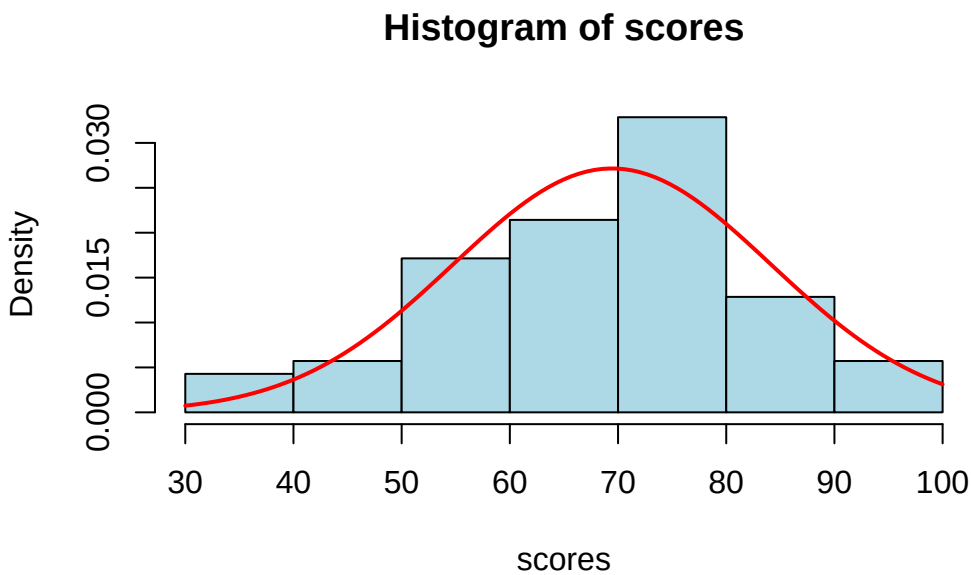
To find the value of $\Phi(z)$, we need to use the normal probability table or statistical software.

**Example 46.1.** Suppose the test score of a class of 50 students is normally distributed with mean 80 and standard deviation 20 (the total mark is 100). A student has scored 90. What is his percentile in the class?

*Solution.* $X \sim N(80, 20)$. We want to find $P(X < 90)$. Standardize the distribution:

$$P(X < 90) = P\left(\frac{X-80}{20} < \frac{90-80}{20}\right) = \Phi(0.5) \approx 0.69.$$

```r
# Exam scores from past students
scores <- read.csv("../dataset/exam.csv")$final

# Histogram of the exam scores
hist(scores, prob = TRUE, col = "lightblue")

# Overlay the normal curve
curve(dnorm(x, mean(scores), sd(scores)), col = "red", lwd = 2, add = TRUE)
```




**Theorem 46.2.** *Suppose $X \sim N(\mu, \sigma^2)$. If $Y = aX + b$, then $Y$ has the Normal distribution $Y \sim N(a\mu + b, a^2\sigma^2)$.*

**Theorem 46.3.** *If the random variables $X_1, \dots, X_k$ are independent and $X_i \sim N(\mu_i, \sigma_i^2)$. Then*

$$X_1 + \cdots + X_k \sim N(\mu_1 + \cdots + \mu_k, \sigma_1^2 + \cdots + \sigma_k^2).$$

**Example 46.2.** Suppose the heights (in centimeters) of women and men independently follow the normal distribution, $X \sim N(165, 25)$, $Y \sim N(170, 25)$. Determine the probability that a randomly selected woman will be taller than a man.

*Solution.* Let $W = Y - X \sim N(170 - 165, 25 + 25)$. Then $W \sim N(5, 50)$. Therefore,

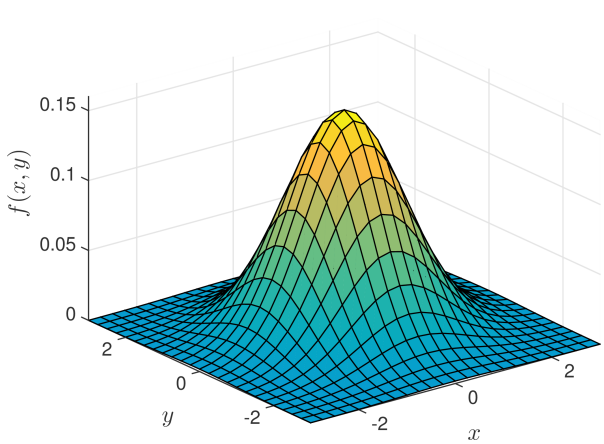
$$P(W < 0) = P\left(\frac{W-5}{\sqrt{50}} < \frac{-5}{\sqrt{50}}\right) = P\left(Z < -\frac{1}{\sqrt{2}}\right) = \Phi(-0.707) \approx 0.24.$$

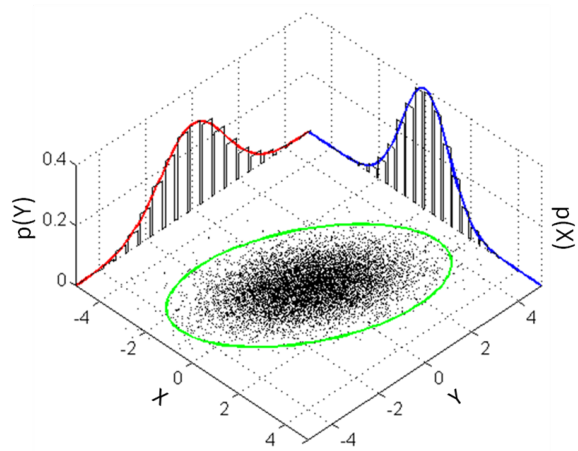
# 47 Multivariate normal

**Definition 47.1** (Bivariate normal distribution)**.** $(X, Y)$ is said to have a Bivariate Normal distribution if the joint PDF satisfies

$$f(x, y) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}(x^2 + y^2 - 2\rho xy)\right)$$

where $\rho \in (-1, 1)$ is the correlation between $X$ and $Y$.



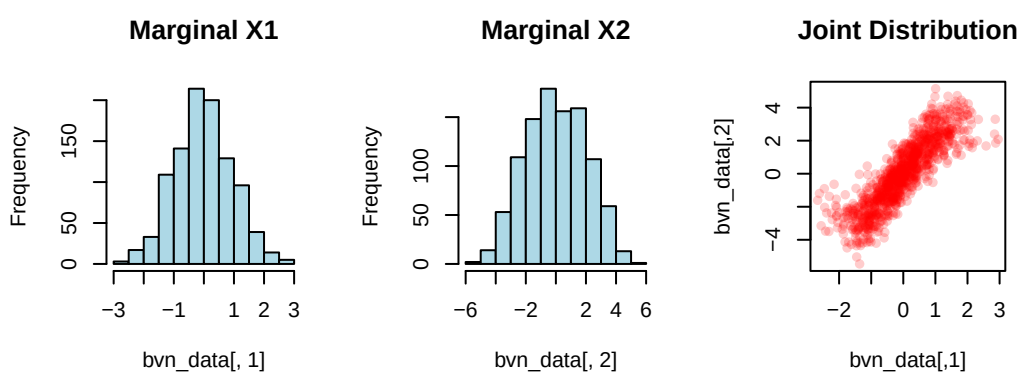




**Multivariate Normal (MVN)** is an extension of the bivariate normal distribution to $n$-dimensional variables. We skip the joint PDF here since it is too complicated. But like the bivariate case, an MVN is fully specified by knowing the mean of each component, the variance of each component, and the covariance between any two components.

> **Marginal normality does not imply joint normality**
>
> If $(X_1, ..., X_k)$ is MVN, then the marginal distribution of every $X_j$ is Normal. However, the converse is false: it is possible to have Normally distributed $X_1, ..., X_k$ such that $(X_1, ..., X_k)$ is not Multivariate Normal.

```
# Load necessary library
library(MASS)
```

```r
# Set seed for reproducibility
set.seed(123)

# Generate bivariate normal data
bvn_data <- mvrnorm(n = 1000,
                    mu = c(0, 0),
                    Sigma = matrix(c(1, 0.5, 0.5, 1), nrow = 2))

# Modify the joint distribution: apply a nonlinear transformation
bvn_data[, 2] <- bvn_data[, 2] + 2 * sin(bvn_data[, 1])

# The marginal distribution remains normal
par(mfrow = c(1, 3))
hist(bvn_data[, 1], main = "Marginal X1", col = "lightblue")
hist(bvn_data[, 2], main = "Marginal X2", col = "lightblue")

# But the joint distribution is not normal
plot(bvn_data, main = "Joint Distribution", pch = 16, col = rgb(1,0,0,.2))
```




**Theorem 47.1.** *A random vector $(X_1, ..., X_k)$ is Multivariate Normal if every linear combination of the $X_j$ has a Normal distribution ($X_j$ do not have to be independent). That is, we require $t_1X_1 + \cdots + t_kX_k$ to have a Normal distribution for any choice of constants $t_1, ..., t_k$.*

**Theorem 47.2.** *In general, uncorrelated does not imply independent. But with an MVN random vector, uncorrelated implies independent. In particular, if $(X, Y)$ is Bivariate Normal and $\rho_{XY} = 0$, then $X$ and $Y$ are independent.*

**Theorem 47.3.** *If $(X, Y)$ is Bivariate Normal, then the conditional expectation satisfies*

$$E(Y|X) = E(Y) + \frac{Cov(X, Y)}{Var(X)}(X - E(X)).$$

*In other words,*

$$E(Y|X) = a + bX$$

*where $b = \frac{Cov(X,Y)}{Var(X)}$ and $a = E(Y) - bE(X)$.*

This is exactly the case in Definition 39.2, where we assume the conditional expectation $E(Y|X)$ is a linear function of $X$. This assumption is true when $(X, Y)$ are jointly normal. Otherwise, the assumption might not be reasonable. In practice, we don't know precisely the joint distribution of variables. The linear model is just a simplified assumption.

# 48 Exponential distribution

Imagine you are a shop owner that waits for your next customer. The customers arrive randomly, with no preference for any specific time interval. What interests us is the waiting time until the next customer arrives. Since the customers arrives randomly, the likelihood of it coming in the next moment is the same whether you've been waiting for one minute or ten minutes. In other words, the waiting time between events that occur randomly and independently over time. The exponential distribution is the mathematical model that best describes such scenarios.

To model the waiting time, let $X$ represent the time until the next event. A crucial feature of this process is that the waiting time has no "memory." That is, no matter how long you've already waited, the probability of waiting an additional amount of time is the same. Mathematically, this memoryless property is expressed as:

$$P(X \geq s + t \mid X \geq s) = P(X \geq t), \quad \text{for all } s, t \geq 0.$$

The conditional probability can be rewritten using the definition of conditional probabilities:

$$P(X \geq s + t \mid X \geq s) = \frac{P(X \geq s + t)}{P(X \geq s)}.$$

Thus, the memoryless property implies:

$$\frac{P(X \geq s + t)}{P(X \geq s)} = P(X \geq t).$$

Let the survival function $S(x)$ represent $P(X \geq x)$ . Substituting $S(x)$ into the equation gives:

$$\frac{S(s + t)}{S(s)} = S(t).$$

This reminds us of the exponential function. In fact, the only continuous and non-negative solution to this equation is:

$$S(x) = e^{-\lambda x}, \quad \lambda > 0,$$

where $\lambda$ is a positive constant. This solution represents the probability that the waiting time exceeds $x$ , and $\lambda$ determines how quickly the probability decreases over time.

The CDF of $X$ is exactly the opposite of $S(x)$:

$$F(x) = 1 - S(x) = 1 - e^{-\lambda x}.$$

Take derivative to get the PDF:

$$f(x) = F'(x) = \lambda e^{-\lambda x}.$$

**Definition 48.1** (Exponential distribution)**.** A random variable $X$ is said to have the Exponential distribution with parameter $\lambda$ if its PDF is

$$f(x) = \lambda e^{-\lambda x}, \qquad x > 0.$$

We denote this as $X \sim \text{Expo}(\lambda)$.$\lambda$ is interpreted as the "rate", i.e. number of events per unit of time.

To compute the expectation and variance, we first standardize the exponential distribution. Let $Y = \lambda X$, then $Y \sim \text{Expo}(1)$, because

$$P(Y \leq y) = P(X \leq y/\lambda) = 1 - e^{-y}.$$

It follows that,

$$E(Y) = \int_0^\infty y e^{-y} dy = [-ye^{-y}]_0^\infty + \int_0^\infty e^{-y} dy = 1;$$
$$Var(Y) = E(Y^2) - (EY)^2 = \int_0^\infty y^2 e^{-y} dy - 1 = 1.$$

For $X = Y/\lambda$, we have $E(X) = \frac{1}{\lambda}$, $Var(X) = \frac{1}{\lambda^2}$.

**Theorem 48.1** (Memoryless property)**.** *If $X$ has the exponential distribution with parameter $\lambda$, and let $t > 0$, $h > 0$, then*

$$P(X \geq t + h | X \geq t) = P(X \geq h).$$

*Proof.* For $t > 0$ we have

$$P(X \geq t) = \int_t^\infty \lambda e^{-\lambda x} dx = e^{-\lambda t}.$$

Hence for each $t > 0$ and each $h > 0$,

$$P(X \geq t + h | X \geq t) = \frac{P(X \geq t + h)}{P(X \geq t)} = \frac{e^{-\lambda(t+h)}}{e^{-\lambda t}} = e^{-\lambda h} = P(X \geq h).$$

□

What are the implications of the memoryless property? If human lifetimes were Exponential, then conditional on having survived to the age of 80, your remaining lifetime would have the same distribution as that of a newborn baby! Clearly, the memoryless property is not an appropriate description for human lifetimes.

The memoryless property is a very special property of the Exponential distribution. In fact, the Exponential is the only memoryless continuous distribution (with support $(0, \infty)$); and Geometric distribution is the only memoryless discrete distribution (with support $0, 1, ...$).

**Example 48.1** (Waiting time)**.** We try to model the waiting time at a bus station. Suppose the bus arrives at random time but on average there will be one bus per 10 minutes. You arrive at the bus stop at a random time, not knowing how long ago the previous bus came. What is the distribution of your waiting time for the next bus? What is the mean waiting time? What is the median waiting time?

*Solution.* Let $X$ be the waiting time and we know it is an Exponential distribution. Since $E(X) = 1/\lambda = 10$, the parameter $\lambda = 1/10$. Thus $X \sim \text{Expo}(0.1)$. By the memoryless property, how much longer the next bus will take to arrive is independent of how long ago the previous bus arrived. The average waiting time is always 10 minutes.

The CDF of $X$ is: $F(x) = 1 - e^{-\lambda x}$. The median $m$ satisfies $F(m) = 1/2$. Thus, $m = \log(2)/\lambda \approx 6.9$ minutes. So the typical waiting experienced by most passengers is less than 10 minutes.

**Theorem 48.2** (Poisson-Exponential connection)**.** *Let $T$ be the time between two consecutive events in Poisson process Pois$(\lambda t)$. Then $T$ follows Exponential distribution $T \sim$ Expo$(\lambda)$.*

*Proof.* The waiting time $T > t$ is equivalent to no event occurred during period $t$. Therefore,

$$P(T > t) = P(N_t = 0) = e^{-\lambda t}\frac{(\lambda t)^0}{0!} = e^{-\lambda t}$$

where $N_t$ is the number of events occurred in $[0, t]$, which follows a Poisson distribution. The CDF of $T$ is

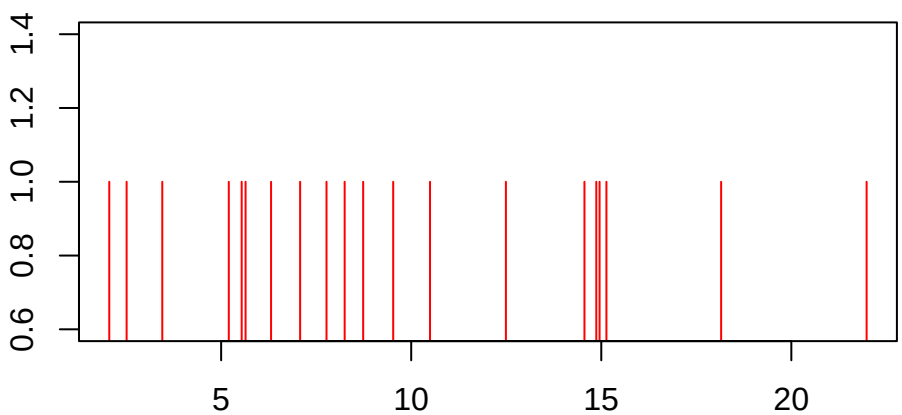
$$F(t) = 1 - P(T > t) = 1 - e^{-\lambda t}$$

The PDF of $T$ is

$$f(t) = F'(t) = \lambda e^{-\lambda t}$$

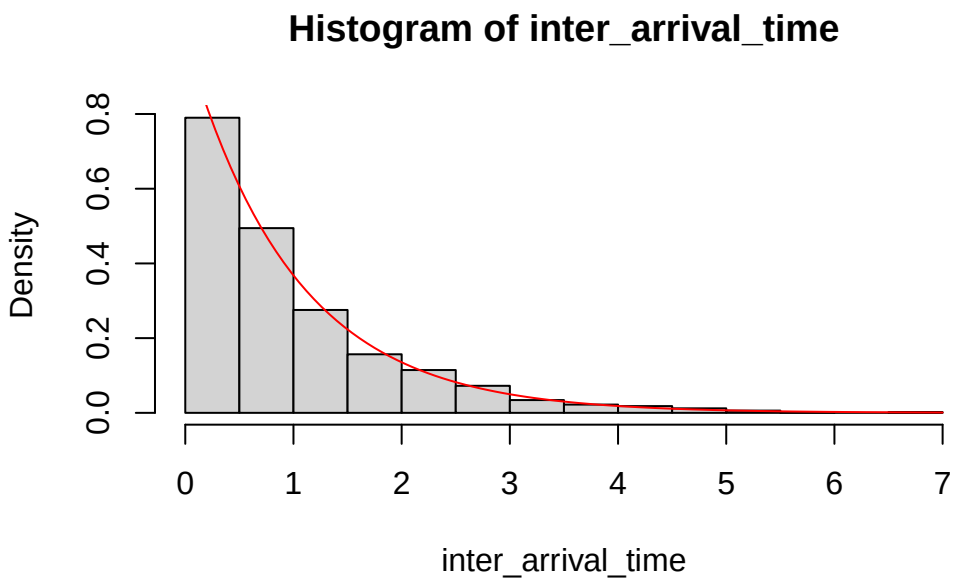
This indicates $T \sim \text{Expo}(\lambda)$. □

```r
#
# Simulate random arrivals and inter-arrival time
#
T <- 1000  # Total time horizon
rate <- 1  # rate of occurrence per unit time

# Total number of arrivals
n_arrivals <- rpois(1, lambda = rate * T)

# Time of each arrivals
t_arrivals <- sort(runif(n_arrivals, min = 0, max = T))

# Plot the timeline of arrivals
plot(t_arrivals[1:20], rep(1, 20), type = "h", col = "red", ann = F)
```




```r
# Compute inter-arrival time
inter_arrival_time <- diff(t_arrivals)

# Plot the distribution of inter-arrival time
hist(inter_arrival_time, prob = TRUE, breaks = 20)

# Overlay the exponential function
```

```r
curve(exp(-x), col = "red", add = TRUE)
```



# 49 Gamma distribution

The Gamma distribution is a continuous distribution on the positive real line; it is a generalization of the Exponential distribution. While an Exponential RV represents the waiting time for the first event to occur, we shall see that a Gamma RV represents the total waiting time for $n$ events to occur.

Let's start with a simple case. Suppose we want to find out the total waiting until the 2nd event occurred. Let $Y = X_1 + X_2$ where $X_1, X_2 \sim \text{Expo}(\lambda)$ independently. If $Y$ is discrete, we have $P(Y = y) = \sum_{k=0}^{y} P(X_1 = k, X_2 = y - k)$. For continuous $y$, we have

$$\begin{aligned} f_Y(y) &= \int_0^y f_X(x) f_X(y-x) dx = \int_0^y \lambda e^{-\lambda x} \lambda e^{-\lambda(y-x)} dx \\ &= \int_0^y \lambda^2 e^{-\lambda y} dx = \lambda^2 e^{-\lambda y} y. \end{aligned}$$

If there is a third variable,

$$\begin{aligned} f_Z(z) &= \int_0^z f_X(x) f_Y(z-x) dx = \int_0^z \lambda e^{-\lambda x} \lambda^2 e^{-\lambda(z-x)} (z-x) dx \\ &= \lambda^3 e^{-\lambda z} \int_0^z (z-x) dx = \lambda^3 e^{-\lambda z} z^2/2. \end{aligned}$$

The general pattern is the Gamma distribution.

**Definition 49.1** (Exponential distribution)**.** An random variable X is said to have the Gamma distribution with parameters $a$ and $\lambda$, $a > 0$ and $\lambda > 0$, if it has the PDF

$$f(x) = \frac{\lambda^a}{\Gamma(a)} x^{a-1} e^{-\lambda x}, \quad x > 0$$

We write $X \sim \text{Gamma}(a, \lambda)$.

Verify this is a valid PDF:

$$\int_0^\infty \frac{1}{\Gamma(a)} (\lambda x)^a e^{-\lambda x} \frac{dx}{x} \overset{u=\lambda x}{=} \frac{1}{\Gamma(a)} \int_0^\infty u^a e^{-u} \frac{du}{u} = \frac{\Gamma(a)}{\Gamma(a)} = 1.$$

Taking $a = 1$, the Gamma$(1, \lambda)$ PDF is $f(x) = \lambda e^{-\lambda x}$, which is the same as Expo$(\lambda)$. So Exponential distribution is a special case of Gamma distribution.

Let's find the expectation and variance of the Gamma distribution. Let $Y \sim$ Gamma$(a, 1)$. Recall $\Gamma$ function has the property $\Gamma(a+1) = a\Gamma(a)$.

$$E(Y) = \int_0^\infty y \cdot \frac{1}{\Gamma(a)} y^{a-1} e^{-y} dy = \frac{1}{\Gamma(a)} \int_0^\infty y^a e^{-y} dy = \frac{\Gamma(a+1)}{\Gamma(a)} = a.$$

Apply LOTUS to evaluate the second moment:

$$E(Y^2) = \int_0^\infty y^2 \cdot \frac{1}{\Gamma(a)} y^{a-1} e^{-y} dy = \frac{1}{\Gamma(a)} \int_0^\infty y^{a+1} e^{-y} dy = \frac{\Gamma(a+2)}{\Gamma(a)} = (a+1)a.$$

Therefore,

$$Var(Y) = (a+1)a - a^2 = a.$$

So for $Y \sim$ Gamma$(a, 1)$, $E(Y) = Var(Y) = a$. For the general case $X \sim$ Gamma$(a, \lambda)$, we now show that $X = \frac{Y}{\lambda}$. Note that

$$\begin{aligned} F_X(x) &= P(X \le x) = P(Y \le x/\lambda) = F_Y(x/\lambda) \\ f_X(x) &= \frac{dF_X}{dx} = \frac{\partial F_Y}{\partial y} \frac{dy}{dx} = f_Y(y)\lambda \end{aligned}$$

Therefore,

$$f_X(x) = \frac{1}{\Gamma(a)} y^{a-1} e^{-y} \lambda = \frac{\lambda^a}{\Gamma(a)} x^{a-1} e^{-\lambda x}.$$

Hence, we have $E(X) = \frac{a}{\lambda}$, $Var(X) = \frac{a}{\lambda^2}$.

**Theorem 49.1** (Exponential-Gamma connection)**.** *Let $X_1, \dots, X_n$ be independent and identical Expo$(\lambda)$. Then*

$$X_1 + \cdots + X_n \sim Gamma(n, \lambda).$$

*Proof.* Let's prove by showing the MGFs are equivalent.

$$M_X(t) = E(e^{tX}) = \int_0^\infty e^{tx} \lambda e^{-\lambda x} dx = \frac{\lambda}{\lambda - t} \quad \text{for } t < \lambda$$

Thus, the MGF of $Y = X_1 + \cdots + X_n$ is $M_Y(t) = (M_X(t))^n = \left(\frac{\lambda}{\lambda - t}\right)^n$. We verify this is the MGF of a Gamma distribution. Suppose $Y \sim$ Gamma$(n, \lambda)$, it has MGF:

$$
\begin{aligned}
M_Y(t) = E(e^{tY}) &= \int_0^\infty e^{ty} \frac{\lambda^n}{\Gamma(a)} y^{n-1} e^{-\lambda y} dy \\
&= \frac{\lambda^n}{(\lambda - t)^n} \int_0^\infty \frac{1}{\Gamma(a)} ((\lambda - t)y)^{n-1} e^{-(\lambda - t)y} (\lambda - t) dy \\
&= \frac{\lambda^n}{(\lambda - t)^n} \int_0^\infty \frac{1}{\Gamma(a)} u^{n-1} e^{-u} du \qquad u = (\lambda - t)y \\
&= \left( \frac{\lambda}{\lambda - t} \right)^n .
\end{aligned}
$$

□

Thus, if $X_i$ represents the i.i.d inter-arrival time. $Y$ has the interpretation of the arrival time until the $n$-th event.

$$
Y = \sum_{i=1}^n X_i = \sum_{i=1}^n (\text{time of the i-th arrival}) \sim \text{Gamma}(n, \lambda).
$$

**Example 49.1** (Service time in a queue)**.** Customer $i$ must wait time $X_i$ for service once reaching the head of the queue. The average service rate is 1 customer per 10 minutes. Assume the service for each customer is independent. If you are the 5th in the queue. What is the expected waiting to be served?

*Solution.* $X_i \sim \text{Expo}(0.1)$. Then $E(X_i) = 10$. Let Y be the time until you are served. Then $Y \sim \text{Gamma}(5, 0.1)$. Thus, $E(Y) = \frac{5}{0.1} = 50$ minutes. The probabilities of some selected values:

$$
P(Y \leq t) = \begin{cases} 5\% & t = 20 \\ 18\% & t = 30 \\ 71\% & t = 60 \end{cases} .
$$

# 50 Beta distribution

The Beta distribution is a continuous distribution on the interval $(0, 1)$. It is a generalization of the Unif$(0, 1)$ distribution, allowing the PDF to be non-constant on $(0, 1)$.

**Definition 50.1** (Beta distribution)**.** A random variable $X$ is said to have the Beta distribution with parameters $a$ and $b$, $a > 0$ and $b > 0$, if its PDF is

$$f(x) = \frac{1}{\beta(a, b)} x^{a-1}(1-x)^{b-1}, \quad 0 < x < 1$$

where the constant $\beta(a, b)$ is chosen to make the PDF integrate to 1. We write this as $X \sim$ Beta$(a, b)$.

The Beta distribution takes different shapes for different $a$ and $b$ values. Here are some general patterns:

- If $a = b = 1$, the Beta$(1, 1)$ PDF is constant on $(0, 1)$, equivalent to Unif$(0, 1)$.
- If $a < 1$ and $b < 1$, the PDF is U-shaped and opens upward. If $a > 1$ and $b > 1$, the PDF opens downward.
- If $a = b$, the PDF is symmetric about $1/2$. If $a > b$, the PDF favors values larger than $1/2$. If $a < b$, the PDF favors values smaller than $1/2$.

To make the PDF integrates to 1, the constant $\beta(a, b)$ has to satisfy

$$\beta(a, b) = \int_0^1 x^{a-1}(1-x)^{b-1} dx.$$

We now try to find this integral:

$$
\begin{aligned}
\beta(a,b) &= \int_0^1 \underbrace{x^{a-1}}_{f} \underbrace{(1-x)^{b-1}}_{g'} dx \\
&= \left[-x^{a-1}\frac{(1-x)^b}{b}\right]_0^1 + \int_0^1 (a-1)x^{a-2}\frac{(1-x)^b}{b}dx \\
&= \frac{a-1}{b}\beta(a-1,b+1) \\
&= \frac{a-1}{b}\cdot\frac{a-2}{b+1}\beta(a-2,b+2) \\
&= \frac{a-1}{b}\cdot\frac{a-2}{b+1}\cdot\frac{a-3}{b+2}\beta(a-3,b+3) \\
&\vdots \\
&= \frac{(a-1)!}{b(b+1)(b+2)\cdots(b+a-2)}\underbrace{\beta(1,a+b-1)}_{\frac{1}{a+b-1}} \\
&= \frac{(a-1)!}{\frac{(b+a-2)!}{(b-1)!}}\cdot\frac{1}{a+b-1} \\
&= \frac{(a-1)!(b-1)!}{(a+b-1)!} \\
&= \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}.
\end{aligned}
$$

**Example 50.1.** Let $X_1, \dots, X_n$ be independent random variables with the uniform distribution on the interval $[0,1]$. Find the distribution of $Y = \max(X_1, \dots, X_n)$.

*Solution.* Let's find the CDF of $Y$:

$$
\begin{aligned}
P(Y \le y) &= P(X_1 \le y \cap X_2 \le y \cap \cdots \cap X_n \le y) \\
&\overset{iid}{=} P(X_1 \le y)P(X_2 \le y)\cdots P(X_n \le y) \\
&= y^n
\end{aligned}
$$

for $y \in [0,1]$. Hence,

$$
F_Y(y) = P(Y \le y) = \begin{cases} 0 & y < 0 \\ y^n & 0 \le y \le 1 \\ 1 & y > 1 \end{cases}
$$

The PDF of $Y$ is

$$
f_Y(y) = F_Y'(y) = \begin{cases} ny^{n-1} & 0 \le y \le 1 \\ 0 & \text{otherwise} \end{cases}
$$

Thus, $Y \sim Beta(n,1)$.

Beta distributions are often used as *priors* for parameters in Bayesian inference. We do not cover Bayesian inference in this book. Nonetheless we illustrate this with an example.

**Example 50.2** (Beta-Binomial conjugacy)**.** We have a coin that lands Heads with probability $p$, but we don't know what $p$ is. Our goal is to infer the value of $p$ after observing the outcomes of $n$ tosses of the coin. The larger that $n$ is, the more accurately we should be able to estimate $p$.

*Solution.* We model the unknown parameter $p$ as a Beta distribution, $p \sim \text{Beta}(a, b)$. Since we are completely ignorant about this $p$, we can also model it as the uniform distribution. But we will see that using the Beta distribution is even simpler than the uniform distribution. Let $X$ be the number of heads in $n$ tosses of the coin. Then

$$X|p \sim \text{Bin}(n, p)$$

Apply the Bayes' rule to inverse the conditioning:

$$\begin{aligned} f(p|X = k) &= \frac{P(X = k|p)f(p)}{P(X = k)} \\ &= \frac{\binom{n}{k}p^k(1-p)^{n-k} \cdot \frac{1}{\beta(a,b)}p^{a-1}(1-p)^{b-1}}{\int_0^1 \binom{n}{k}p^k(1-p)^{n-k}f(p)dp} \\ &\propto p^{a+k-1}(1-p)^{b+n-k-1} \end{aligned}$$

This the kernel of $\text{Beta}(a + k, b + n - k)$. The rest is just a normalizing constant. Therefore,

$$p|X = k \sim \text{Beta}(a + k, b + n - k).$$

The *posterior* distribution of $p$ after observing $X = k$ is still a Beta distribution! This is a special relationship between the Beta and Binomial distributions called *conjugacy*: if we have a Beta prior distribution on $p$ and data that are conditionally Binomial given $p$, then when going from prior to posterior, we don't leave the family of Beta distributions. We say that the Beta is the conjugate prior of the Binomial.

# 51 Multivariate problems

We extend the concepts of joint, marginal and conditional distribution to continuous random variables.

| | Discrete | Continuous |
|---|---|---|
| Joint CDF | $F_{XY}(x,y) = P(X \leq x, Y \leq y)$ | $F_{XY}(x,y) = P(X \leq x, Y \leq y)$ |
| Joint PMF/PDF | $p_{XY}(x,y) = P(X = x, Y = y)$ | $f_{XY}(x,y) = \frac{\partial^2}{\partial x \partial y} F_{XY}(x,y)$ |
| | $\sum_x \sum_y P(X = x, Y = y) = 1$ | $\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f_{XY}(x,y) dx dy = 1$ |
| Marginal PMF/PDF | $P(X = x) = \sum_y P(X = x, Y = y)$ | $f_X(x) = \int_{-\infty}^{+\infty} f_{XY}(x,y) dy$ |
| Conditional PMF/PDF | $P(X = x \vert Y = y) = \frac{P(X=x,Y=y)}{P(Y=y)}$ | $f_{X\vert Y}(x\vert y) = \frac{f_{XY}(x,y)}{f_Y(y)}$ |
| Independence | $P(X = x, Y = y) = P(X = x)P(Y = y)$ | $f_{XY}(x,y) = f_X(x) f_Y(y)$ |
| | $P(X = x \vert Y = y) = P(X = x)$ | $f_{X\vert Y}(x\vert y) = f_X(x)$ |
| | $F_{XY}(x,y) = F_X(x) F_Y(y)$ | $F_{XY}(x,y) = F_X(x) F_Y(y)$ |
| Bayes' rule | $P(Y = y \vert X = x) = \frac{P(X=x\vert Y=y)P(Y=y)}{P(X=x)}$ | $f_{Y\vert X}(y\vert x) = \frac{f_{X\vert Y}(x\vert y) f_Y(y)}{f_X(x)}$ |
| LOTP | $P(X = x) = \sum_y P(X = x \vert Y = y) P(Y = y)$ | $f_X(x) = \int_{-\infty}^{+\infty} f_{X\vert Y}(x\vert y) f_Y(y) dy$ |
| LOTUS | $E(g(X,Y)) = \sum_x \sum_y g(x,y) P(X = x) P(y = y)$ | $E(g(X,Y)) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x,y) f_{XY}(x,y) dx dy$ |

**Example 51.1.** Suppose $X$ and $Y$ are uniformly distributed on a disk $\{(x,y) : x^2 + y^2 \leq 1\}$. Find the joint PDF, marginal distributions and conditional distributions. Are $X$ and $Y$ independent?

*Solution.* The area of the disk is $\pi$, therefore

$$f(x,y) = \begin{cases} \frac{1}{\pi} & x^2 + y^2 \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

The marginal distributions are

$$f_X(x) = \int_{-\sqrt{1-x^2}}^{\sqrt{1+x^2}} \frac{1}{\pi} dy = \frac{2}{\pi}\sqrt{1-x^2}, \qquad -1 \le x \le 1$$
$$f_Y(y) = \int_{-\sqrt{1-y^2}}^{\sqrt{1+y^2}} \frac{1}{\pi} dx = \frac{2}{\pi}\sqrt{1-y^2}, \qquad -1 \le y \le 1$$

The conditional distributions are

$$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)} = \frac{\frac{1}{\pi}}{\frac{2}{\pi}\sqrt{1-x^2}} = \frac{1}{2\sqrt{1-x^2}}$$

Therefore, $Y|X \sim \text{Unif}(-\sqrt{1-x^2}, \sqrt{1-x^2})$.

Since $f(x,y) \neq f_X(x)f_Y(y)$, $X$ and $Y$ are not independent. This is because knowing the value of $X$ constrains the value of $Y$.

**Example 51.2.** Suppose $X, Y \overset{iid}{\sim} \text{Unif}(0,1)$. Find the probability $P\left(Y \le \frac{1}{2X}\right)$.

*Solution.* The joint distribution is

$$f(x,y) = \begin{cases} 1 & 0 \le x \le 1, 0 \le y \le 1 \\ 0 & \text{otherwise} \end{cases}$$

$$P\left(Y \le \frac{1}{2X}\right) = \int_0^{1/2}\int_0^1 1 dy dx + \int_{1/2}^1 \int_0^{1/2x} 1 dy dx = \frac{1}{2} + \int_{1/2}^1 \frac{1}{2x} dx = \frac{1}{2} + \ln\sqrt{2}.$$

**Example 51.3.** For $X, Y \overset{iid}{\sim} \text{Unif}(0,1)$, find $E(|X-Y|)$.

*Solution.* Apply 2D LOTUS:

$$\begin{aligned} E(|X-Y|) &= \int_0^1\int_0^1 |x-y| dx dy \\ &= \int_0^1\int_y^1 (x-y) dx dy + \int_0^1\int_0^y (y-x) dx dy \\ &= 2\int_0^1\int_y^1 (x-y) dx dy \\ &= \frac{1}{3}. \end{aligned}$$

**Example 51.4.** $X, Y \overset{iid}{\sim} N(0,1)$, find $E(|X-Y|)$.

*Solution.* Since the sum or difference of independent Normal variables is also Normal, $X-Y \sim N(0,2)$. Let $Z = X-Y$. Then $Z \sim N(0,1)$, and $E(|X-Y|) = \sqrt{2}E(|Z|)$. Apply LOTUS,

$$E(|Z|) = \int_{-\infty}^{\infty} |z| \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz = 2\int_{0}^{\infty} z \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz = \sqrt{\frac{2}{\pi}},$$

Therefore, $\mathbb{E}(|X-Y|) = \frac{2}{\sqrt{\pi}}$.

# 52 Transformation

**Example 52.1** (Min/max of random variables)**.** Let $X_1, X_2, \dots, X_n$ be i.i.d random variables, each following a uniform distribution on the interval $[0, 1]$. Find the distribution of $\max(X_1, X_2, \dots, X_n)$.

*Solution.* Let $M = \max(X_1, X_2, \dots, X_n)$. The CDF of $M$, denoted $F_M(m)$, is the probability that $M \leq m$. For $M \leq m$ to hold, all $X_i$ must satisfy $X_i \leq m$. Since the $X_i$ are independent and identically distributed:

$$\begin{aligned} F_M(m) = P(M \leq m) &= P(X_1 \leq m, X_2 \leq m, \dots, X_n \leq m) \\ &= P(X_1 \leq m) \cdot P(X_2 \leq m) \cdots P(X_n \leq m) \end{aligned}$$

For a uniform distribution on $[0, 1]$, $P(X_i \leq m) = m$ for $0 \leq m \leq 1$. Thus:

$$F_M(m) = m^n.$$

The PDF of $M$, denoted $f_M(m)$, is the derivative of the CDF:

$$f_M(m) = \frac{d}{dm} F_M(m) = \frac{d}{dm}(m^n) = nm^{n-1}.$$

**Example 52.2** (Chi-square PDF)**.** Let $X \sim N(0, 1)$, $Y = X^2$. The distribution of $Y$ is an example of a Chi-Square distribution. Find the PDF of $Y$.

*Solution.* Again, we try to find the CDF first, and differentiate to the PDF.

$$F_Y(y) = P(X^2 \leq y) = P(-\sqrt{y} \leq X \leq \sqrt{y}) = \Phi(\sqrt{y}) - \Phi(-\sqrt{y}) = 2\Phi(\sqrt{y}) - 1$$

Therefore,

$$f_Y(y) = 2\varphi(\sqrt{y}) \cdot \frac{1}{2} y^{-1/2} = \varphi(\sqrt{y}) y^{-1/2}, \quad y > 0.$$

**Theorem 52.1** (Transformation)**.** *Let $X$ be a continuous r.v. with PDF $f_X$, and let $Y = g(X)$, where $g$ is differentiable and <u>strictly increasing</u> (or <u>strictly decreasing</u>). Then the PDF of $Y$ is given by*

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|,$$

*where $x = g^{-1}(y)$.*

*Proof.* Let $g$ be strictly increasing. The CDF of $Y$ is

$$F_Y(y) = P(Y \le y) = P(g(X) \le y) = P(X \le g^{-1}(y)) = F_X(g^{-1}(y)) = F_X(x)$$

By the chain rule, the PDF of $Y$ is

$$f_Y(y) = f_X(x)\frac{dx}{dy}.$$

If $g$ is strictly decreasing,

$$F_Y(y) = P(Y \le y) = P(g(X) \le y) = P(X \ge g^{-1}(y)) = 1 - F_X(g^{-1}(y)) = 1 - F_X(x)$$

Then the PDF of $Y$ is

$$f_Y(y) = -f_X(x)\frac{dx}{dy}.$$

But in this case, $dx/dy < 0$. So taking absolute value covers both cases. □

**Example 52.3** (Log-Normal PDF)**.** Let $X \sim N(0,1)$, $Y = e^X$. Then the distribution of $Y$ is called the Log-Normal distribution. Find the PDF of $Y$.

*Solution.* Since $g(x) = e^x$ is strictly increasing. Let $y = e^x$, so $x = \log y$ and $dy/dx = e^x$. Then

$$f_Y(y) = f_X(x)\left|\frac{dx}{dy}\right| = \varphi(x)\frac{1}{e^x} = \varphi(\log y)\frac{1}{y}, \quad y > 0.$$

Note that after applying the change of variables formula, we write everything on the right-hand side in terms of $y$, and we specify the support of the distribution. To determine the support, we just observe that as $x$ ranges from $-\infty$ to$\infty$, $e^x$ ranges from 0 to $\infty$.

**Theorem 52.2** (Transformation of multi-variables)**.** *Let $\mathbf{X} = (X_1, \dots, X_n)$ be a continuous random vector with joint PDF $f_{\mathbf{X}}$, and let $\mathbf{Y} = g(\mathbf{X})$ where $g$ is an invertible function from $\mathbb{R}^n$ to $\mathbb{R}^n$. Let $\mathbf{y} = g(\mathbf{x})$. Define the Jacobian matrix:*

$$\frac{\partial \mathbf{x}}{\partial \mathbf{y}} = \begin{pmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} & \cdots & \frac{\partial x_1}{\partial y_n} \\ \vdots & \vdots & & \vdots \\ \frac{\partial x_n}{\partial y_1} & \frac{\partial x_n}{\partial y_2} & \cdots & \frac{\partial x_n}{\partial y_n} \end{pmatrix}.$$

*Also assume that the determinant of the Jacobian matrix is never 0. Then the joint PDF of $\mathbf{Y}$ is*

$$f_{\mathbf{Y}}(\mathbf{y}) = f_{\mathbf{X}}(\mathbf{x})\left|\frac{\partial \mathbf{x}}{\partial \mathbf{y}}\right|,$$

*where $\left|\frac{\partial \mathbf{x}}{\partial \mathbf{y}}\right|$ is the absolute value of the determinant of the Jacobian matrix.*

**Example 52.4.** Suppose $X, Y \overset{iid}{\sim} Expo(1)$. Find the distribution of $X/(X+Y)$.

*Solution.* Let $U = \frac{X}{X+Y}$, $V = X + Y$. Then $X = UV$, $Y = V - UV$. The determinant of the Jacobian matrix is

$$\left|\frac{\partial(x,y)}{\partial(u,v)}\right| = \begin{vmatrix} v & u \\ -v & 1-u \end{vmatrix} = v$$

Thus, the joint distribution of $(U, V)$ is

$$f_{UV}(u,v) = f_{XY}(x,y)|v| = f_X(x)f_Y(y)v = e^{-(x+y)}v = e^{-v}v.$$

The distribution of $X/(X+Y)$ is equivalent to the marginal distribution of $U$:

$$f_U(u) = \int_0^\infty e^{-v}vdv = 1$$

for $0 \leq u \leq 1$. Hence $U$ is a Uniform distribution over $[0, 1]$.

# Part VI

# Sampling distributions

# 53 Law of large numbers

We now introduce two important theorems describing the behavior of the sample mean as the sample size grows. Throughout this section and the next, we assume $X_1, X_2, \dots, X_n$ are i.i.d RVs drawn from a population with mean $\mu$ and variance $\sigma^2$. The sample mean is defined as

$$\bar{X}_n = \frac{X_1 + \cdots + X_n}{n}.$$

As we have discussed previously, the sample mean is itself a random variable with mean and variance:

$$\begin{aligned} E(\bar{X}_n) &= \frac{1}{n} E(X_1 + \cdots + X_n) = \frac{1}{n}(E(X_1) + \cdots + E(X_n)) = \mu, \\ Var(\bar{X}_n) &= \frac{1}{n^2} Var(X_1 + \cdots + X_n) \overset{iid}{=} \frac{1}{n^2}(Var(X_1) + \cdots + Var(X_n)) = \frac{\sigma^2}{n}. \end{aligned}$$

The law of large numbers (LLN) says that as n grows, the sample mean $\bar{X}_n$ converges to the true mean $\mu$.

**Theorem 53.1.** *(Strong law of large numbers). The sample mean $\bar{X}_n$ converges to the true mean $\mu$ point-wise as $n \to \infty$, with probability 1. In other words, the event $\bar{X}_n \to \mu$ has probability 1.*

**Theorem 53.2.** *(Weak law of large numbers). For all $\epsilon > 0$, $P(|\bar{X}_n - \mu| > \epsilon) \to 0$ as $n \to \infty$. (this is known as converge in probability).*

We don't need a rigorous proof here. But an intuitive proof is obvious. As $n \to \infty$, $Var(\bar{X}_n) = \frac{\sigma^2}{n} \to 0$. The random variable $\bar{X}_n$ becomes fixed at $\mu$ as $n$ becomes large. Thus, it converges to $\mu$ in a probabilistic sense.

It seems that the LLN just states the obvious. But it has wide applications in daily time that you might not even realize. What it says is essentially this: the uncertainty at the individual level becomes certain when aggregating together; the risks that are unmanageable at the individual level becomes manageable collectively. Think about a rare disease, it happens at 1 out of a million probability. For each individual, no one knows if they will get the disease or not. But as the sample size gets large, suppose we have one billion population, the LLN says

the sample mean will be very close the true mean. That is, there will be almost surely 1000 people being infected by the disease. We provide two more examples.

**Example 53.1.** (Lottery). A lottery company is designing a game with a 6-digit format. Each time someone buys a ticket, they receive a randomly generated 6-digit number. Only one number will win the grand prize of 10 million dollars. What should the company charge per ticket to break even?

*Solution.* The probability of winning the game is $p = 1/10^6$. Suppose the company has sold $n$ tickets. The price for each ticket is $x$. The revenue for the company is therefore $xn$. By the LLN, the cost of the company should be very close to $10^7np$. The break even point is $xn = 10^7np$. So $x = 10^7p = 10$. Therefore, if the company sells each ticket above 10 dollars. The business is surely profitable as long as $n$ is large. If the company is a monopoly, it can reap as much profit as it desires as long as they know the basic probability theory! The same can be said about gambling companies.

**Example 53.2.** (Insurance). Insurance is anther great application of the LLN. It is essentially the same as the the lottery game but most people do not realize it. Suppose there is a disease with infection rate of 1 out of 1 million. The medical expenditure to cure the disease is 10 million dollars. How much the insurance company should charge per customer to cover this disease?

*Solution.* The solution is essentially the same as above. Suppose the premium for the insurance product is $x$. The revenue of the company by selling the premium is $xn$. The cost is — when one customer is infected, the company has to pay the medical cost —$10^7np$. The break even price for the insurance premium is thus 10 dollars.

What is the implication of this insurance? Without the insurance, each individual either chooses to set aside 10 million dollars pre-cautiously for the disease (if he is rich enough) or be exposed to the risk completely uncovered. The insurance product enables everyone to get covered at a cost of just 10 dollars. It is a typical example that the unmanageable risk at the individual level becomes manageable collectively.

# 54 Central limit theorem

The LLN shows the convergence of the sample mean to the population mean. What about the entire sample distribution? This is addressed by the central limit theorem (CLT), which, as its name suggests, is a limit theorem of central importance in statistics.

The CLT states that for large $n$, the distribution of $\bar{X}_n$ after standardization approaches a standard Normal distribution, regardless of the underlying distribution of $X_i$. By standardization, we mean that we subtract $\mu$, the expected value of $\bar{X}_n$, and divide by $\sigma/\sqrt{n}$, the standard deviation of $\bar{X}_n$.

**Theorem 54.1.** *(Central limit theorem). As $n \to \infty$,*

$$\sqrt{n}\left(\frac{\bar{X}_n - \mu}{\sigma}\right) \to N(0,1) \text{ in distribution.}$$

*In other words, the CDF of the left-hand side approaches the CDF of the standard normal distribution.*

*Proof.* We will prove the CLT assuming the MGF of the $X_i$ exists, though the theorem holds under much weaker conditions. Without loss of generality let $\mu = 1, \sigma^2 = 1$ (since we standardize it anyway). We show that the MGF of $\sqrt{n}\bar{X}_n = (X_1 + \cdots + X_n)/\sqrt{n}$ converges to the MGF of the $N(0,1)$.

The MGF of $N(0,1)$ is

$$\begin{aligned}
E(e^{tX}) &= \int_{-\infty}^{\infty} e^{tx} \cdot \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx \\
&= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-x^2/2+tx} dx \\
&= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-t)^2+\frac{1}{2}t^2} dx \\
&= e^{\frac{t^2}{2}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-t)^2} dx \\
&= e^{t^2/2}
\end{aligned}$$

Compute the MGF of $\sqrt{n}\bar{X}_n$:

$$\begin{aligned}
E(e^{\sqrt{n}\bar{X}_n}) &= E(e^{t(X_1+\cdots+X_n)/\sqrt{n}}) \\
&= E(e^{tX_1/\sqrt{n}})E(e^{tX_2/\sqrt{n}})\cdots E(e^{tX_n/\sqrt{n}}) \\
&= \left[E(e^{tX_i/\sqrt{n}})\right]^n \qquad \text{since } i.i.d \\
&= \left[E\left(1+\frac{tX_i}{\sqrt{n}}+\frac{t^2X_i^2}{2n}+o(n^{-1})\right)\right]^n \\
&= \left[1+\frac{t}{\sqrt{n}}E(X_i)+\frac{t^2}{2n}E(X_i^2)+o(n^{-1})\right]^n \\
&= \left[1+\frac{t^2}{2n}+o(n^{-1})\right]^n \\
&= \left[1+\frac{t^2/2}{n}+o(n^{-1})\right]^n \\
&\to e^{t^2/2} \qquad \text{as } n\to\infty
\end{aligned}$$

Therefore, the MGF of $\sqrt{n}\bar{X}_n$ approaches the MGF of the standard normal. Since MGF determines the distribution, the distribution of $\sqrt{n}\bar{X}_n$ also approaches the standard normal distribution. □

The CLT tells us about the limiting distribution of $\bar{X}_n$ as $n \to \infty$. That means, we can reasonably approximate the distribution $\bar{X}_n$ with normal distribution when $n$ is a finite large number.

$$\bar{X}_n \approx N(\mu, \sigma^2/n) \quad \text{for large } n.$$

The Central Limit Theorem was first proved by Pierre-Simon Laplace in 1810. Let's take a moment to admire the generality of this result. The distribution of the individual $X_i$ can be anything in the world, as long as the mean and variance are finite. This does mean the distribution of $X_i$ is irrelevant, however. If the distribution is fairly close to normal, the result would hold for smaller $n$. If the distribution is far away from normal, it would take larger $n$ to converge.

The CLT gives the distribution of the sample mean regardless of the underlying distribution. This allows to assess the "quality" of the sample mean — how close it is to the true mean. The LLN tells us the larger the sample, the closer the sample mean to the population mean. The CLT tells us the distribution of the sample mean for sample size $n$. For smaller $n$, the distribution is more spread-out (a normal distribution with large $\sigma^2$); hence the uncertainty is huge, other values are more likely. For larger $n$, the uncertainty is reduced, most values would be centered around the true mean. We will delve deeper into this when we get to hypothesis testing.

**Example 54.1.** Suppose that a fair coin is tossed 900 times. Approximate the probability of obtaining more than 395 heads.

*Solution.* Let $H = \sum_{i=1}^{900} X_i$ be the number of heads, where $X_i \sim \text{Bern}(\frac{1}{2})$. We could compute the probability by

$$P(H > 495) = \sum_{k=496}^{900} \binom{900}{k} \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{900-k}$$

But this is quite tedious. Because $n = 900$ is reasonably large, we can apply the CLT:

$$\begin{aligned} \frac{1}{n}\sum_{i=1}^{900} X_i &\sim N(\mu, \sigma^2/n) \quad \text{or} \\ \sum_{i=1}^{900} X_i &\sim N(n\mu, n\sigma^2) \end{aligned}$$

We know $\mu = E(X_i) = \frac{1}{2}$, $\sigma^2 = Var(X_i) = \frac{1}{4}$. Thus $H \sim N(450, 225)$. Therefore,

$$P(H > 495) = 1 - P(H \leq 495) \approx 1 - \Phi\left(\frac{495 - 450}{15}\right) = 0.0013.$$

# 55 Samples and statistics

We model real-world uncertain events with random variables. We have also introduced various distributions suitable to model different kinds of events. However, we never observe the full distribution or the true parameters of the assumed distribution. Instead, we only observe a sample of that random variable. We can only infer the properties of the distribution from a limited sample. For example, suppose we model the height of an Asian women with a normal distribution. But we never know exactly what the mean and variance are. We can only observe a sample of the distribution.

In statistics, the conceptual distribution $F$ is called the **population distribution**, or just the **population**.[1] It is tempting to think of the population as all the observations (e.g. all the population on the planet), but this is not exactly correct. The population distribution is more of a mathematical abstraction or an assumption. Suppose we are modeling the height of human being, even if we have all the observations on the planet, that does not include the people that have died or yet to be born. Thus, it is still a sample of the assumed distribution.

A collection of random variables $\{X_1, X_2, \dots, X_n\}$ is a **random sample** from the population $F$ if $X_i$ are **independent and identically distributed (*i.i.d*)** with distribution $F$. What we mean by $i.i.d$ is that $X_1, \dots, X_n$ are mutually independent and have exactly the same distribution $X_i \sim F$. Survey sampling is an useful metaphor to understand random sampling, in which we randomly select a subset of the population with equal probability. The **sample size** $n$ is the number of individuals in the sample.

A **data set** is a collection of numbers, typically organized by observation. We sometimes call a data set also as a sample. But it should not be confused with the random sample defined above. As the former is a collection of random variables, whereas the latter is one **realization** of the random variables.

Typically, we will use $X$ without the subscript to denote a random variable or vector with distribution $F$, $X_i$ with a subscript to denote a random observation in the sample, and $x_i$ or $x$ to denote a specific or realized value.

The problem of **statistical inference** is to learn about the underlying process — the population distribution or data generating process — by examining the observations. In most cases, we assume the population distribution and want to learn about the its parameters (e.g. $\mu$ and $\sigma^2$ in the normal distribution). As a convention, we use greek letters to denote population parameters.

[1] This section is based on Bruce Hansen's *Probability and Statistics for Economists*.

A **statistic** is a function of the random sample $\{X_1, X_2, \dots, X_n\}$. Recall that there is a distinction between random variables and their realizations. Similarly there is a distinction between a statistic as a function of a random sample — and is therefore a random variable as well — and a statistic as a function of the realized sample, which is a realized value. When we treat a statistic as random we are viewing it is a function of a sample of random variables. When we treat it as a realized value we are viewing it as a function of a set of realized values. One way of viewing the distinction is to think of "before viewing the data" and "after viewing the data". When we think about a statistic "before viewing" we do not know what value it will take. From our vantage point it is unknown and random. After viewing the data and specifically after computing and viewing the statistic the latter is a specific number and is therefore a realization. It is what it is and it is not changing. The randomness is the process by which the data was generated — and the understanding that if this process were repeated the sample would be different and the specific realization would be therefore different. The distribution of a statistic is called the **sampling distribution**, since it is the distribution induced by sampling.

An **estimator** $\widehat{\theta}$ for a population parameter $\theta$ is a statistic intended to infer $\theta$. It is conventional to use the hat notation $\widehat{\theta}$ to denote an estimator. Note that $\widehat{\theta}$ is a statistic and hence also a random variable. We call $\widehat{\theta}$ an **estimate** when it is a specific value (or realized value) calculated in a specific sample.

A standard way to construct an estimator is by the analog principle. The idea is to express the parameter $\theta$ as a function of the population $F$, and then express the estimator $\widehat{\theta}$ as the analog function in the sample.

For example, suppose we want to construct an estimator for the population mean $\mu = E(X)$. By definition, if each value of $X$ is of equal probability, $\mu$ is simply the average. By analogy, we construct the **sample mean** as $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$. It is conventional to denote a sample average by the notation "X bar". Because it is an estimator for $\mu$, we also denote it as $\widehat{\mu} = \bar{X}_n$. Note that from different samples we calculate different estimates. In one sample, $\widehat{\mu} = 6.5$; in another sample, $\widehat{\mu} = 6.7$. All of them are erroneous estimate of the true parameter $\mu$. The question is therefore how close they are to the true parameter. To answer this question, we need to study the distribution of the sample mean.

# 56 Estimator accuracy

The central question of statistics is we want to learn about the population from a finite sample. We know sample mean is different from the population mean. But we also want to know how large the error could be, that is, how far or close the sample mean is from the true population mean. The question is exceedingly difficult to answer because the population mean is unknown. Fortunately, with the help of the CLT, we can say more about the distribution of the sample mean. This chapter bridges our probability theory with statistics. We use the theorems we have derived to infer the properties of a statistic.

As the purpose of statistics is to learn about the population, we want our sample estimator to be as good as possible. But what is a "good" estimator? This section we discuss two properties that we usually demand from a good estimator, namely, unbiased and consistency. Next section will tackle the more challenging concept of confidence interval.

**Definition 56.1.** The bias of an estimator $\hat{\theta}$ of a parameter $\theta$ is

$$\text{Bias}[\hat{\theta}] = E(\hat{\theta}) - \theta.$$

We say that an estimator is **biased** if its sampling is incorrectly centered. We say that an estimator is **unbiased** is the bias is zero.

**Theorem 56.1.** *$\bar{X}_n$ is **unbiased** for $\mu = E(x)$ if $E(X) < \infty$.*

*Proof.*

$$E(\bar{X}_n) = E\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n}\sum_{i=1}^{n} E(X_i) = \frac{1}{n}\sum_{i=1}^{n} \mu = \mu.$$

□

**Theorem 56.2.** *If $\hat{\theta}$ is an unbiased estimator of $\theta$, then $\hat{\beta} = a\hat{\theta} + b$ is an unbiased estimator of $\beta = a\theta + b$.*

But obtaining an unbiased estimator is not always as straightforward as it seems. Consider the sample variance as an estimator for the population variance. By the analog principle, the sample variance should be

$$
\begin{aligned}
\hat{\sigma}^2 &= \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2 \\
&= \frac{1}{n}\sum_{i=1}^{n}(X_i - \mu + \mu - \bar{X}_n)^2 \\
&= \frac{1}{n}\sum(X_i - \mu)^2 + \frac{2}{n}\sum(X_i - \mu)(\mu - \bar{X}_n) + \frac{1}{n}\sum(\mu - \bar{X}_n)^2 \\
&= \frac{1}{n}\sum(X_i - \mu)^2 + 2(\bar{X}_n - \mu)(\mu - \bar{X}_n) + (\mu - \bar{X}_n)^2 \\
&= \frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)^2 - (\bar{X}_n - \mu)^2 \\
&= \tilde{\sigma}^2 - (\bar{X}_n - \mu)^2
\end{aligned}
$$

We know that

$$
E(\tilde{\sigma}^2) = \frac{1}{n}\sum_{i=1}^{n} E(X_i - \mu)^2 = \sigma^2
$$

Thus, if we compute the bias of this estimator:

$$
\begin{aligned}
E[\hat{\sigma}^2] &= \sigma^2 - \frac{\sigma^2}{n} = \left(1 - \frac{1}{n}\right)\sigma^2 \\
\text{Bias}[\hat{\sigma}^2] &= -\frac{\sigma^2}{n} \neq 0
\end{aligned}
$$

Therefore, the estimator $\hat{\sigma}^2$ is a biased estimator for $\sigma^2$! To correct the bias, we divide the sample sum of squares by $(n-1)$.

$$
s^2 = \frac{n}{n-1}\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2.
$$

It is straightforward to see that $s^2$ is an unbiased estimator for $\sigma^2$. We call $s^2$ the **bias-corrected variance estimator**.

**Theorem 56.3.** *$s^2$ is an unbiased estimator for $\sigma^2$ if $E(X^2) < \infty$.*

**Definition 56.2.** The mean square error of an estimator $\hat{\theta}$ for $\theta$ is

$$
\text{MSE}[\hat{\theta}] = E\left[(\hat{\theta} - \theta)^2\right].
$$

By expanding the square we find that

$$\begin{aligned}
\text{MSE}[\hat{\theta}] &= E\left[(\hat{\theta}-\theta)^2\right] \\
&= E\left[(\hat{\theta}-E[\hat{\theta}]+E[\hat{\theta}]-\theta)^2\right] \\
&= E\left[(\hat{\theta}-E[\hat{\theta}])^2\right] + 2E(\hat{\theta}-E[\hat{\theta}])(E[\hat{\theta}]-\theta) + (E[\hat{\theta}]-\theta)^2 \\
&= Var[\hat{\theta}] + (\text{Bias}[\hat{\theta}])^2.
\end{aligned}$$

Thus the MSE is the variance plus the squared bias. The MSE as a measure of accuracy combines the variance and bias.

**Theorem 56.4.** *For any estimator with a finite variance, we have*

$$MSE[\hat{\theta}] = Var[\hat{\theta}] + (Bias[\hat{\theta}])^2.$$

**Definition 56.3.** An estimator is consistent if $\text{MSE}[\hat{\theta}] \to 0$ as $n \to \infty$.

Bias is the property of an estimator for finite samples. Consistency is the property of an estimator when the sample size gets large. It means that for any given data distribution, there is a sample size $n$ sufficiently large such that the estimator $\hat{\theta}$ will be arbitrarily close to the true value $\theta$ with high probability. In practice, we usually do not know how large this $n$ has to be. But it is a desirable property for an estimator to be considered a "good" estimator.

For unbiased estimator, MSE is solely determined by the variance of the estimator. Recall that the variance for the sample mean is $Var(\bar{X}_n) = \sigma^2/n$. But this is not a very useful formula because the it depends on unknown parameter $\sigma^2$. We need to replace these unknown parameters by estimators. To put the latter in the same units as the parameter estimate we typically take the square root before reporting. We thus arrive at the following concept.

**Definition 56.4.** A standard error of an estimator $\hat{\theta}$ is defined as

$$SE(\hat{\theta}) = \hat{V}^{1/2}$$

where $\hat{V}$ is the estimator for $Var[\hat{\theta}]$.

**Definition 56.5.** The standard error for $\bar{X}_n$ is

$$SE(\bar{X}_n) = \frac{s}{\sqrt{n}}$$

where $s$ is the bias-corrected estimator for $\sigma$.

Note the difference between **standard error** and **standard deviation**. Standard deviation describes the dispersion of a distribution. Standard error is the standard deviation of an *estimator*. It indicates the "precision" of the estimator, thereby carrying a sense of "error". The smaller the standard error, the more precise the estimator.

# 57 Confidence intervals

Confidence intervals provide a method of adding more information to an estimator $\hat{\theta}$ when we wish to estimate an unknown parameter $\theta$. We can find an interval $(A, B)$ that we think has high probability of containing $\theta$. The length of such an interval gives us an idea of how closely we can estimate $\theta$.

**Definition 57.1.** A $100(1-\alpha)\%$ confidence interval (CI) for $\theta$ is an interval $[L(\theta), U(\theta)]$ such that the probability that the interval contains the true $\theta$ is $(1-\alpha)$.

Due to randomness we rarely seek a confidence interval with 100% coverage as this would typically need to be the entire parameter space. Instead we seek an interval which includes the true value with reasonably high probability. Standard choices are $\alpha = 0.05$ and 0.10, corresponding to 95% and 90% confidence.

Confidence intervals are reported to indicate the degree of precision of our estimates. The narrower the confidence interval, the more precise the estimate. Because a small range of values contains the true parameter with high probability.

With the help of the CLT, it is not hard to find the CI for the sample mean $\bar{X}_n$. Let's set $\alpha = 5\%$, that is, we are trying to find the CI that contains the true mean 95% of the times. Assume our sample size $n$ is large enough to invoke the CLT, we thus have

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

Let's find the interval $[a, b]$ such that

$$P\left(a \leq \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \leq b\right) = 1 - 2\Phi(L) = 0.95$$

since the normal distribution is symmetric, $b = -a$. By looking at the CDF of standard normal, we get $a = -1.96$, $b = 1.96$. Thus,

$$P\left(-1.96 \leq \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \leq 1.96\right) = 0.95$$

With a little rearrangement, we have

$$P\left(\bar{X}_n - 1.96\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X}_n + 1.96\frac{\sigma}{\sqrt{n}}\right) = 0.95$$

Therefore, the interval $\left[\bar{X}_n - 1.96\frac{\sigma}{\sqrt{n}}, \bar{X}_n + 1.96\frac{\sigma}{\sqrt{n}}\right]$ contains the true mean 95% of the times.

**Theorem 57.1.** *The* $100(1-\alpha)\%$ *confidence interval for the sample mean* $\bar{X}_n$ *is* $\bar{X}_n \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$, *where* $z_{\alpha/2}$ *is the critical value such that* $\Phi(z_{\alpha/2}) = \frac{\alpha}{2}$.

In practice, because we do not know $\sigma/\sqrt{n}$, we replace it with the standard error $s/\sqrt{n}$. Thus, we compute the confidence interval as $\bar{X}_n \pm z_{\alpha/2}SE$. However, this replacement is not without risk. When the sample size is small, $s$ is a very poor estimate of $\sigma$. For the approximation to be valid, we require either the sample size is large enough ($n \geq 30$ at least) or the population distribution is nearly normal. Some commonly used confidence levels:

- 90% CI: $\alpha = 0.1$, $z_{0.05} = 1.645$
- 95% CI: $\alpha = 0.05$, $z_{0.025} = 1.96$
- 99% CI: $\alpha = 0.01$, $z_{0.005} = 2.58$

We go through some common misunderstandings about confidence intervals through an example. Suppose we have a sample fo size 50 with mean 3.2 and standard deviation 1.74. We construct the 95% confidence interval as

$$\bar{X} \pm 1.96 \times \frac{1.74}{\sqrt{50}} \approx 3.2 \pm 0.5 = (2.7, 3.7).$$

Now check the following interpretations (true or false):

1. We are 95% confident that the sample mean is between 2.7 and 3.7.
   False. The CI definitely contains the sample mean $\bar{X}$.
2. 95% of the population observations are in 2.7 to 3.7.
   False. The CI is about covering the population mean, not for covering 95% of the entire population.
3. The true mean falls in the interval (2.7, 3.7) with probability 95%.
   False. The true mean $\mu$ is a fixed number, not a random one that happens with a probability.
4. If a new random sample is taken, we are 95% confident that the new sample mean will be between 2.7 and 3.7.
   False. The confidence interval is for covering the population mean, not for covering the mean of another sample.

5. This confidence interval is not valid if the population or sample is not normally distributed.
   False. The construction of the CI only uses the normality of the sampling distribution of the sample mean (by the CLT). Neither the population nor the sample is required to be normally distributed.

So what is exactly the thing that has a 95% change to happen? <u>It is the procedure to construct the 95% interval.</u> About 95% of the intervals constructed following the procedure will cover the true population mean $\mu$. After taking the sample and an interval is constructed, the constructed interval either covers $\mu$ or it doesn't. But if we were able to take many such samples and reconstruct the interval many times, 95% of the intervals will contain the true mean.

# 58 Hypothesis testing

Confidence interval allows us to construct an interval estimate of a population parameter. Hypothesis testing allows us to test specific hypothesis about a population parameter with the evidence obtained from a sample. The earliest use of statistical hypothesis testing is generally credited to the question of whether male and female births are equally likely (null hypothesis), which was addressed in the 1700s by John Arbuthnot and later by Pierre-Simon Laplace.

Let $p$ be the population ratio (defined as the ratio of boys to the total number of babies). We hypotheses that

$$H_0 : p = 0.5$$

This is called the **null hypothesis**, which is the hypothesis we want to test. If the null hypothesis is false, we have

$$H_1 : p \neq 0.5$$

This is called the **alternative hypothesis**. How am I able to test which hypothesis is true? I can answer this question by collecting a small sample. Suppose I have collected a sample of 50 babies computed a sample ratio of $\hat{p} = 0.55$. Does it prove or disprove the hypothesis?

Note that the ratio $\hat{p}$ can be regarded as a sample mean. Let $X_i$ be a random variable that equals 1 if the $i$-th baby is a boy and 0 otherwise. Then, $\hat{p} = \frac{1}{n}\sum_{i=1}^{n} X_i$. The variance of $\hat{p}$ is given by

$$Var(\hat{p}) = \frac{1}{n^2}\sum_{i=1}^{n} Var(X_i) = \frac{p(1-p)}{n}$$

since $X_i$ is a Bernoulli random variable. By the Central Limit Theorem, we have

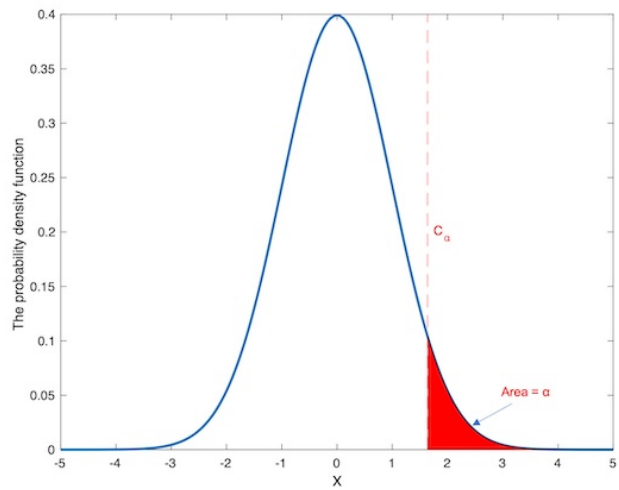
$$\frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}} \to N(0,1)$$

Suppose $H_0$ is true, then we know the distribution of $\hat{p}$. In particular, there is 95% chance that $\hat{p}$ would be in the interval

$$p \pm 1.96\sqrt{\frac{p(1-p)}{n}} = 0.5 \pm 0.14$$

Our observed sample mean $\hat{p} = 0.55$ is not outrageous. It is well within this interval. <u>That means the evidence is not against the null hypothesis.</u> It does not mean $H_0$ is true. But it is reasonable given we have observed a sample mean $\hat{p} = 0.55$.

Suppose we have observed $\hat{p} = 0.65$. This piece of evidence does not seem to be consistent with the null hypothesis. Because if $H_0$ is true, we only have less than 5% chance of observing this sample mean. It is extremely unlikely. Based on this sample, we are more inclined to reject the $H_0$. Rejecting the null hypothesis does not mean it is false, <u>but it means our evidence does not support this hypothesis.</u>




$p$**-value**: the probability of obtaining test results at least as extreme as the result actually observed, under the assumption that the null hypothesis is correct. A very small $p$-value means that such an extreme observed outcome would be very unlikely under the null hypothesis. Thus, <u>The smaller the $p$-value, the stronger the evidence against the $H_0$.</u>

In some studies, we can simply report the $p$-value and let people judge whether the evidence is strong enough. In other studies, we prefer to select a cut-off value $\alpha$, call the **significance level**, and follow the rule:

- If the $p$-value $< \alpha$, reject $H_0$;
- If the $p$-value $> \alpha$, do not reject $H_0$.

Commonly used significance levels: 0.05 and 0.01. And we like to use the word "significant" to describe the test result:

- A test with $p$-value $< 0.05$ is said to be (statistically) **significant**;
- A test with $p$-value $< 0.01$ is said to be highly **significant**.

When we make a decision about accepting or rejecting a hypothesis, there are chances that we make a mistake. There are two types of mistakes: **Type 1 error** and **Type 2 error**.

| | Decision | |
|---|---|---|
| | Reject $H_0$ | Fail to reject $H_0$ |
| $H_0$ is true | Type 1 error | ✓ |
| $H_0$ is false | ✓ | Type 2 error |

**Type 1 error** is rejecting the $H_0$ when it is true. **Type 2 error** is failing to reject the $H_0$ when it is false. Usually, it is more important to control the Type 1 error than the the Type 2 error. That is, we want to minimize the chance of falsely rejecting the null hypothesis.

In the example above, we reject the null hypothesis on the ground that there is only 2.3% of the chance that we could observe this sample. Therefore, the probability of Type 1 error is only 2.3%.

If we make decisions based on a significance level, the significance level is the Type 1 error rate. In other words, when using a 5% significance level, there is 5% chance of making a Type 1 error.

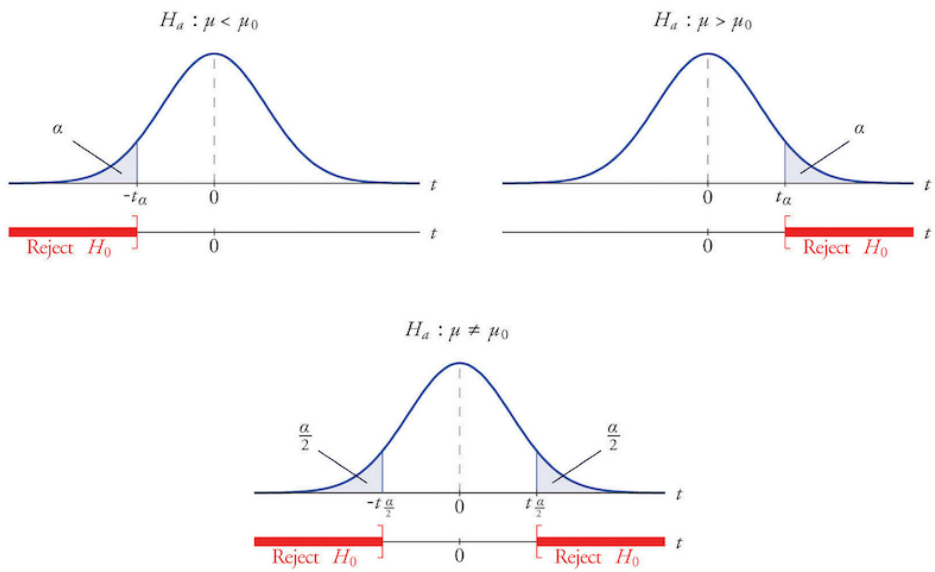
$$P(\text{Type 1 error}|H_0 \text{ is true}) = \alpha$$

This is why we prefer small values of $\alpha$—smaller $\alpha$ reduces the Type 1 error rate. However, significance level doesn't control Type 2 error rate.

### Hypothesis testing with $z$-statistics

We may have noticed that, in the above example, the assumption that the population $\sigma$ is known is unrealistic. In practice, we approximate it with the standard error $s/\sqrt{n}$. The approximate is valid if the the sample size is large enough or the underlying distribution is nearly normal. If this is not the case, we would opt for a $t$-test. Here we summarize the steps of testing for a population mean with $z$-statistics.

We notice that the **two-sided** hypothesis tests are very closed related to the concept of confidence intervals. A two-sided test means we are interested in rejection regions on both sides of the tail distribution. Typically, the alternative hypothesis is $H_1 : \mu \neq \mu_0$.




Suppose we are doing a hypothesis test under the significance level $\alpha$, the region of accepting the $H_0$ is

$$-z_{\alpha/2} \leq \frac{\bar{X} - \mu}{SE} \leq z_{\alpha/2}$$

such that the rejection region ($p$-value) has probability $\alpha$. This is equivalent to

$$\bar{X} - z_{\alpha/2}SE \leq \mu \leq \bar{X} + z_{\alpha/2}SE$$

which is exactly the $100(1-\alpha)\%$ confidence interval of $\bar{X}$. Therefore, for a two-sided test, we have the rule:

- Reject $H_0$ if $\mu$ is not in the $100(1-\alpha)\%$ CI: $\bar{X} \pm z_{\alpha/2}SE$

We conclude this chapter by reiterating a couple of critical points that could be easily misunderstood.

Rejecting $H_0$ doesn't means we are 100% sure that $H_0$ is false. We might make Type 1 errors. Setting a significance level just guarantee we won't make Type 1 error too often.

Failing to reject $H_0$ does not necessarily mean $H_0$ is true. We could make a type 2 error when failing to reject $H_0$. Moreover, unlike type 1 error rate is controlled at a low level, type 2 error rate is usually quite high. When we fail to reject $H_0$, it just means the data are not able to distinguish between $H_0$ and $H_1$. That's why we say *fail to reject*. <u>$p$-value is not the probability that the $H_0$ is true.</u>

Saying that results are statistically significant just informs the reader that the findings are unlikely due to chance alone. However, it says nothing about the practical importance of the finding. For example, rejecting the $H_0$: $\mu = \mu_0$ does not tell us how big the difference $|\mu - \mu_0|$ is. Mostly in practice we care more about the magnitude of this difference, rather than the fact that they are indeed different. It is possible that the difference is too small to be relevant even if it is significant.

### Hypothesis testing with $t$-statistics

When the sample size is small, we opt for $t$-test for more reliable hypothsis testing. Define test statistics

$$T = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

where $s$ is the sample standard deviation. For small samples, this test statistics follows a Student $t$-distribution with $n$ degrees of freedom, $T \sim t(n)$.

Why Student-$t$ distribution? Recall the definition of Student-$t$ distribution: when the underlying distribution of $X_1, X_2, \dots, X_n$ is Normal, sample variance $s^2$ follows a $\chi^2$ distribution. $T$ follows $t$ distribution by definition regardless of the sample size. However, if the underlying distribution is not normal, this argument loses ground. We use $t$-test mainly as a convention. But $t$ distribution has heavier tails than standard normal, meaning that we are more likely to reject a hypothesis based on $t$ distribution. In other words, $t$-test is a more conservative choice than $z$-test for small samples.

| one-tail $\alpha$ | 0.05 | 0.025 | 0.005 |
|---|---|---|---|
| two-tail $\alpha$ | 0.10 | 0.05 | 0.01 |
| d.f. | | | |
| 10 | 1.812 | 2.228 | 3.169 |
| 20 | 1.725 | 2.086 | 2.845 |
| 30 | 1.697 | 2.042 | 2.750 |
| $z$ value | 1.645 | 1.960 | 2.576 |

The table shows a few critical values for $t$-test with different degrees of freedom (d.f.). We can see as the sample size gets larger, $t$ distribution converges to standard normal.